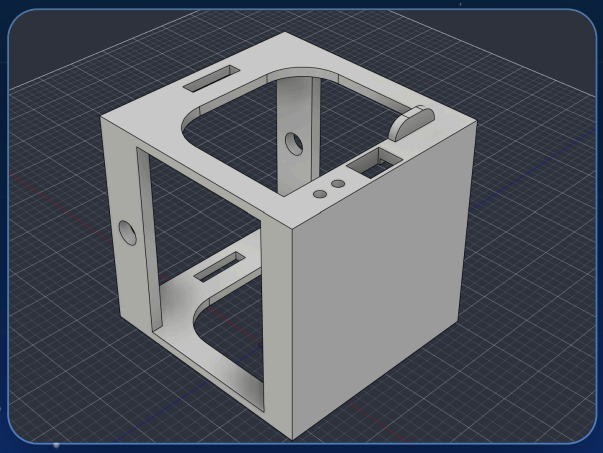
FINAL PROJECT REPORT

Team Singularity CAN-Team-25

Design, simulation, flight software, verification and flight analysis of a LoRa-telemetry CanSat that is launched from a height of about 100 ft and brought back to the ground under a 6 ft parachute — two terrace-launched flights flown at the competition on 30 September 2026.



2 flights
FLOWN AND ANALYSED

1.9–2.3 m/s
STEADY DESCENT RATE, LIMIT 5 M/S

3.09 Hz
TELEMETRY RATE IN FLIGHT, 41 PACKETS RECEIVED

6,132 checks
AUTOMATED TESTS, ALL PASSING

Contents

1 Executive summary 7

1.1 The vehicle in one page 7
1.2 The competition launch, 30 September 2026 7
1.3 Five ideas that shaped the design 7
1.4 Where each result is found 7

2 Mission and requirements 12

2.1 The mission 12
2.2 The rulebook's hard numbers 12
2.3 Requirement-to-evidence matrix 12
2.4 What the rulebook says about packet format, and how the vehicle matches it 12

3 Concept of operations 17

3.1 The flight, event by event 17
3.2 The command window and arming 17
3.3 Detecting launch and landing 17
3.4 Launch-day operating sequence 17
3.5 How the competition flights were launched 17

4 System architecture 22

4.1 The layer model 22
4.2 The end-to-end data path 22
4.3 Rates: each stage runs at what its physics allows 22
4.4 The seven design rules 22
4.5 How a claim becomes trustworthy 22
4.6 The software at a glance 22

5 Components 27

5.1 Bill of materials 27
5.2 The flight computer — Raspberry Pi Pico (RP2040) 29
5.3 The radio — SX1278 RA-02 30
5.4 The inertial sensor — MPU-6500 30
5.5 The barometer — BMP280 31
5.6 The additional sensors 31
5.7 Storage — microSD 32
5.8 Power 32
5.9 Telemetry antenna and structure parts 33

6 Electrical design 34

6.1 The power path 34
6.2 One rail, answered by measurement 34
6.3 The current budget 34
6.4 Buses, and the lesson each one taught 34
6.5 Decoupling, sized against a failure that actually happened 34
6.6 Pin assignment 34
6.7 Why a hand-assembled universal board 34

7 Mechanical design 40

7.1 The design 40
7.2 Material — white PETG 40
7.3 The mass budget 40
7.4 Egg chamber and recovery hardware 40
7.5 The built vehicle 40
7.6 Fabrication 40

8 Structural simulation 46

8.1 What was simulated 46
8.2 Material properties, and derating for a printed part 46
8.3 The studied loads against the loads the vehicle actually met 46
8.4 The Fusion 360 results 46

9 Descent system and simulation 52

9.1 What the simulation computes 52
9.2 Two answers from one model: the minimum canopy, and the canopy flown 52
9.3 The 6 ft canopy across the mass band 52
9.4 Three constraints on the canopy that are not the diameter 52
9.5 The flights against the model 52

10 Flight software 56

10.1 The flight loop 56
10.2 Sensor acquisition 56
10.3 The mission state machine 56
10.4 Start-up calibration 56
10.5 Telemetry generation and the radio 56
10.6 The fault model 56
10.7 Code originality and independence 56
10.8 Design rules, restated as behaviour 56

11 Telemetry protocol and the radio link 66

11.1 The packet 66
11.2 Two packet shapes and a 200-byte ceiling 66
11.3 The radio configuration 66
11.4 Airtime decides the rate 66
11.5 The link budget, against the flights 66

12 Onboard logging and the ground station 71

12.1 Onboard logging 71
12.2 Ground station 71
12.3 The web console at the competition 71

13 Verification and bring-up 76

13.1 The host test suite 76
13.2 Bring-up measurements 76
13.3 The measurement that characterises the vehicle 76
13.4 The documentation checker 76

14 Flight results and data analysis

14.1 What the ground station recorded 80

14.2 The telemetry stream itself 80
14.3 Flight 1 80
14.4 Flight 2 80
14.5 The two flights compared 80
14.6 Attitude and stability 80
14.7 The pressure–altitude law, and why the transmitted altitude is slightly small 80
14.8 Descent physics: drag, energy and what the vehicle felt on arrival 80
14.9 Correlations between the channels 80
14.10 The radio link in flight 80
14.11 The sound sensor in flight 80
14.12 GPS in flight 80
14.13 The post-landing session: a free calibration experiment 80
14.14 Predictions against flights 80
14.15 Summary of the flight analysis 80

15 Engineering discoveries 105

15.1 What running the vehicle taught that reading it could not 106
15.2 The defects a host suite cannot reach 106
15.3 How each discovery changed the design 107

16 Design decision register 108

16.1 How to read the register 108
16.2 System and software 108
16.3 Radio and protocol 108
16.4 Hardware, structure and recovery 108
16.5 Decisions about how the work was done 108

17 Project timeline 113

17.1 The schedule 113
17.2 The flight-day analysis was ready before the flights 113

18 Lessons and conclusion 116

18.1 Lessons learned 116
18.2 Conclusion 116

A Appendices 119

A.1 Flight configuration constants 119
A.2 Repository map 119
A.3 The complete flight dataset 119
A.4 References 119
A.5 Glossary 119

About this report

This is the complete record of the CanSat built by **Team Singularity** (packet identity **CAN-Team-25**) for the SVNIT Physics Club CanSat Competition 2026. It covers the vehicle from the first requirement to the last flight packet: what was built, why every important decision was taken the way it was, how the software thinks, what the simulations predicted, what was measured on the bench, and what the vehicle did on 30 September 2026.

Everything in one repository
Firmware, ground station, CAD, structural studies, simulations, the flight log and its analysis, and this report's source:
`https://github.com/KunjSorathiya/CanSat-Team-Singularity`

Design

Requirements, architecture, components, electrical, mechanical, structural and descent simulation — each with the reasoning behind it.

Code

Flight loop, state machine, calibration, telemetry, logging and the ground station, with the logic drawn as flowcharts and the key routines quoted.

Flights

Two descents analysed packet by packet: altitude, temperature, pressure, acceleration, orientation, descent rate, radio link, correlations.

How to read it

- **Chapters 1–4** give the mission, the requirements and the shape of the whole system.
- **Chapters 5–9** are the hardware: components, electrical design, structure, simulation and the descent system.
- **Chapters 10–13** are the software and the radio link, ground station, logging and the testing that backs them.
- **Chapter 14** is the flight analysis — the results — and **Chapter 15** collects what was learned building the vehicle.
- **Chapter 16** is the decision register: one table listing every major design decision, the alternatives, the reason, and the evidence.
- The appendices hold the pin map, the flight constants, the repository map, the complete flight dataset, references and a glossary.

Conventions

- **Times** in the flight chapter are IST (UTC + 5:30); the ground station logged UTC. **Mission time** is the vehicle's own clock, which restarts at every power-up.
- **Altitude** is the vehicle's transmitted `A-` field unless stated otherwise: pressure altitude relative to the ground baseline the vehicle sets itself at calibration. Where a corrected height is used (hypsometric, from the measured pressure and temperature) it is labelled.
- **Figures** with a blue frame and a title are drawn from the flight log by `analysis/flight-2026-09-30/launch_analysis.py`; every number in the flight chapter is in `results.json` beside it.
- The CanSat, its firmware, the ground station, the simulations, the CAD model and this report's source are all in one repository: **github.com/KunjSorathiya/CanSat-Team-Singularity**.

Data sources for the flight chapter

Source	What it is
`Team-25.xlsx`	The organizers' ground-station export: 180 rows, each a received LoRa packet with host timestamp, packet text, RSSI and SNR
Repository parser	Every row was parsed with the same `parse_packet()` the ground station uses — one packet format, one definition

Source	What it is
`flight_data_clean.csv`	The 102 distinct packets that remain after the repeated exports are removed, with derived channels
`results.json`	Every statistic quoted in Chapter 14

1

CHAPTER 1

Executive summary

One page of answers: what the CanSat is, what it did at the competition launch on 30 September, and the five ideas that shaped it.

1.1 The vehicle in one page

1.2 The competition launch, 30 September 2026

1.3 Five ideas that shaped the design

1.4 Where each result is found

1.1 The vehicle in one page

The CanSat is a can-sized satellite built around a Raspberry Pi Pico. It is launched from 100 ft (30.48 m) — the rulebook plans a drone release, and at the competition launch on 30 September the height was a building terrace and the vehicle was thrown by hand — deploys a 6 ft parachute, descends well under the 5 m/s limit, and streams rulebook-format telemetry over 433 MHz LoRa from the moment it is switched on until well after it lands. Everything it senses is also written to an onboard microSD card.

29.4 m
launch height above the ground-floor pad, Flight 1 — 96 ft; the arc of the throw peaks at 100.8 ft

29.6 m
launch height above the landing point, Flight 2 — 97 ft

2.27 m/s
steady descent rate, Flight 1 (± 0.05) — 2.2× inside the 5 m/s limit

1.88 m/s
steady descent rate, Flight 2 (± 0.02) — 2.7× inside the limit

3.09 Hz
telemetry rate in the armed flight configuration; 1.43 Hz in the pre-arm configuration; rulebook floor 1 Hz

41
packets received by the organizers' station during Flight 1, and 41 more after landing

≥ 14 dB
weakest link margin over receiver sensitivity (RSSI −109 … −79 dBm)

12.95 s
of telemetry heard after landing — 2.6× the 5 s requirement

The system has four parts that were designed together:

1. **A flight computer** — one RP2040, running a single non-blocking loop that reads an MPU-6500 IMU, a BMP280 barometer, a NEO-6M GPS and a microphone, calibrates itself on the pad, detects its own launch and landing, and transmits a rulebook packet every 0.97 s (three packets) or 0.70 s.
2. **A radio link** — an SX1278 (RA-02) LoRa module at 433 MHz, SF7 / 125 kHz, with the official launch sync word, shared by the vehicle and by the team's own ground bridge.
3. **A structure and recovery system** — a 3D-printed PETG frame with an egg chamber, a 6 ft (1.83 m) sewn canopy, sized by a closed-form descent model, a manual switch and a power LED.
4. **A ground station and an analysis pipeline** — a bridge Pico, a CRC-framed USB link, a parsing/validating Python pipeline, a single-file web console, and the post-flight analysis that produced Chapter 14.

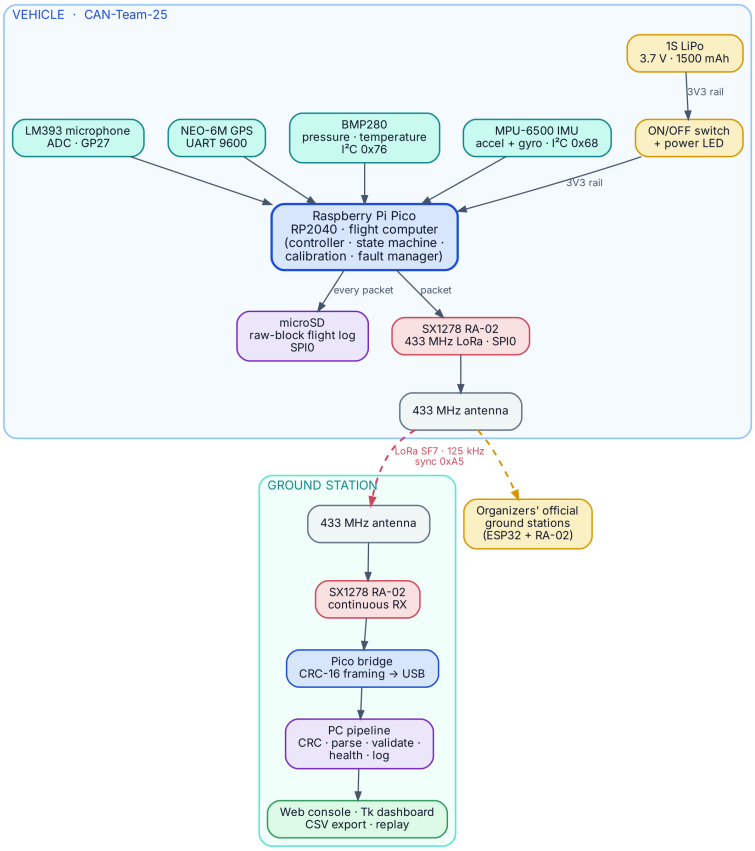


Figure 1 · The whole system: the vehicle, the radio link, and the team's ground station beside the organizers' official stations.

1.2 The competition launch, 30 September 2026

Both competition flights were launched by hand from a building terrace at about 30 m — the height the rulebook equates to an eight-storey building — rather than from a drone. The organizers' ground station recorded the two descents, plus two shorter captures. All four are analysed in Chapter 14.

Table 1 · **The four captures in the organizers' log**

Capture	Packets	What it shows
Flight 1 — 18:25 IST	41 (P-1585 … P-1625)	Powered on at the ground floor, carried up the building, held at the terrace edge at 29 m, **thrown**, canopy opening, steady descent to within a few metres of the landing surface. The vehicle reports `FLIGHT` throughout. **41 packets received**
Flight 1, after landing	41 (P-001 … P-041)	13 s on the ground after landing: the flight computer re-initialised itself, resumed telemetry at the max rate within about 2 s, re-calibrated in 5.5 s and re-armed
Flight 2 — 18:46 IST	18 received (P-148 … P-171)	Powered on at the terrace, thrown, and a 29.6 m descent, 15.4 s long, from the first packet to touchdown
Pad capture — 17:55 IST	2 (P-420, P-421)	Two packets from a vehicle standing on the ground at 0 m

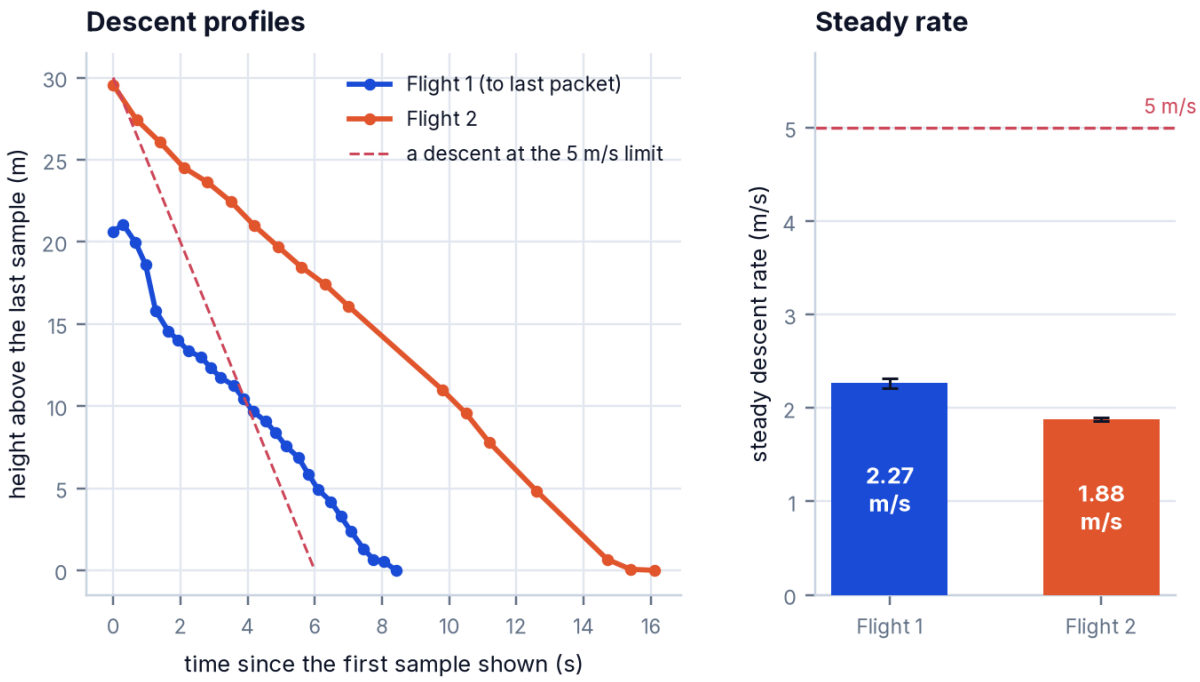


Figure 2 · The two descents on one axis, with the 5 m/s limit drawn as a reference line, and the steady rates.

THE HEADLINE

Both flights descended at roughly 38–45 % of the permitted rate. Flight 1 settled at 2.27 m/s about 1.3 s after the apex of the throw and held it to the end of the record; Flight 2 held 1.88 m/s from its first sample to touchdown. The slower descent buys a longer, gentler flight — 15.4 s for the 29.6 m of Flight 2 — and a landing equivalent to a fall of only 18–26 cm.

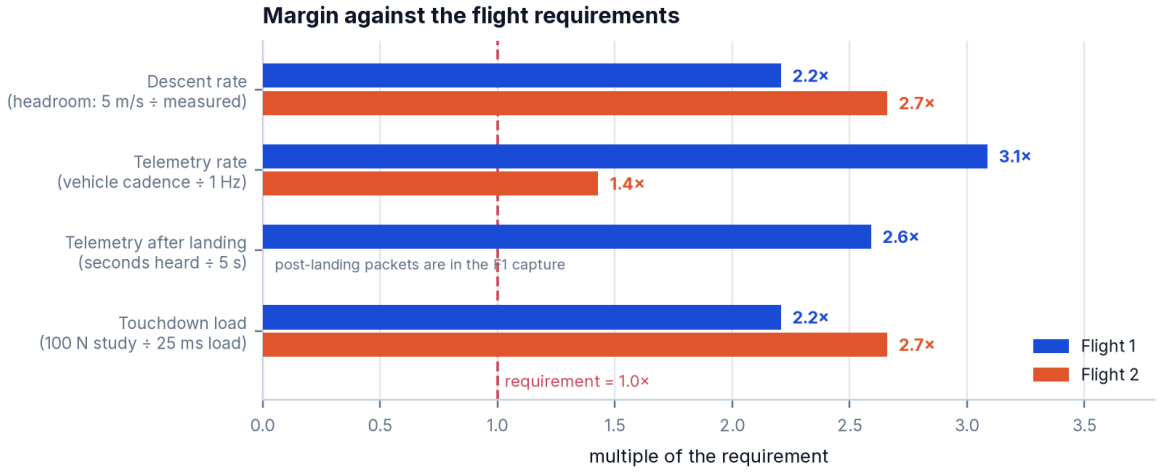


Figure 3 · Each measured quantity as a multiple of the requirement it answers. 1.0 is the requirement.

1.3 Five ideas that shaped the design

1. Telemetry never stops. No peripheral failure, no mission state — not even `FAULT` — can silence the radio. The loop never blocks on hardware, every recovery path is bounded, and a 2 s hardware watchdog restarts the flight computer if anything hangs. The flight data shows this working: after landing, the vehicle restarted itself and was transmitting again within about two seconds.

2. Wrong data is worse than no data. A mandatory field that cannot be trusted suppresses the packet instead of sending a plausible-looking wrong value — and a suppressed packet does not consume a packet number, so a gap at the ground station always means radio loss and never a sensor decision.

3. The flight core knows no hardware. All mission logic compiles on a laptop against six abstract interfaces. That is why 5,485 C++ assertions could exercise the entire flight before a single component had been unpacked, and why the logic met real hardware only after thousands of assertions had already exercised it.

4. Every number says where it came from. Rulebook, datasheet, textbook, or measurement — named in a comment at the definition. A documentation checker fails the build when a number in the prose disagrees with the source that defines it.

5. Predictions are written down before measurements. The bring-up record lists 94 predicted values; each was entered before the instrument was touched. Chapter 14 closes the same loop for the flight: the descent model, the link budget, the airtime and the structural load cases are each compared with what the flights measured.

1.4 Where each result is found

Question	Where
What does the mission require, and how is each requirement met?	Chapter 2
What does the vehicle do, minute by minute?	Chapter 3
Why these components?	Chapters 5 and 16
How is the structure strong enough?	Chapter 8
Why does the parachute work the way it does?	Chapter 9
How does the software decide that it has launched or landed?	Chapters 3 and 10
What exactly goes over the air?	Chapter 11
What did the flights show?	**Chapter 14**

2

CHAPTER 2

Mission and requirements

The rulebook turned into thirty-odd testable requirements — each with a design response and the evidence that it holds.

2.1 The mission

2.2 The rulebook's hard numbers

2.3 Requirement-to-evidence matrix

2.4 What the rulebook says about packet format, and how the vehicle matches it

2.1 The mission

The CanSat is powered on at ground level and begins transmitting at once. It is lifted to **100 ft (30.48 m) — the rulebook plans a drone, which releases it; at the competition launch the height was a building terrace and the vehicle was thrown by hand** — and launched. It deploys a parachute, descends at **no more than 5 m/s**, transmits throughout, keeps transmitting for **at least five seconds after impact**, and is then recovered.

Flight is **fully autonomous.** There is no launch command, no arming button and no manual trigger anywhere in the firmware: the vehicle powers on, calibrates itself, arms itself, detects its own launch and its own landing, and keeps talking through every failure it can survive. The reason is simple — an operator cannot be inside a loop that closes in under fifteen seconds, whether the launch is a drone release or a throw, and a design with a human trigger has a human-sized failure mode.

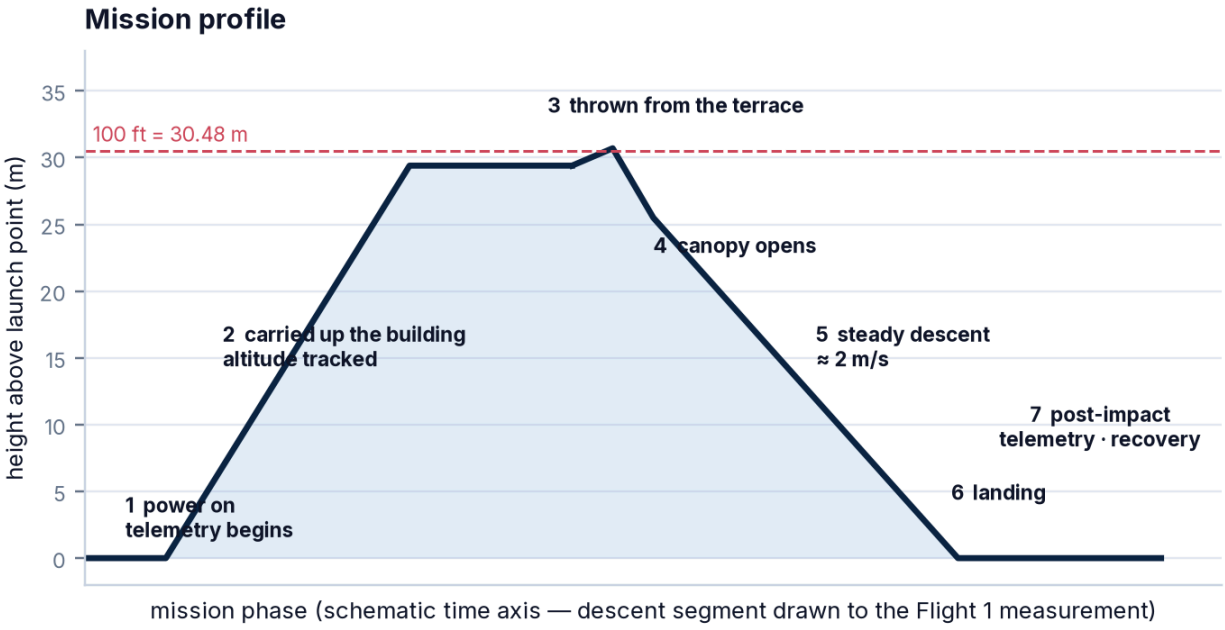


Figure 4 · The mission profile. The descent segment is drawn to the Flight 1 measurement (held at 29.4 m, thrown, canopy opening, steady descent).

2.2 The rulebook's hard numbers

Table 2 · The quantities the rulebook fixes, with the requirement that carries each

Quantity	Value	Requirement	The design's figure
Body envelope	21 cm (+ 7 cm egg chamber) × 12 cm	GEN-004	**118.5 × 115.0 × 110.0 mm** — 91.5 mm of height unused
Mass	500 g ± 10 % (450–550 g)	GEN-005	In the band; 280 g bare structure + avionics, trimmed to the band
Launch altitude	100 ft / 30.48 m (drone release; equivalent to an eight-storey building)	MIS-001	Flights launched by hand from a terrace at **29.4 m and 29.6 m** (96–97 ft)
Descent rate	≤ 5 m/s	REC-005	**2.27 and 1.88 m/s** steady, measured
Telemetry rate	≥ 1 packet per second	TEL-005	**1.43 Hz** pre-arm, **3.09 Hz** armed

Quantity	Value	Requirement	The design's figure
Post-impact telemetry	≥ 5 s	REC-008	**12.95 s** heard after Flight 1's landing
Radio	433 MHz LoRa; sync `0xF3` test / `0xA5` launch	TEL-023/024	`0xA5` on both ends, always
Packet	`CAN-Team-XX; P-XXX; Ti-…; A-…; …`	TEL-009 … 020	Byte-exact; 180 / 180 logged rows parse

Three rulebook ambiguities, each resolved before it could matter

The original guidelines contradicted themselves on dimensions and launch altitude. Neither was guessed at locally; both were escalated to the organizers, and the 2026 revision settled both:

Was ambiguous	Resolution
Dimensions	**21 cm (+ 7 cm for the egg chamber) × 12 cm**, now stated identically in two places
Launch altitude	**100 ft, released from a drone** ("equivalent to an 8-story building"), now stated identically in two places
"12 cm across" — a width or a diameter?	For a prismatic body the two differ by 33 %. The organizers confirmed on 9 September 2026 that **a 12 cm sided box is acceptable**, so the section limit is a 120 mm square

> **WHY A BOX RATHER THAN A CYLINDER**
> A 120 mm circular bore would cap a flat 100 × 100 mm board's corner-to-corner diagonal (141 mm) well outside the circle, forcing the electronics to stand on edge. A sided box lets the board lie flat, which is the layout with the shortest wires, the best access for assembly and the most predictable centre of mass. The written confirmation from the organizers settled the question before the CAD was committed.

2.3 Requirement-to-evidence matrix

Thirty requirements were extracted from the rulebook; each has a design response, a verification method and a recorded status in `documentation/requirements/requirements.md`. The matrix below is its summary. **Evidence** names the measurement or test; flight evidence refers to Chapter 14.

Table 3 · Requirement-to-evidence matrix

Requirement	Design response	Evidence
GEN-002/003 Self-built, no kits	Every module bought individually and identified on arrival; board hand-soldered; structure printed; canopy sewn	Receiving-inspection record; photographs of every part
GEN-004 Envelope	`Cansat_D1` : 118.5 × 115.0 × 110.0 mm, read from the STEP file by a script that fails the build on any disagreement	`tools/cad_dimensions.py`
GEN-005 Mass band	Mass budget from weighed parts (110.6 g board, 40.7 g battery); bare vehicle 280 g, trimmed into 450–550 g	Scale readings 9 and 12 Sep
MIS-003 Altitude ≈ 0 on the pad	Pad calibration averages 80 barometer samples into a ground baseline	Resting altitude −0.2 … +0.2 m (G1 session); −0.5 … +0.1 m over 62 bench packets
MIS-004 Telemetry reflects the lift	Altitude in every packet, from the same compensated reading the flight core uses	Flight 1: 27.6–27.9 m reported while held at the terrace edge

Requirement	Design response	Evidence
MIS-005 Parachute deployment	6 ft (1.83 m) canopy, external / semi-exposed so it opens as soon as the vehicle is launched	Flight 1: 7 m/s transient → 2.3 m/s in about 1 s, 2.1 g opening load
PAY-002 Egg chamber	Cushioned chamber designed into the frame within the +7 cm allowance	CAD; 128.7 g printed structure including the chamber
PWR-001/002 Switch and LED	Switch in the battery lead; power LED on the regulated 3.3 V rail, not a GPIO	Rocker switch on short leads outside the frame, reachable without opening the vehicle (Figure Figure 15)
PWR-004 Auto-transmit on power-up	No arming step anywhere between power and the first packet	Telemetry runs from mission time 0 (post-landing session: P-001 at `Ti-00:00:00:000`)
TEL-001…004 Continuous telemetry	Scheduler independent of mission state; `FAULT` does not stop transmission	Flight 1: 41 packets received; after landing 41
TEL-005 ≥ 1 Hz	700 ms period (1.43 Hz); 3-slot pattern (3.11 Hz) after the command window	Measured 1.43 Hz and 3.09 Hz in the flights
TEL-006/010 Team identity	Identity is configuration; the placeholder `CAN-Team-XX` is refused at start-up	180 / 180 rows carry `CAN-Team-25`
TEL-007/008 `P-001` start, sequential	Counter starts at 1; suppression does not consume a number	G1 session starts at P-001
TEL-009…020 Field order and precision	Byte-exact formatter; C++, Python and JavaScript parsers held to one fixture file	180 / 180 rows parse under the strict parser
TEL-021 No corrupted mandatory fields	Per-field validity flags; an incomplete set yields no packet	Test suite; no invalid packet in the log
TEL-023/024 Sync words	`0xA5` for launch; both Picos fly it always	Organizers' station received the flights
SEN-001…010 Mandatory sensing	BMP280 (pressure, temperature, altitude), MPU-6500 (3-axis accelerometer, 3-axis gyro), Mahony attitude filter	All channels present and consistent in every flight packet
SEN-011 Additional sensors	NEO-6M GPS (position, altitude), LM393 microphone (acoustic level)	`GP-` and `SN-` fields in the rich packets
REC-005 ≤ 5 m/s	Model minimum 80 cm (5 m/s at 550 g, 35 °C); **6 ft canopy flown** (model: 1.9–2.2 m/s)	**2.27 and 1.88 m/s** steady
REC-006 Stable descent	6 ft canopy; attitude filter reports roll/pitch/yaw each packet	Swing ≤ 18° (F1) and ≤ 46° (F2); upright throughout both descents
REC-007 Structure intact	Open-frame PETG; structural safety factors 7.6–40.9	Touchdown load 1.6 g measured
REC-008 ≥ 5 s post-impact	`LANDED` holds a 5 s window; telemetry never stops	12.95 s of packets heard after Flight 1 landing

2.4 What the rulebook says about packet format, and how the vehicle matches it

The rulebook gives one packet format and one rule above all others — *mandatory fields first, optional fields only if bandwidth allows.* The vehicle's packet is that format, byte for byte, with three optional groups appended in priority order: GPS, then sound, then a status field.

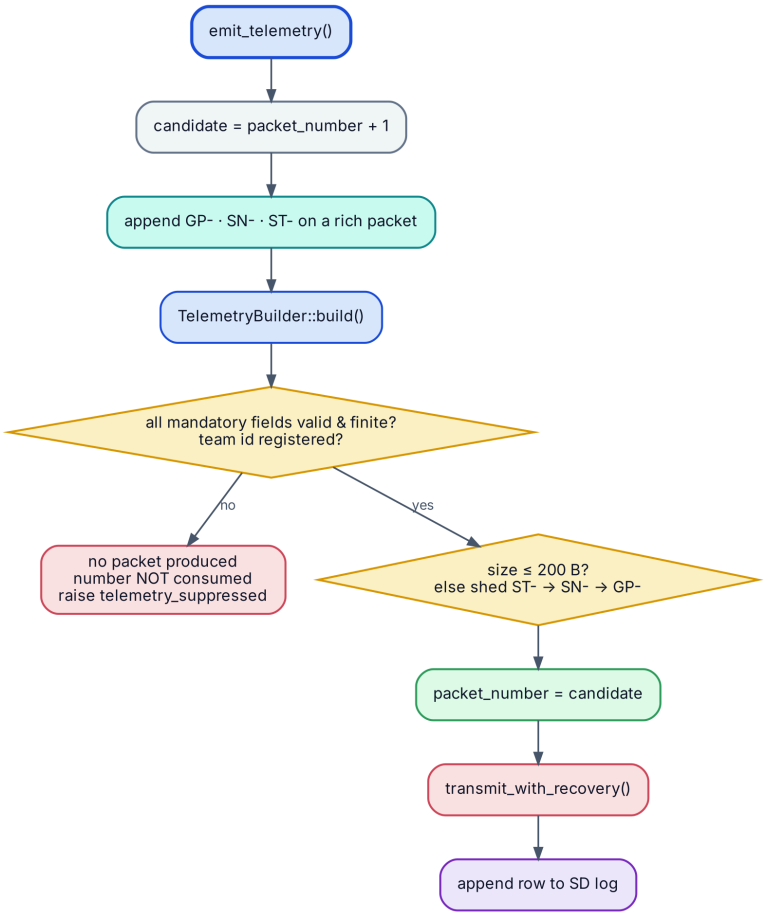


Figure 5 · The path from a snapshot to a packet. Nothing is built if any mandatory field is invalid, and the packet number is consumed only when a packet is actually produced.

Section 11 specifies the packet in full; the flight chapter reproduces real packets from both flights.

3

CHAPTER 3

Concept of operations

The flight as the vehicle lives it: power-on, the command window, arming, launch, deployment, landing — and the logic that detects each.

3.1 The flight, event by event

3.2 The command window and arming

3.3 Detecting launch and landing

3.4 Launch-day operating sequence

3.5 How the competition flights were launched

3.1 The flight, event by event

Table 4 · **Mission milestones and where each number comes from**

Milestone	When	Where the number comes from
Telemetry begins	Immediately at power-on, at 1.43 Hz	No trigger exists in the firmware
Command window	0 – 300 s	`command_window_ms` . The vehicle listens between packets and does not yet arm
Window closes	At 300 s, or on an accepted `MAX_RATE`	Whichever is first
Max rate	From the close	`auto_max_rate` — **3.09–3.11 Hz measured**
Armed	3 s after the close, once re-calibration settles	`arming_delay_ms` ; visible as `ST-R11…`
`READY → FLIGHT`	15 m above the pad, held 300 ms (or > 30 m/s² boost) — met while the vehicle is carried up	`launch_altitude_gain_m`
Launch altitude	30.48 m	Rulebook MIS-001 — flown from a terrace at 29.4 / 29.6 m
Landing detected	Descent observed, then 3 s at rest	`landing_confirm_ms`
Post-impact window	5.0 s, then `RECOVERY`	`post_impact_transmission_ms`

The pad phase is up to five minutes; the flight is under fifteen seconds of it. That ratio shaped the design: almost everything the vehicle has to get right — calibration, arming, link health — happens while it is sitting still on the ground, where there is time, and almost nothing has to be decided in the air.

3.2 The command window and arming

A vehicle that must transmit from power-on has one awkward property: it cannot tell the ground station anything until something is listening, and it cannot be told anything unless it listens. The sealed flight image resolves this with a five-minute **command window**: after power-on the vehicle transmits at 1.43 Hz so that a command can be heard in the gap between packets. When the window closes — or when an authorised `MAX_RATE` command closes it early — the vehicle moves to its fastest transmit pattern by itself, **discards its power-on calibration, re-calibrates where it now sits (on the pad, after the operator has finished with it)**, and arms.

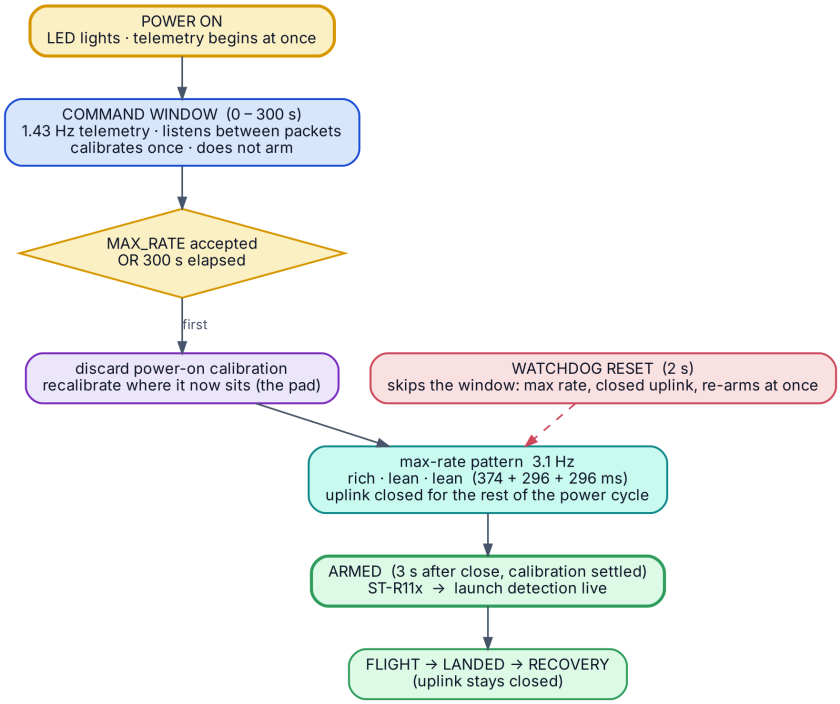


Figure 6 · The power-on sequence. The watchdog path skips the window: a reset may come mid-flight, and five minutes unarmed and listening would be five minutes without launch or landing detection.

WHY A WINDOW, AND WHY RE-CALIBRATE WHEN IT CLOSES

- **Why a window at all.** The rulebook forbids a manual trigger for telemetry, but the team still needs to change the transmit pattern on the pad without re-flashing. A listening window that closes by itself keeps both rules: nothing is required of an operator, and an operator can still intervene.
- **Why 1.43 Hz inside it.** Listening needs gaps between packets. At 700 ms a 324 ms packet leaves 376 ms of receive time per cycle, and the rate stays above the 1 Hz floor with 43 % margin.
- **Why re-calibrate at the close.** The power-on calibration was taken wherever the vehicle happened to be switched on. The pad is where the baseline must be zero, so the vehicle re-takes it there — gyro bias, accelerometer offset and the barometric ground reference.
- **Why the uplink then closes for good.** Once armed, the vehicle is in every state that holds a log that cannot be recreated. Closing the receiver removes a whole class of ways to disturb it.

The operator sees arming in the `ST-` status field on every rich packet — `ST-R004` (ready, not armed, 4 faults counted) becomes `ST-R113` (ready, **armed**, **calibrated**) — and in the packet rate rising from 1.43 Hz to about 3.1 Hz. The post-landing session in Chapter 14 shows the sequence exactly: `R004` for the first 5.5 s, then `R113` once calibration settled.

3.3 Detecting launch and landing

The state machine has to answer two questions with only a barometer and an accelerometer: *has the vehicle been launched?* and *has it landed?* Both look simple. Neither is.

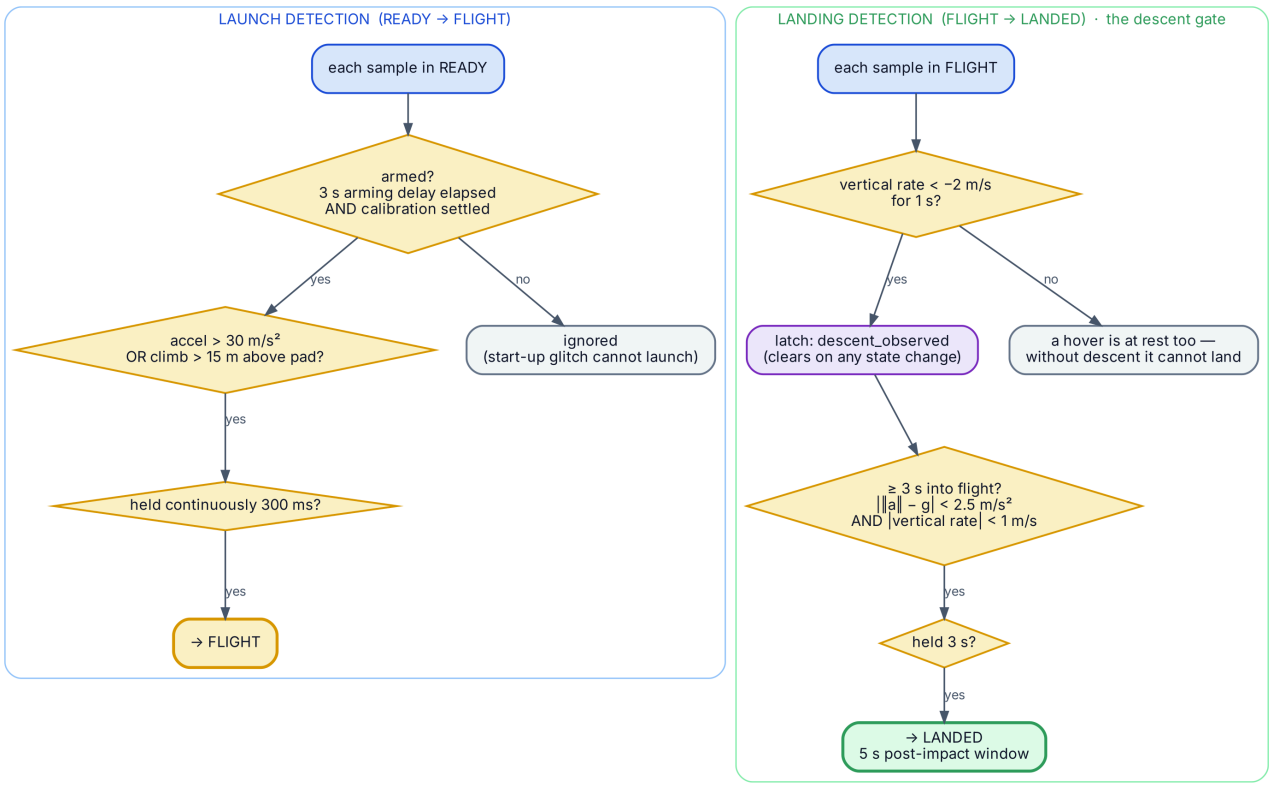


Figure 7 · Launch detection (left) and landing detection with the descent gate (right).

Launch

`READY → FLIGHT` requires the vehicle to be **armed** *and* either a boost above 30 m/s² *or* a climb of more than 15 m above the pad, **held for 300 ms**. Three guards sit in that sentence:

- **Armed** — arming needs the 3 s delay and a settled calibration, so a start-up glitch cannot launch the state machine.
- **Either condition** — a drone lift, or being carried up a building, is a slow climb with almost no acceleration; a throw is a short sharp boost (5.2 g in Flight 1). Both must be recognised, so either may trigger.
- **Held 300 ms** — a knock on the table is shorter than 300 ms; a launch is not.

Landing, and the problem that the descent gate solves

A vehicle descending under a parachute at a steady rate has no net acceleration: the accelerometer reads about 1 g, exactly as it does sitting on the ground. Landing therefore cannot be an accelerometer test. It is the **vertical rate** that discriminates — a real descent is several metres per second against a 1 m/s threshold.

Writing this concept of operations exposed a subtler problem. A vehicle **hovering under a drone** — or **held still at the edge of a terrace** — also reads 1 g and also has near-zero vertical speed. Run against the real state machine with a 3 m/s climb and a 20 s hover, the original logic produced this:

```
t= 10362 ms  alt=  16.1 m  rate= +3.0 m/s   READY → FLIGHT
t= 18018 ms  alt=  30.0 m  rate= +0.0 m/s   FLIGHT → LANDED     ⟵ still under the drone
t= 23034 ms  alt=  30.0 m  rate= +0.0 m/s   LANDED → RECOVERY   ⟵ 12 s before release
```

The fix is physical rather than tuned: **a vehicle cannot land without descending first.** The state machine latches `descent_observed_` once the vertical rate has been below −2 m/s for one second, and refuses `FLIGHT → LANDED` until it is set. The latch belongs to one `FLIGHT` and clears on any state change; `validate_config()` refuses a

descent threshold at or below the at-rest threshold. No hover, however long or gentle, can satisfy it.

> **WHY THIS MATTERED AT THE COMPETITION**
>
> Flight 1 was held still at the terrace edge, at 29.4 m, for at least 4.3 s before the throw (the first 14 recorded packets) — exactly the condition of a vehicle hovering under a drone: 1 g and no vertical motion. The log shows the vehicle reporting `FLIGHT` throughout the hold and the descent — it never mistook the hold for a landing. The gate is not a theoretical nicety: it is the reason the state field is correct for the whole of Flight 1.

3.4 Launch-day operating sequence

Step	Operator action	What the vehicle does	What confirms it
1	Close the power switch on the ground floor	LED lights at once; telemetry begins; pad calibration	Power LED; packets on the ground station
2	Wait for the command window to close (or send `MAX_RATE`)	Re-calibrates on the pad, then arms	`ST-R11x`; rate rises to ≈ 3.1 Hz
3	Hand over to the drone (rulebook plan) or carry the vehicle up the building (competition launch)	Telemetry continues; altitude tracks the lift	Altitude rising in every packet
4	Drone releases at 100 ft (rulebook plan) / thrown from the terrace (competition launch)	`READY → FLIGHT` (during the lift); canopy deploys	Altitude peak, acceleration spike, descent rate
5	Vehicle lands	Descent gate opens; 3 s at rest → `LANDED`; 5 s window; `RECOVERY`	Altitude flat at the ground; packets keep arriving
6	Recover the vehicle; read the card	The SD log holds every packet and the full-precision GPS	`tools/read_flight_log.py`

Operating rule: never connect USB while the battery is connected. USB and the battery are alternative sources for the same Pico supply input, so the vehicle is programmed and bench-tested on USB with the battery disconnected, and flown on the battery alone.

3.5 How the competition flights were launched

The rulebook plans a drone release; at the competition launch on 30 September the same 100 ft was obtained from a building terrace, and the vehicle was **thrown by hand like a projectile**, parachute deploying behind it. The procedure was the same as in Section 3.4 with the lift and the release replaced by a staircase and a throw:

Flight	Power-on	Carried / held	Launch
1	At the ground floor, 18:14 IST; command window closed and the vehicle armed at 18:19	Up the building: the climb past 15 m met the launch condition and the state field read `FLIGHT` for the rest of the record; held still at the terrace edge for at least 4.3 s	Thrown at 18:25:51: a 5.2 g impulse, an upward speed of about 5 m/s, an arc 1.4 m above the hold point, then the canopy
2	On the terrace, 18:44 IST — so its altitude zero is the terrace and the descent reads −26.9 m	Held on the terrace; still inside the five-minute command window (`ST-R003`, 1.43 Hz)	Thrown at about 18:45:50; the log opens just above the terrace level (+1.0 m), at the top of the throw

A hand throw is a demanding test of the recovery system: the vehicle leaves the hand moving upward at about 5 m/s with no drone to carry it clear, so the canopy must open from a ballistic start — and it did, within about a second of the apex of the throw.

4

CHAPTER 4

System architecture

Layers, data path, rates, and the seven rules the code is built around — and why each exists.

4.1 The layer model

4.2 The end-to-end data path

4.3 Rates: each stage runs at what its physics allows

4.4 The seven design rules

4.5 How a claim becomes trustworthy

4.6 The software at a glance

4.1 The layer model

The system is layered, and each layer depends only on the one beneath it. **The flight core never includes a Pico SDK header.** Hardware reaches it only through six abstract interfaces — `Imu`, `Barometer`, `Gps`, `Radio`, `SdLogger`, `BoardIo`. Tests substitute mocks; the vehicle substitutes `flight::pico::*`.

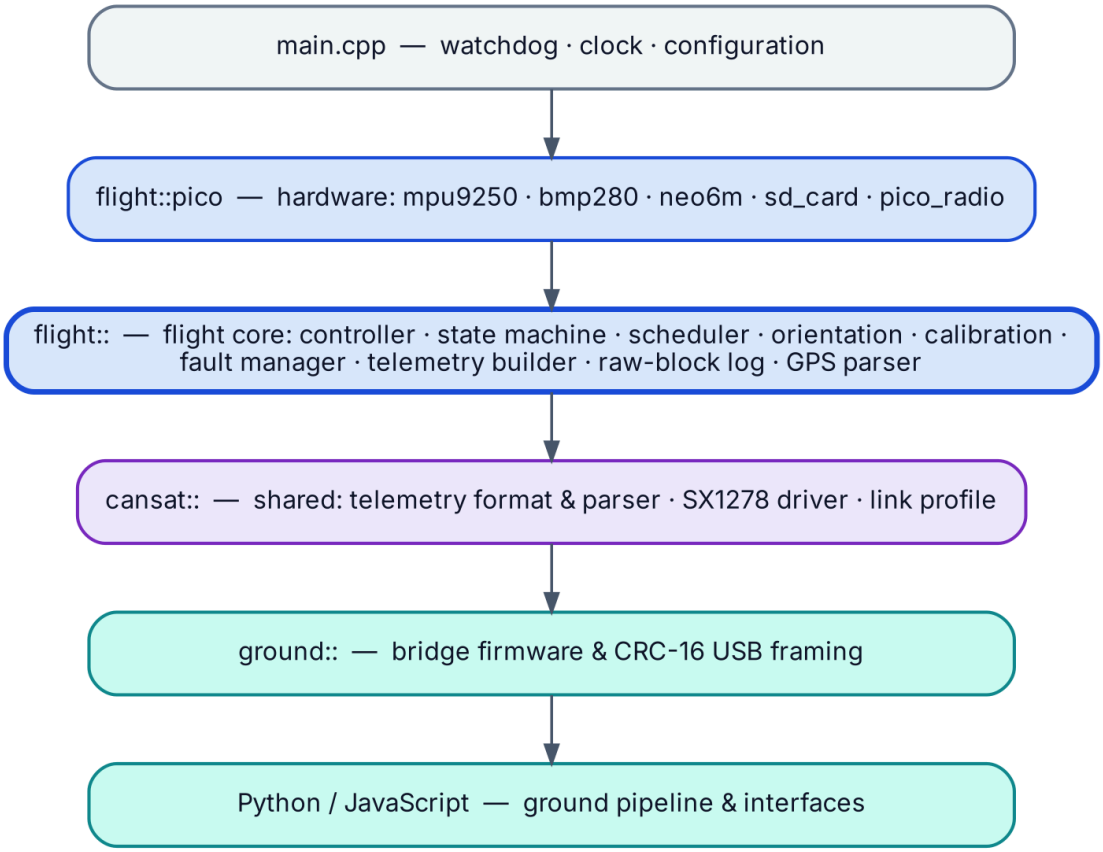


Figure 8 · Each layer depends only on the layer beneath it. The flight core (bold) is hardware-free.

WHY THIS RULE

A flight computer has a hardware-dependent part (pins, buses, timing) and a logic part (what a launch looks like, when to transmit). If the two are mixed, the logic can only be tested with the hardware attached — which means late, slowly, and never exhaustively. Separated by six small interfaces, the logic runs on a laptop at full speed: **5,485 C++ assertions exercise the entire mission logic with no board present**, including failure cases (a sensor that stops, a radio that refuses to transmit, a card that fills up) that would be dangerous or impossible to produce on demand with hardware. The software was therefore complete and tested before any component had been unpacked, and the hardware work could concentrate on what only hardware can reveal.

4.2 The end-to-end data path

```
sensors → controller → plausibility gate → bias correction → orientation
        → AGL and vertical rate
        → TelemetryBuilder   (nothing is built if any mandatory field is invalid)
        → packet string → SX1278 → 433 MHz LoRa → SX1278 → bridge Pico
        → CRC-framed USB serial
        → transport → CRC check → parse → validate → health → log
        → dashboard, web console, CSV export
```

A packet becomes a telemetry point only if it survives every stage, **and each stage's rejection is counted separately** — so a transport fault is never mistaken for a sensor fault.

4.3 Rates: each stage runs at what its physics allows

The radio is the slowest stage by two orders of magnitude and nothing else is tied to it.

Table 5 · The rate split

Stage	Rate	Bounded by	Why this value
Main loop tick	500 Hz (2 ms)	Scheduling jitter and the GPS UART FIFO	At 9600 baud the RP2040's 32-byte FIFO fills in 33 ms; a 2 ms tick drains it long before, and jitter stays under 6 % of the sensor period
Sensor acquisition, attitude, vertical rate	**30 Hz** (33 ms)	Barometer conversion time	The BMP280 delivers ~83 Hz; sampling at 30 Hz stays inside 2.4× margin so every sample is a fresh conversion
Mission state machine	30 Hz	Fed each acquisition	A 33 ms decision latency against a 300 ms launch confirmation
Battery sample, health refresh	1 Hz	Nothing meaningful changes faster	
SD flush	0.5 Hz	Write latency	Appends happen per packet; the flush is the sync
RF telemetry	**1.43 Hz → 3.11 Hz**	**LoRa airtime**	Chapter 11

30 Hz of **RF** telemetry is physically impossible with this radio and packet — the best case in the airtime table (SF7 at 250 kHz) reaches 5 Hz at 100 % channel occupancy. 30 Hz of **acquisition and state estimation** is possible, and that is what the loop does. Decoupling the two rates rather than faking one is the architecture's central timing decision.

4.4 The seven design rules

These are the invariants the code is built around. Breaking one is a design change, not a refactor.

#	Rule	Reason
1	**Telemetry never stops.** No peripheral failure and no state, `FAULT` included, suppresses telemetry that can still be produced correctly	The telemetry link is the one thing every other judgement about the flight depends on
2	**Wrong data is worse than no data.** Mandatory fields that cannot be trusted suppress the packet rather than send a plausible wrong value	A silent wrong value cannot be filtered downstream; a gap can
3	**Sequential numbering is over transmitted packets.** Suppression does not consume a number	A gap at the receiver then means radio loss and nothing else

#	Rule	Reason
4	**The flight core knows no hardware**	Testability — Section 4.1
5	**Nothing in the flight loop blocks or allocates.** Fixed-size buffers, bounded loops	A blocking call or a heap failure in a 2 ms loop is a mission failure; bounded code has bounded worst cases
6	**Every constant says where it came from** — rulebook, datasheet, textbook or engineering choice	A number with no source cannot be reviewed, and a tuned value with no record is lost
7	**Predictions stay labelled as predictions** until a measurement replaces them	It keeps the model honest — the analysis in Chapter 14 can only compare predictions to flights because the predictions were recorded first

4.5 How a claim becomes trustworthy

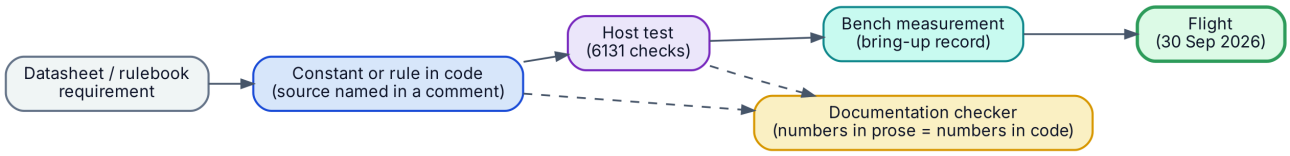


Figure 9 · A requirement is traced from the rulebook to flight. The documentation checker ties the numbers in the prose to the source that defines them.

`tools/check_doc_claims.py` reads numbers out of the documentation and checks them against the source that defines them — pin assignments, telemetry rates, watchdog periods, packet sizes, test counts, rulebook constants, the generated netlist, the descent model's own output, the mass-budget arithmetic, and every relative link and heading anchor across 53 documents. It runs in the same build as the firmware tests, so **a document that drifts from the code fails the build.**

4.6 The software at a glance

Table 6 · **The flight computer's modules**

Module	Responsibility
`controller.cpp`	The flight loop orchestrator — the one place mission behaviour is composed
`state_machine.cpp`	`INIT → SELF_TEST → READY → FLIGHT → LANDED → RECOVERY`, plus `FAULT`
`scheduler.cpp`	`PeriodicTask` — fixed-period, non-allocating, stall-tolerant timers
`orientation.cpp`	Mahony complementary filter on a unit quaternion: gyro propagates, accelerometer corrects roll and pitch
`sensor_math.cpp`	Accelerometer/gyro scaling, Bosch BMP280 compensation, barometric altitude
`startup_calibration.cpp`	Pad calibration: gyro bias, a rotation-invariant accelerometer scale, barometric reference
`telemetry_builder.cpp`	Snapshot → record → packet string → SD CSV row
`fault_manager.cpp`	Fixed-size fault store indexed by enum; never grows
`raw_block_log.cpp`	Append-only 512-byte-block log, no filesystem
`gps_parser.cpp`	Streaming NMEA-0183 (GGA and RMC) with checksum validation
`sx1278.cpp` (shared)	Register-level LoRa driver, driven through a callback struct so it runs on the vehicle, the bridge, or a fake register bank

Module	Responsibility
`telemetry.cpp` (shared)	Rulebook formatter and strict parser

5

CHAPTER 5

Components

Every part on the vehicle: what it is, what it does, why it was chosen over the alternatives, and how it was verified.

5.1 Bill of materials

5.2 The flight computer — Raspberry Pi Pico (RP2040)

5.3 The radio — SX1278 RA-02

5.4 The inertial sensor — MPU-6500

5.5 The barometer — BMP280

5.6 The additional sensors

5.7 Storage — microSD

5.8 Power

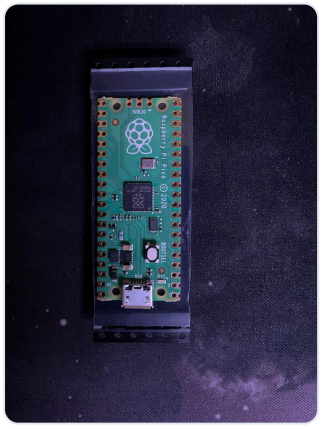
5.9 Telemetry antenna and structure parts

5.1 Bill of materials

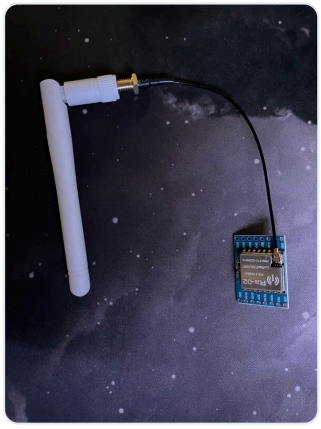
Every module was bought individually, photographed on arrival, and identified from the board and — where the silkscreen was not enough — from a register read over the bus. **The datasheet describes a chip; a photograph and a register read describe the part that was delivered.**

Table 7 · **Bill of materials**

Subsystem	Component	Qty	Purpose
Flight computer	Raspberry Pi Pico (RP2040)	2	Vehicle flight computer + ground bridge
Telemetry	SX1278 RA-02 433 MHz LoRa module	2	Vehicle + ground radio
Telemetry	433 MHz whip antenna, SMA	2	Radiating element on each end
Telemetry	10 cm IPEX-to-SMA RG1.13 pigtail	2	Radio-to-antenna feed, panel-mounted
Sensors	MPU-6500 six-axis IMU on a GY-6500/9250 breakout	1	3-axis acceleration, 3-axis angular rate
Sensors	GY-BMP280-3.3 barometer	1	Pressure, altitude, temperature
Sensors	NEO-6M GPS with EEPROM, active patch antenna	1	Position and time
Sensors	LM393 sound module, 4-pin	1	Acoustic level during descent
Storage	microSD card reader, 3.3 V	1	Onboard flight log
Power	1S LiPo, 3.7 V, 1500 mAh, 25C	1	Primary power
Power	ON/OFF switch; red power LED + 1 kΩ; green status LED + 1 kΩ	1 each	Mandatory switch and indicator; mission-state blink
Structure	`Cansat_D1` , white PETG, 3D printed, with egg chamber	1	Airframe
Recovery	**6 ft (1.83 m) diameter** sewn canopy, lines and harness	1	Descent at ≤ 5 m/s
Integration	10 × 10 cm universal prototype board, 2.54 mm pitch	2	One built, one spare
Passives	470 µF + 100 nF (microSD), 10 µF + 100 nF (radio), 33 kΩ/33 kΩ divider	—	Decoupling and battery sensing



Raspberry Pi Pico (RP2040). Two were fitted with headers and used: vehicle and ground bridge.



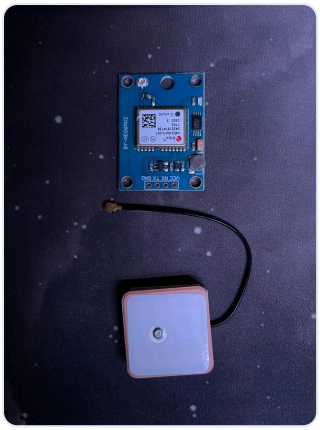
SX1278 RA-02 with its antenna mated through the IPEX-to-SMA pigtail.



IMU breakout. The device answers as an MPU-6500 at address 0x68.



GY-BMP280-3.3 barometer module, six-pin I²C/SPI variant.



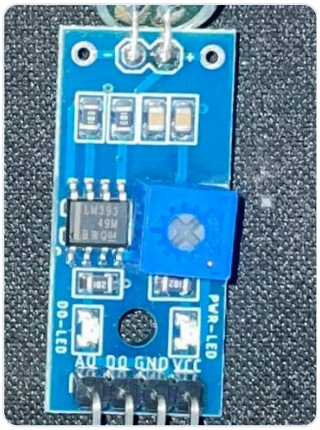
NEO-6M GPS with EEPROM and active patch antenna.



3.3 V microSD reader: four 10 kΩ pull-ups and two capacitors, no regulator.



1S 1500 mAh LiPo with JST-RCY and JST-XH balance leads.



LM393 sound module: electret capsule, comparator and gain trimmer.

5.2 The flight computer — Raspberry Pi Pico (RP2040)

Cores / clock	Dual Arm Cortex-M0+, 133 MHz
Memory	264 KB SRAM, 2 MB flash
Peripherals used	I²C0 (IMU + barometer), SPI0 (radio + card), UART0 (GPS), ADC0/ADC1 (battery, microphone), hardware watchdog, 1 µs timer
Supply	1.8 – 5.5 V at VSYS; on-board regulator provides 3.3 V

WHY THE PICO

- **Enough of everything, nothing in excess.** 264 KB of RAM holds every fixed buffer in the flight core many times over; the loop uses a small fraction of one core at a 2 ms tick, so the second core is spare.
- **A free, hardware watchdog and a 12-bit ADC** — the two peripherals this design leans on hardest. The watchdog is what makes "telemetry never stops" a property of the hardware and not only of the code; the ADC reads the microphone and the battery without extra parts.
- **A C/C++ SDK with a stable build.** The firmware is C++17 with no third-party libraries in the flight path; a toolchain that builds the same code on a laptop and on the target is what makes the host-tested core trustworthy.
- **No radio of its own.** A Pico W would add a 2.4 GHz radio the competition does not use, more current, and a second source of interference beside the GPS. The plain Pico leaves the 433 MHz LoRa module as the only transmitter on the board.
- **Cost and availability.** Two were needed — one flies, one is the ground bridge — and both were identical, so any firmware or driver verified on one is verified on the other.

Alternatives considered: an ESP32 (Wi-Fi/Bluetooth radio noise, higher idle current, and a flash cache that complicates hard real-time behaviour), an Arduino-class AVR (too little RAM for the buffers and no hardware watchdog with a configurable period), and a bare STM32 (no ready-to-solder board in the available time).

Verified: both Picos identified by silkscreen (`RP2-B2` stepping) and exercised end to end; headers fitted and soldered; the vehicle Pico's pins pass through the board and are soldered directly, so there is no socket to work loose under launch vibration.

5.3 The radio — SX1278 RA-02

Band / modulation	433 MHz ISM, LoRa (chirp spread spectrum)
Configured	SF7, 125 kHz, coding rate 4/5, preamble 8, CRC on, +17 dBm on PA_BOOST
Interface	SPI; `RST`, `DIO0`, `DIO1` wired
Supply	3.3 V directly (no regulator on the carrier), 87 mA at +17 dBm TX, 12 mA RX

WHY THIS RADIO, AND WHY THESE MODEM SETTINGS

- **The organizers' ground station is an SX1278 RA-02 on 433 MHz.** The rulebook offers two compatible radios (SX1278 LoRa, nRF24L01). Choosing the LoRa module means the vehicle and the scoring receiver are the same silicon, so compatibility is by construction.
- **LoRa over nRF24L01.** The nRF24L01 is a 2.4 GHz short-range link; LoRa at 433 MHz penetrates structures and ground clutter far better and has tens of dB more link margin than a 30 m descent needs.
- **SF7 is chosen for airtime, not range.** A ~190-byte packet needs 943 ms at SF9/125 kHz and about 302 ms at SF7/125 kHz; only the latter meets 1 Hz with margin. The link budget still closes with ≥ 14 dB of margin on the weakest flight packet (Chapter 11).
- **17 dBm, not 20.** +17 dBm is the PA_BOOST maximum without enabling the high-power DAC; +20 dBm would draw 120 mA instead of 87 mA and stress the 3.3 V rail for a link that does not need it.
- **One modem definition for both ends.** The vehicle and the bridge read the same `link_profile.hpp`, because before it existed the two carried separate copies that agreed only by coincidence.

5.4 The inertial sensor — MPU-6500

Measures	3-axis acceleration (±16 g configured), 3-axis angular rate (±2000 °/s configured)
Filters	Digital low-pass 21.2 Hz (accelerometer) and 20 Hz (gyro); 200 Hz internal data rate
Bus	I²C0 at `0x68`
Supply	3.3 V

WHY A SIX-AXIS IMU, AND WHY THESE RANGES

- **The rulebook's reference IMU is the six-axis MPU-6050.** The MPU-6500 is its successor in the same family — same measurements, same register style, lower noise — so it meets the "gyroscope + accelerometer" requirement exactly.
- **±16 g and ±2000 °/s.** The loads in this mission are the throw (measured 5.2 g in Flight 1), a canopy opening (2.1 g) and a touchdown (1.6 g). ±16 g leaves a factor of three of headroom over the worst of them; ±2000 °/s covers a vehicle tumbling at several revolutions per second without saturating the integrator.
- **A 20 Hz filter against a 30 Hz sample rate.** The filter bandwidth is set so that the loop's 30 Hz sampling does not alias vibration into the attitude estimate.
- **Yaw is a relative angle.** Roll and pitch are referenced to gravity through the accelerometer and are absolute. Yaw is integrated from the gyro and zeroed at calibration; the telemetry declares it as such. Firmware for a magnetometer-referenced yaw is also present and tested against simulated fields for a nine-axis part.
- **Bias is measured, not assumed.** Gyro bias per axis was measured on the bench at X −3.30, Y +0.86, Z −0.06 °/s; at every power-up the pad calibration re-measures it, which is why the post-landing session in Chapter 14 shows yaw frozen to ±0.1° after calibration.

5.5 The barometer — BMP280

Measures	Pressure 300 – 1100 hPa, temperature −40 … +85 °C
Output rate	≈ 83 Hz at the configured oversampling (measured 83.0 Hz)
Bus	I²C0 at `0x76`
Compensation	Bosch 64-bit integer path; reproduces the datasheet's own reference vector

WHY THE BMP280, AND WHY THE ALTITUDE IS COMPUTED THE WAY IT IS

- **The rulebook names a BMP as the altitude sensor** and lists pressure and temperature as mandatory fields. One chip supplies all three.
- **Resolution fits the task.** The pressure noise measured on the ground is 1.15 Pa (σ), which is 0.10 m of altitude — one tenth of a metre against a 30 m flight.
- **Baseline-relative altitude.** The firmware converts pressure to altitude with the international standard atmosphere formula and subtracts a ground baseline averaged over 80 samples at calibration, so the packet reads ≈ 0 m on the pad as the rulebook asks, whatever the weather. Chapter 14 shows the formula reproduces the transmitted altitude to ±0.03 m.
- **Why the temperature field is the barometer's.** The chip measures its own die temperature for compensation, so it is free, always valid and in the same burst read as the pressure.

5.6 The additional sensors

GPS — NEO-6M

u-blox NEO-6 receiver on a GY-NEO6MV2 board with a 24C32A EEPROM, backup cell and active patch antenna on a u.FL lead. UART0 at 9600 baud; all six NMEA sentences, zero checksum errors in bench logs. Parsed by the project's own streaming GGA/RMC parser, which **refuses a fix below 4 satellites or above HDOP 5.0**. Printed on

the air as 5-decimal latitude/longitude (1.1 m resolution against the receiver's ~2.5 m error) and whole-metre altitude; the SD log keeps full precision plus satellite count and HDOP.

Why GPS: it is the textbook additional sensor for a descent — position, drift and an independent altitude — and in the flight it measured the vehicle's ground track under canopy (Chapter 14).

Microphone — LM393 sound module

An electret capsule with an LM393 comparator and gain trimmer; the analogue output goes to ADC1 on GP27. Each loop tick samples a burst of conversions and reduces it to a peak-to-peak envelope; a second channel records what fraction of the window exceeded a threshold. Resolution is 3.3 V / 4096 = 0.806 mV per count.

> WHY A MICROPHONE
>
> **A microphone on a descending probe is a flight-proven atmospheric instrument.** Mars 2020 *Perseverance* carried one dedicated to entry, descent and landing; the *Huygens* probe carried an acoustic sensor through Titan's atmosphere in 2005; *Venera 13* and *14* recorded wind noise on Venus. Acoustics is one of the cheapest ways to instrument a descent. In order of usefulness it measures: **impulsive events** (the throw is the loudest packet of Flight 1 — 36.3 mV p-p), **flow noise** against descent speed, and canopy dynamics. Two channels are recorded deliberately — *how loud* and *what fraction of the time* — because a sharp crack and a steady roar can reach the same peak and mean opposite things. The level is a relative envelope in millivolts, comparable within a flight at one gain setting.

5.7 Storage — microSD

3.3 V reader (no regulator, no level shifter — the host is 3.3 V, so none is needed), on the shared SPI0 bus with its own chip select (GP6). The project's driver implements the SD protocol from CMD0/CMD8/ACMD41 initialisation through SDHC/SDSC block addressing. Measured: 100/100 writes; 2.68 ms mean block write; ~300 writes/s sustained.

Why a card at all: the radio link is lossy by nature; the SD log is the record that cannot be. It holds every packet the radio sent, plus fields that never go on the air.

5.8 Power

Battery	1S LiPo, 3.7 V nominal (4.2 V full), 1500 mAh, 25C, JST-RCY discharge lead
Rail	Pico's on-board regulator: 3V3(OUT) — measured **3.28–3.29 V under 45 back-to-back radio transmits** and **3.28–3.30 V at 100 % microSD write duty**
Switch	In the battery positive lead, ahead of everything
Power LED	Red, 1 kΩ, on the regulated 3.3 V rail
Battery sense	GP26 (ADC0) through a 33 kΩ / 33 kΩ divider

> WHY NO EXTERNAL REGULATOR
>
> An AMS1117-3.3 was assessed and rejected: at a full cell of ≈ 4.2 V it does not clear its high-load dropout. The Pico's own regulator was then **measured** carrying every load on the vehicle, and it held 3.28 V or better in the worst cases. One rail, no external part, one fewer thing to fail. Chapter 6 gives the current budget.

5.9 Telemetry antenna and structure parts

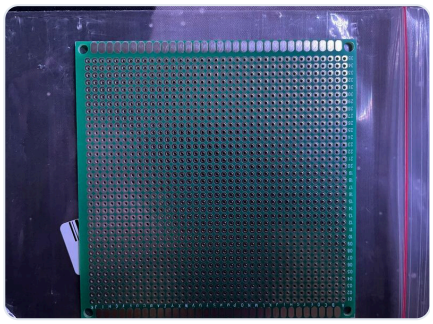
The whip antenna and its IPEX-to-SMA pigtail were verified to mate hand-tight with no adapter; the pigtail is a bulkhead type, so its nut and star washer clamp it through the airframe wall, which is also its strain relief. The GPS patch antenna sits skyward and the LoRa antenna is routed away from it, because a +17 dBm transmitter beside a receiver working near −130 dBm can overload its front end.



433 MHz whip antenna on a hinged base.



IPEX-to-SMA bulkhead pigtail with panel nut and star washer.



10 × 10 cm universal board that carries the vehicle electronics.

6

CHAPTER 6

Electrical design

One battery, one rail, two shared buses — the power path, the current budget, the pin map, and the decoupling that keeps the radio and the card from disturbing each other.

6.1 The power path

6.2 One rail, answered by measurement

6.3 The current budget

6.4 Buses, and the lesson each one taught

6.5 Decoupling, sized against a failure that actually happened

6.6 Pin assignment

6.7 Why a hand-assembled universal board

6.1 The power path

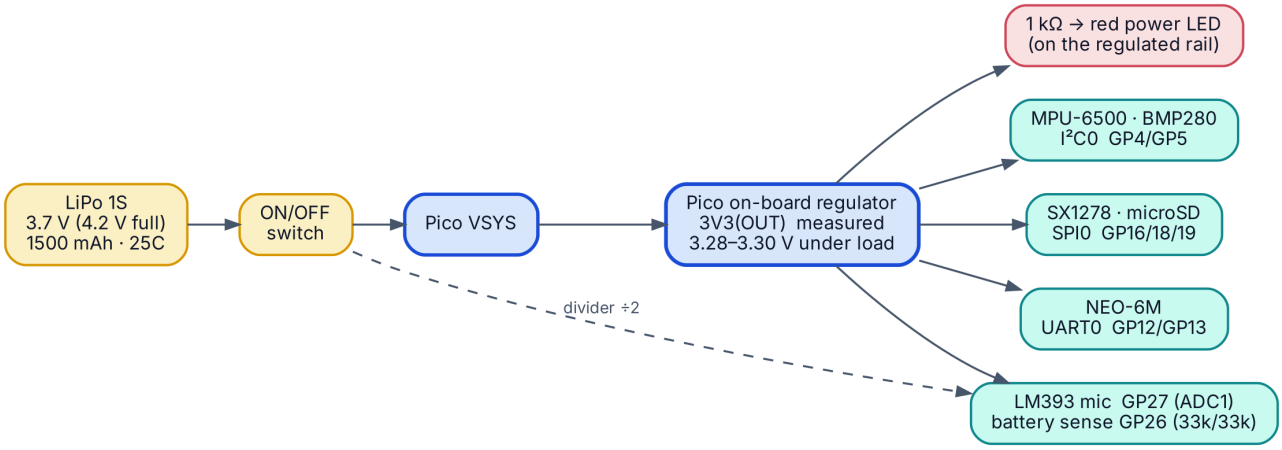


Figure 10 · From battery to every load. The switch sits in the battery lead ahead of everything; every peripheral runs from the Pico's own 3.3 V output.

The power LED hangs off the regulated rail through a series resistor, not off a GPIO. The rulebook requires the LED to light *immediately* on power-up; a firmware-driven LED would wait for the boot. The firmware separately drives a green status LED on GP14 whose blink rate encodes the mission state.

DESIGN DECISION — WHICH NODE THE POWER LED HANGS OFF

Hung off the switched battery	Hung off the regulated 3.3 V rail — chosen
Means "the battery is connected"	Means "the system is actually powered"
Stays lit with a dead flight computer	Goes dark if the regulator fails — the honest signal
3.4 – 4.2 V, so it dims visibly as the cell drains	Regulated: constant brightness and current

A red LED at 1 kΩ from 3.3 V draws 1.30 mA — bench-bright and cheap in current, which is the right direction for a battery vehicle.

6.2 One rail, answered by measurement

Three hardware questions held the design up for days. All three closed by measurement, and how they closed is more useful than the fact that they did.

Table 8 · **The 3.3 V rail under every load**

Condition	Rail voltage
Idle	3.30 V
45 back-to-back radio transmits (+17 dBm)	**3.28 – 3.29 V**
100 % microSD write duty	**3.28 – 3.30 V**
Radio at 100 % TX duty	3.26 – 3.27 V

No external regulator is needed, and the microSD reader turned out to be a **3.3 V board**: its supply pin is printed `3V3` and its entire parts list is four 10 kΩ pull-ups and two capacitors. A listing that described a 4.5–5.5 V board with an on-board regulator described a different product; the delivered board is simpler and runs from the same rail as everything else.

6.3 The current budget

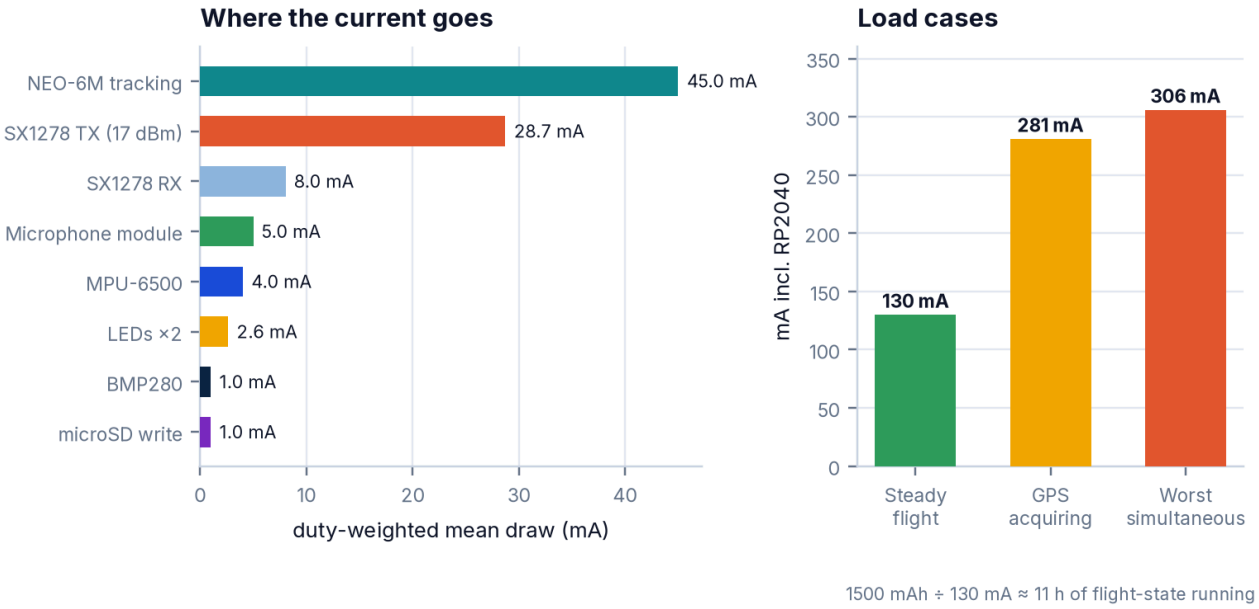


Figure 11 · Duty-weighted current by device (left) and the three load cases (right).

Table 9 · **Per-device worst-case draw**

Device	Worst case	Condition	Duty in flight
SX1278 RA-02	**87 mA**	TX at +17 dBm (PA_BOOST)	33 % (333.7 ms of every second)
	12 mA	RX continuous	the other 67 %
microSD + reader	~100 mA	block write	~1 % (two writes ≈ 10 ms per telemetry second)
NEO-6M	~70 mA / ~45 mA	cold acquisition / tracking	start-up / continuous
MPU-6500	~4 mA	gyro + accel active	continuous
BMP280	~1 mA	83 Hz, high oversampling	continuous
Microphone module	~5 mA	continuous	continuous
LEDs	2.6 mA	both lit	power LED continuous; status LED ≤ 50 %

Steady-state flight draw is about 130 mA including the RP2040 (TX at 33 % duty, RX otherwise, GPS tracking). A 1500 mAh cell therefore carries roughly eleven hours of flight-state running — against a mission whose pad phase is five minutes and whose flight is seconds. The worst simultaneous case (TX, SD write, GPS acquiring, all sensors, both LEDs) is 306 mA; it is brief and the measured rail holds through it.

6.4 Buses, and the lesson each one taught

Two shared buses carry everything except the GPS:

- **I²C0 (GP4/GP5)** carries the IMU at `0x68` and the barometer at `0x76`. Both verified together on the soldered board, 400 kHz.
- **SPI0 (GP16/GP18/GP19)** carries the radio and the microSD, with separate chip selects on GP17 (radio) and GP6 (card).

> **DESIGN DECISION — CHIP-SELECT DISCIPLINE ON THE SHARED SPI BUS**
>
> Running the sealed flight image on the soldered board revealed a defect that no host test could have found. The radio's chip select (GP17) was configured only by the radio driver, which initialises *second*. Through the whole SD-card initialisation GP17 was an unconfigured pad, and an RP2040 pad resets with its pull-down enabled — which holds it **low**, which on the RA-02 means *selected*. The radio drove MISO for the entire card sequence.
>
> The fix is a rule: **both chip selects are driven high together with the bus, before any device on it is touched.** A SPI bus is only shared safely if every device on it is deselected by construction rather than by the order in which drivers happen to start.

6.5 Decoupling, sized against a failure that actually happened

Table 10 · **Decoupling capacitors and what each is for**

Where	Value	What it is for
microSD reader, across its own 3V3 and GND	**470 µF** electrolytic	The write-current spike, ~100 mA for a few ms; it must not reach the regulator and stack on the radio's peak
Same pins	**100 nF** ceramic	Fast edges the electrolytic is too slow for
RA-02, across its 3.3 V and GND	**10 µF**	PA key-up: 1.5 mA standby to 87 mA in microseconds, 45 times a minute
Same pins	**100 nF** ceramic	As above

Why the card gets the largest capacitor: on the breadboard a long 3.3 V jumper let the card initialise, read its filesystem — then go silent on the first write, five runs running. A short jumper fixed it outright (100/100). The write-current spike arrives down whatever wire the supply has; reads never draw enough current to expose it. On the soldered board the supply became a short track and a local 470 µF bulk capacitor. The same fault explained a radio that transmitted 5/5 and then 0/5: **two intermittents, one cause, on two different modules.**

6.6 Pin assignment

`BoardPins` in `firmware/flight-computer/include/flight/config.hpp` is the single source of truth. The netlist at `electrical/schematics/vehicle-netlist.tsv` is **generated from it**, and the generator refuses to run if the two disagree; the wiring diagrams come from the same source. A pin cannot drift between the code and the documentation.

Table 11 · **Pin assignment (vehicle Pico)**

GPIO	Function	Device
GP4 / GP5	I²C0 SDA / SCL	MPU-6500 (`0x68`) + BMP280 (`0x76`)
GP16 / GP18 / GP19	SPI0 MISO / SCK / MOSI	RA-02 + microSD
GP17	Chip select	RA-02
GP6	Chip select	microSD

GPIO	Function	Device
GP20 / GP21 / GP22	RESET / DIO0 / DIO1	RA-02
GP12 / GP13	UART0 TX / RX	NEO-6M
GP7	Interrupt	MPU-6500 INT, wired
GP14	Status LED	Green LED, blink rate encodes mission state
GP15	Comparator input	LM393 `DO`
GP26	ADC0	Battery sense, divider 33 kΩ / 33 kΩ (ratio 2.0)
GP27	ADC1	LM393 `AO` (microphone)

6.7 Why a hand-assembled universal board

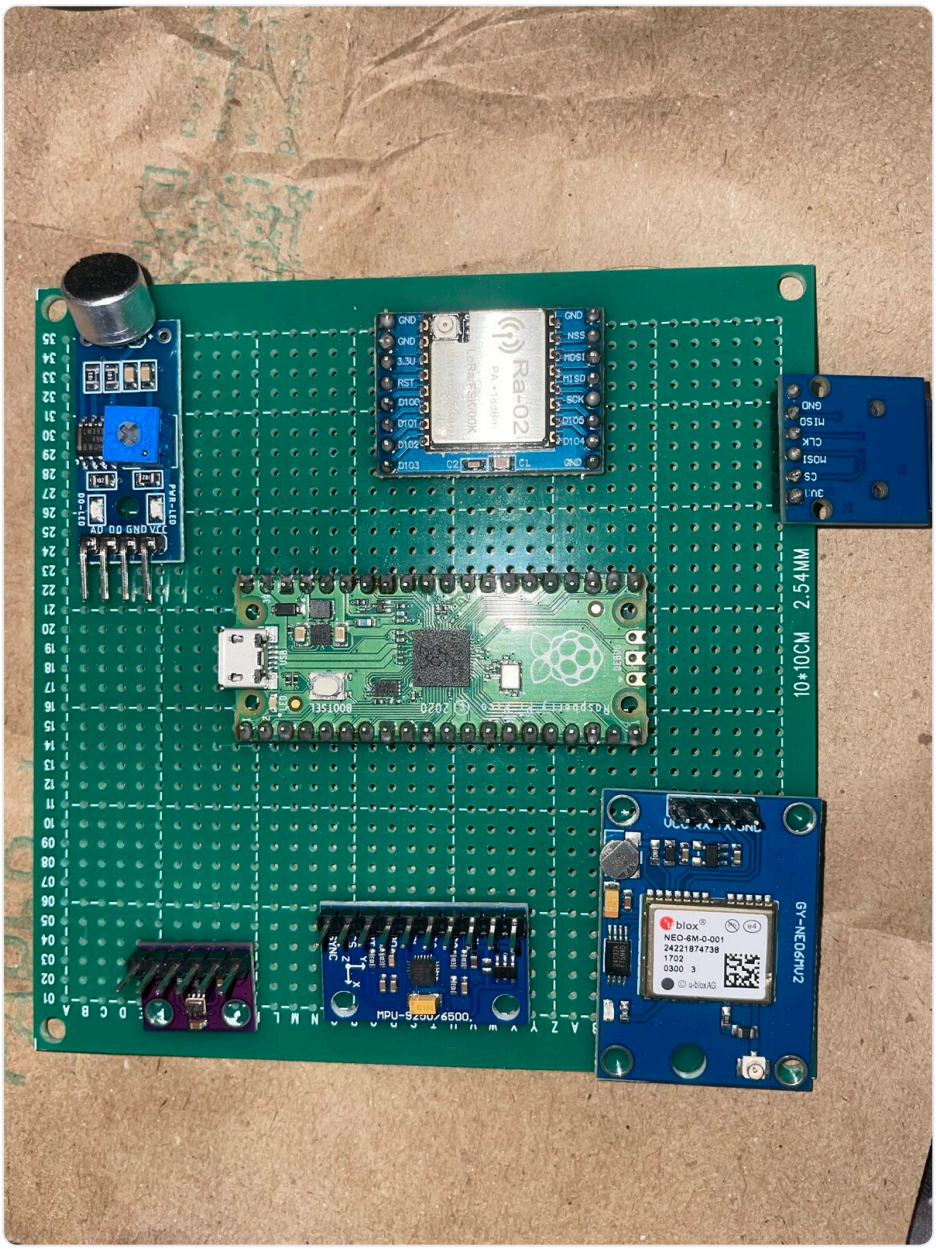


Figure 12 · **The vehicle board, component side.** 10 × 10 cm universal board with the microphone module (top left), RA-02 radio (top), microSD reader (right edge), Raspberry Pi Pico (centre), GPS module (lower right), IMU (lower centre) and BMP280 (lower left), laid out to keep the radio and card close to the Pico and the sensors on short I²C runs.

A 10 × 10 cm single-sided universal board carries the electronics, with every module's pins soldered directly (no sockets). The choice was deliberate:

- **The pin map was frozen and hardware-verified before the board was built**, so the wiring is known rather than discovered; the build followed a sixteen-step gated assembly plan, each step with a measurement.
- **Short, direct runs.** Radio, card and Pico sit within a few centimetres of each other, so the high-current supply runs are short tracks and the two noisiest loads have their own local capacitors.
- **Fast iteration for a twelve-day schedule.** A custom board needs a fabrication round trip the schedule could not spare; the universal board let the supply wiring be changed the same day a bench finding called for it — the short card supply in Section 6.5 is exactly such a change.
- **Photographable and inspectable.** Every joint is visible, which made the bring-up measurements — and the board photographs in the report — straightforward.

7

CHAPTER 7

Mechanical design

`Cansat_D1`: an open PETG frame with an egg chamber — geometry, material, mass budget and the reasoning behind each.

7.1 The design

7.2 Material — white PETG

7.3 The mass budget

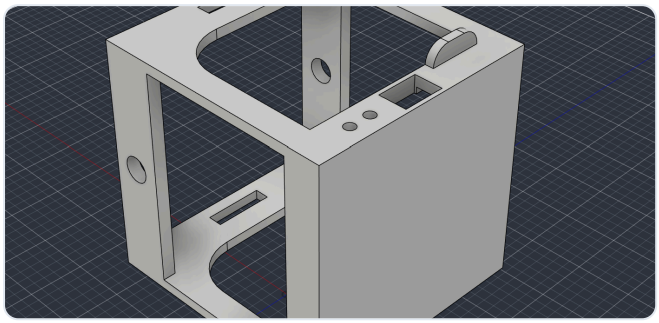
7.4 Egg chamber and recovery hardware

7.5 The built vehicle

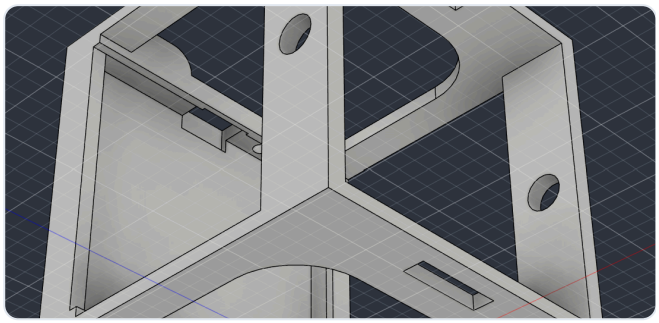
7.6 Fabrication

7.1 The design

`Cansat_D1` is modelled in Fusion 360 and exported to STEP. **Every dimension quoted in this report is read out of the STEP file by `tools/cad_dimensions.py`, and the build fails if the documentation and the model disagree.**



Render, three-quarter view from above: switch cut-out and two LED holes on the upper face.



Render from below: arched openings, central spine and harness slots.

Table 12 · Cansat_D1 geometry, read from the STEP file

Solids	one, `Body1`, 56 faces
Bounding box	**118.5 × 115.0 × 110.0 mm**
Cross-section diagonal	159.1 mm
Height against the 210 mm allowance	**91.5 mm unused**
Clearance per side against the 120 mm section	2.5 mm and 5.0 mm

It is an **open box frame**: two solid side panels, two faces opened out with large arched cutouts, a central vertical spine, harness slots top and bottom, round holes in the side panels and spine, and a rectangular cut-out for the switch with two small holes beside it for the LEDs.

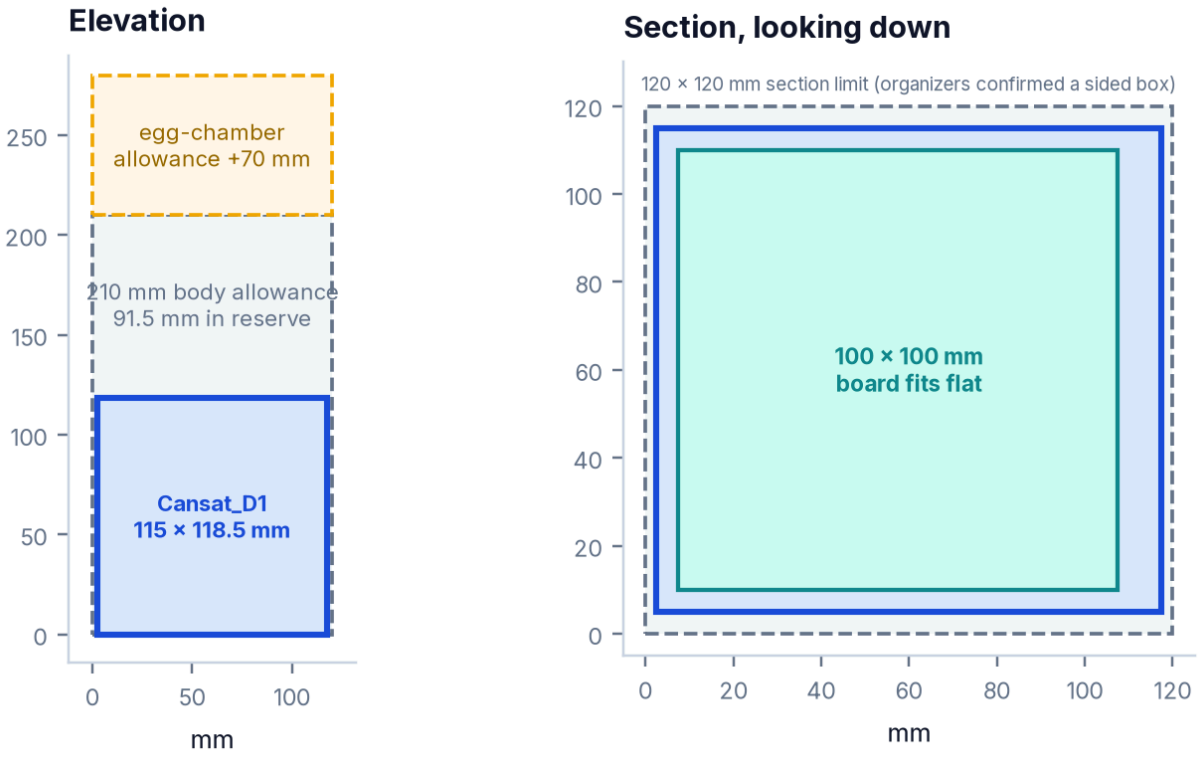


Figure 13 · The vehicle against the rulebook envelope: elevation (left) and the section looking down (right).

WHY AN OPEN FRAME

- **Material goes where the load path is.** The solid side panels and central spine carry load; everything else is air. The three structural studies (Chapter 8) confirm the load path with a safety factor of 7.6 or more even after worst-case derating.
- **The barometer must breathe.** A sealed body would lag the vehicle's true pressure during a fast descent; open arched faces keep the BMP280 in moving ambient air and keep the altitude channel honest.
- **Access.** The board is reached from two sides without disassembly; the antenna and GPS patch are on the outside faces and the switch is on leads outside the frame.
- **Mass.** The printed frame weighs ≈ 128.7 g including the egg chamber — light enough that the vehicle is trimmed up to the mass band rather than fighting its upper edge.
- **A prismatic form is the most efficient way to carry a flat board.** The 100 × 100 mm board fits flat inside the 115 × 110 mm section, with 15 and 10 mm to spare for walls and standoffs.

7.2 Material — white PETG

The rulebook permits any outer material and offers a bonus for sustainable or unconventional ones; PETG was chosen and the reasoning recorded.

Table 13 · Why PETG

Criterion	PETG	PLA	ABS
Toughness under one impact	Deforms before it fractures	Brittle	Tough
Printable without an enclosure	Yes, little warping	Yes	No — warps, fumes
Heat in a sunlit pad	Safe to ~70 °C	Softens near 55 °C	Safe

Criterion	PETG	PLA	ABS
Interlayer adhesion	Good	Fair	Good
Recyclable / widely available	Yes	Yes	Less so

- **Impact energy goes into deformation rather than fracture** — a bent frame is a recovered vehicle, a shattered one is not.
- **Print orientation was decided before printing and recorded**: as modelled, sitting on its base. It was the single free variable that changes a printed part's strength, it costs nothing at slicing time, and it cannot be changed afterwards. Layers stack vertically, so the weak directions are tension normal to the layers and interlayer shear — which is why the structural studies are derated for it.
- **White** shows a clean print rather than hiding a poor one, photographs cleanly against any background, and runs cooler in sunlight on a pad than a dark part.

7.3 The mass budget

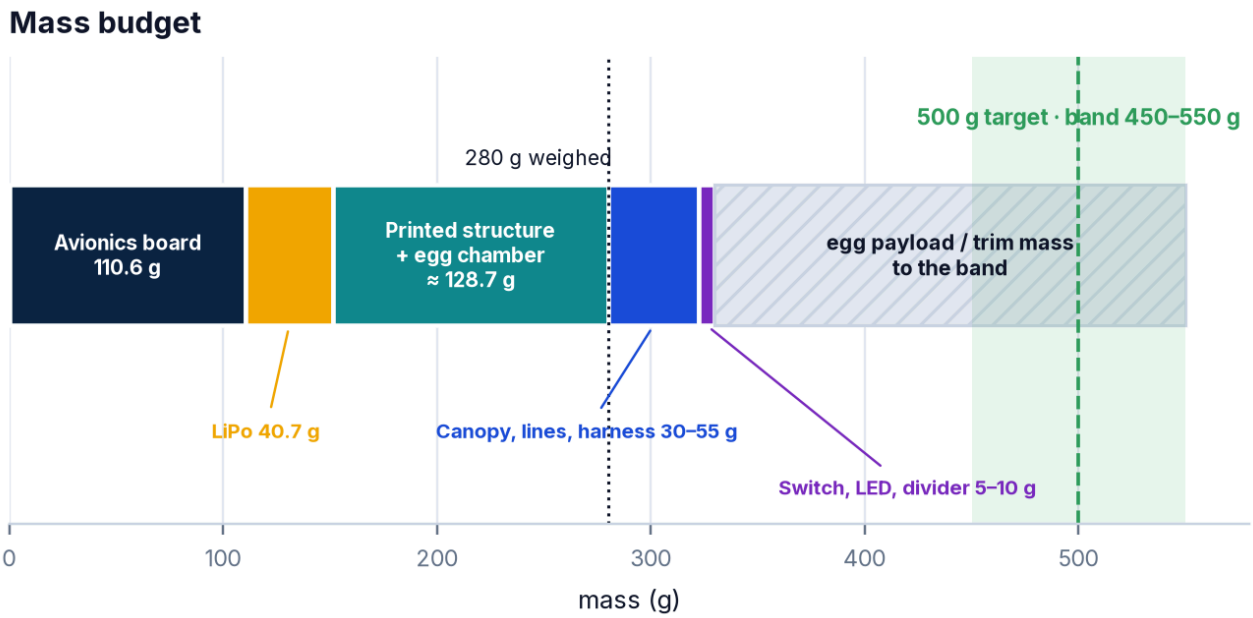


Figure 14 · From weighed parts to the mass band. The electronics were weighed separately (110.573 g board, 40.726 g battery); the 280 g bare vehicle was weighed after assembly; the structure is the difference.

Table 14 · Mass budget

Item	Mass	How
Assembled vehicle board, no battery	110.573 g	Weighed 9 Sep
Battery — 1500 mAh 1S LiPo	40.726 g	Weighed 9 Sep
Electronics, all-up	**151.299 g**	Sum
Printed structure + egg chamber	**≈ 128.7 g**	By difference: 280 − 151.299
Assembled vehicle, no canopy	**280 g**	Weighed 12 Sep
6 ft canopy, lines, harness	30 – 55 g (estimate)	Fitted after that reading
Switch, power LED, divider	5 – 10 g	Fitted after that reading
Egg payload / trim mass	to the band	Added to bring the vehicle into 450 – 550 g

Item	Mass	How
Flight mass	**450 – 550 g**	Within the rulebook's 500 g ± 10 %

What was measured and what was derived. The scale figure is the 280 g assembled vehicle; the electronics were weighed separately three days earlier; the structure line is arithmetic on two measurements. The submitted mass lies in the 450–550 g band, and the descent system was sized for the top of it (Chapter 9).

The structure came in 33 % lighter than the estimate

Predicted — solid volume × 1.27 g/cm³	193.0 g
Printed — and that includes an egg chamber the estimate did not	≈ 128.7 g
Difference	−64 g, −33 %

Infill is the whole of it: the 193 g assumed no infill saving; a real slice of a thin-walled open frame is mostly perimeter and air. The estimate was labelled an upper bound and behaved like one. **Both estimates in the project — the avionics estimate was 31 g low and the structure estimate 64 g high — were corrected by a scale in one minute**, which is why every mass in the budget above is a weighed number.

7.4 Egg chamber and recovery hardware

- **Egg chamber.** A cushioned, secure chamber is part of the printed structure and sits inside the +7 cm allowance the rulebook gives it; it holds the egg against the descent and landing loads and is included in the 128.7 g printed mass.
- **Parachute placement.** The rulebook requires the canopy to deploy instantly on launch and forbids burying it tightly inside the structure. The canopy is therefore carried external / semi-exposed, attached through the frame's harness slots, so the first moments of free fall pull it open (Flight 1: canopy fully loaded 1.0 s after the apex of the throw).
- **Antenna and power hardware.** The pigtail's panel nut clamps through the frame wall; the rocker switch sits on short leads outside the frame, and the LED positions are in the upper face; both are reachable on the pad without opening the vehicle.

7.5 The built vehicle

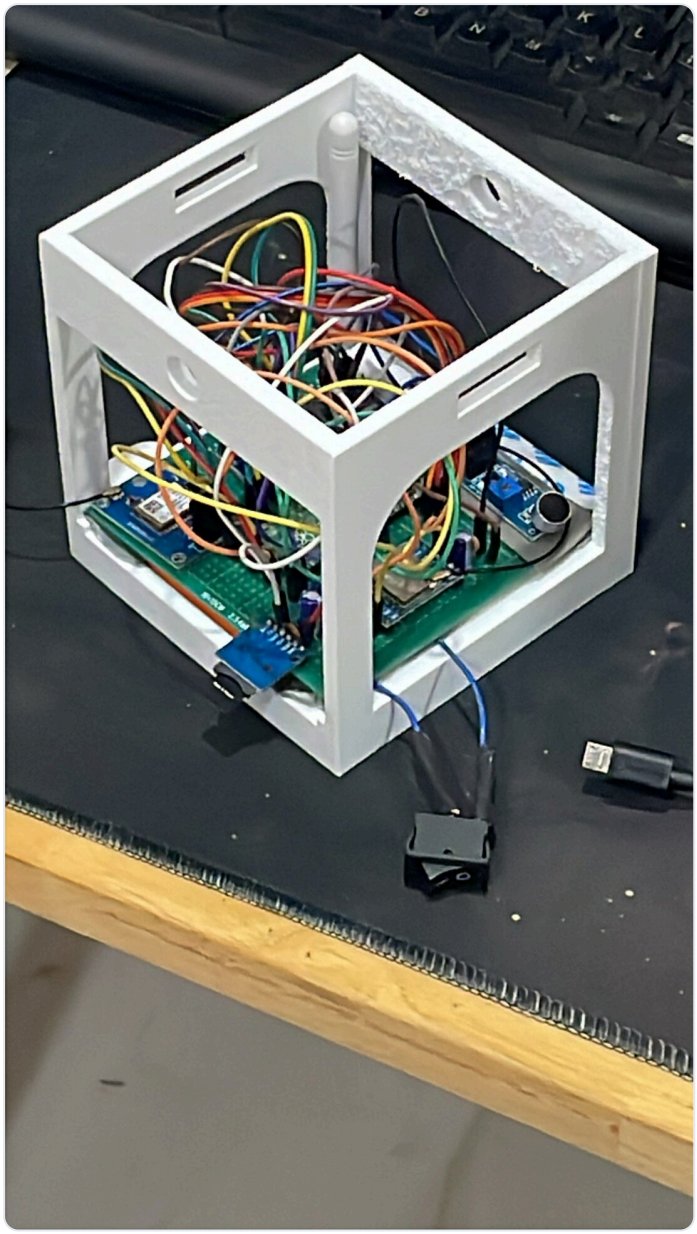


Figure 15 · **The assembled CanSat.** The white PETG frame carries the universal board with the GPS module, microphone module, radio and sensor modules; the 433 MHz antenna stands up through the frame; the rocker ON/OFF switch is on short leads outside, and the USB lead (right) is used for programming and bench work only.

7.6 Fabrication

The structure is original work printed from the team's own CAD model — not an off-the-shelf enclosure. The frame, the egg chamber, the harness points, the switch and LED positions and the antenna clamp are all features of the one model, so nothing is bolted on afterwards that was not planned. The vehicle was printed, assembled with the electronics and egg chamber, and weighed on 12 September 2026; the canopy, switch and LED were fitted and the mass trimmed into the band on 13–14 September.

8

CHAPTER 8

Structural simulation

Three static-stress studies on the printed frame — what was loaded, what the stresses were, and how the margin holds up against the real flight loads.

8.1 What was simulated

8.2 Material properties, and derating for a printed part

8.3 The studied loads against the loads the vehicle actually met

8.4 The Fusion 360 results

8.1 What was simulated

Three static-stress studies were run in Fusion 360 on `Cansat_D1` on 9 September 2026: static stress, one fixed constraint, parabolic elements.

Table 15 · The three studies

Study	Load	Mesh	Peak von Mises	Yield ÷ stress
1 · Horizontal	30 N on +Z	5,838 nodes / 2,619 elements	2.885 MPa	**18.9**
2 · Tearing	30 N on −X	5,838 nodes / 2,619 elements	1.330 MPa	**40.9**
3 · Impact	100 N on −X	7,148 nodes / 3,299 elements	2.345 MPa	**23.2**

In all three, Fusion's minimum safety-factor plot reads its display ceiling of 15 everywhere on the part: **at or above 15 across the whole frame** in every load case. The three yield-derived figures above are the sharper statement, because the plot saturates.

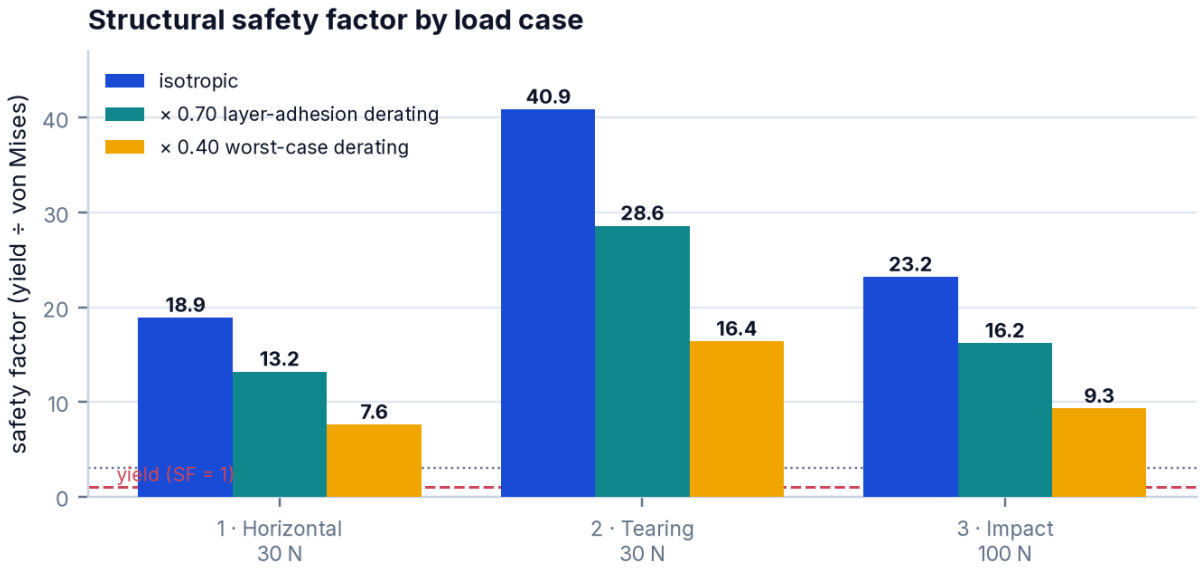


Figure 16 · Safety factor by load case: isotropic, and derated for the layered structure of a printed part.

WHY THESE THREE LOAD CASES

- **Horizontal (30 N)** — the side load of a gust, a harness pull or a lift-off scrape.
- **Tearing (30 N)** — a pull on the harness points that tries to peel the frame apart.
- **Impact (100 N)** — the touchdown. 100 N is a 500 g vehicle at 5 m/s (2.5 N·s) stopped in 25 ms: a landing on grass or soil with the canopy and the structure doing part of the work. The stopping time was chosen to represent ordinary ground, and the bar is deliberately set at the rulebook's 5 m/s cap rather than at the vehicle's real descent rate.

8.2 Material properties, and derating for a printed part

The studies use Fusion's PET plastic (E = 2758 MPa, yield 54.4 MPa). Bulk PETG is slightly softer and about 10 % lower in yield (E ≈ 2000–2100 MPa, ≈ 50 MPa), and an FDM part is anisotropic: interlayer adhesion is typically 40–70 % of in-plane strength. Print orientation was therefore fixed in advance (sitting on its base, as modelled) and the safety factors are reported with the derating applied:

Table 16 · **Safety factor after derating for the printed layer structure**

Study	Isotropic	× 0.70	× 0.55	× 0.40
Horizontal	18.9	13.2	10.4	**7.6**
Tearing	40.9	28.6	22.5	16.4
Impact	23.2	16.2	12.8	**9.3**

Even at the pessimistic 40 % factor the smallest margin is 7.6 — and taking the plot's capped value of 15 instead of the yield-derived number gives an effective floor of 6. The structure is designed with margin to spare, not argued into compliance.

8.3 The studied loads against the loads the vehicle actually met

The 100 N touchdown assumption can now be set against the flights. From the measured steady rates, the momentum of the vehicle at arrival and the force for a 25 ms stop are:

Table 17 · **Touchdown load: the study against the flights, for a vehicle in the 450–550 g band**

Case	Mass	Rate	Momentum	Force at 25 ms	at 10 ms
Study 3's assumption	500 g	5.00 m/s	2.50 N·s	**100 N**	250 N
Flight 1, measured	450 – 550 g	2.27 m/s	1.02 – 1.25 N·s	**41 – 50 N**	102 – 125 N
Flight 2, measured	450 – 550 g	1.88 m/s	0.85 – 1.03 N·s	**34 – 41 N**	85 – 103 N

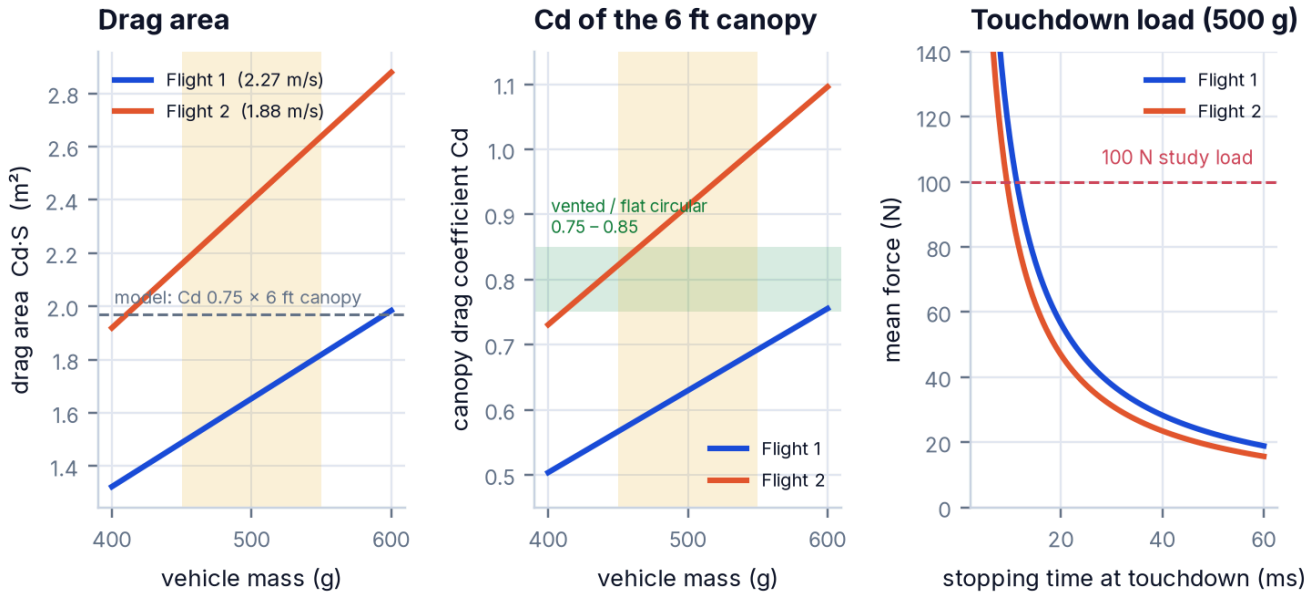


Figure 17 · Left: drag area (Cd·S) implied by each flight's steady rate against vehicle mass, with the model's value for the 6 ft canopy. Centre: the drag coefficient of the 6 ft canopy that each flight implies. Right: mean touchdown force against stopping time, with the 100 N study load.

The flights landed with 40–50 % of the momentum the structure was analysed for. The accelerometer agrees: Flight 2's touchdown packet read 1.57 g and the steady-descent swing peaks stayed below 2 g in both flights. Even in the unlikely case of a 10 ms stop the force (≈ 85 – 125 N) is the same order as the 100 N the study covers — against an isotropic safety factor of 23.

> **CONCLUSION OF THE STRUCTURAL WORK**
>
> Under every load the studies applied, the stress is a small fraction of yield, and the real flight loads are a fraction of the studied loads. The frame has margin in the studied directions by a factor of 7.6 to 40.9, and the measured landing momentum is below half of what the impact study assumed.

8.4 The Fusion 360 results

The three figures below are the result plots exported by Fusion 360 itself (the full HTML reports are in `mechanical/simulation/`): von Mises stress, the deformed shape drawn magnified, the safety-factor plot, and the constraint and load set-up. In every study one face is held fixed (blue) and the load acts on the opposite structure.

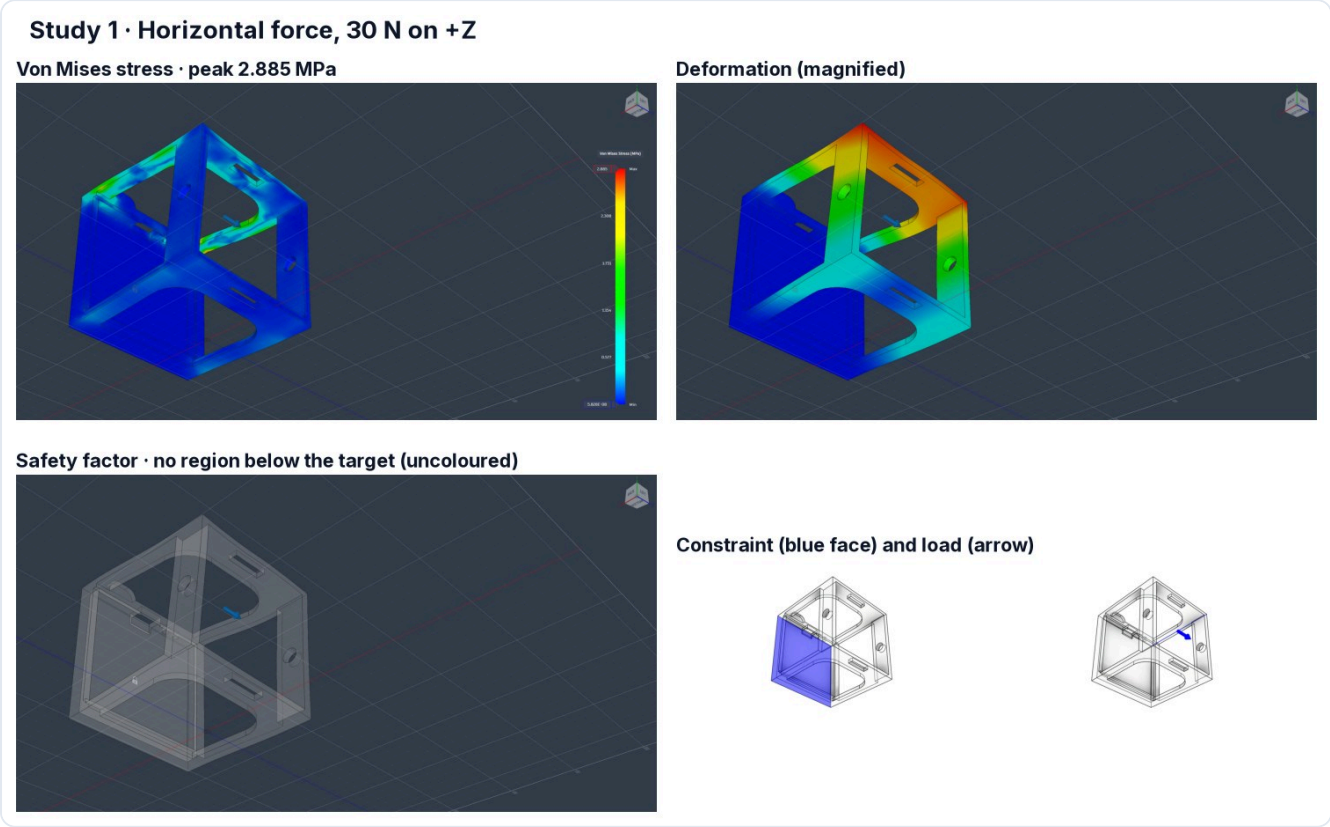


Figure 18 · **Study 1 — horizontal force, 30 N on +Z.** The stress concentrates around the arched cut-outs and where the central spine meets the top rail, peaking at 2.885 MPa; the safety-factor plot shows no region below the study target (the frame is left uncoloured).

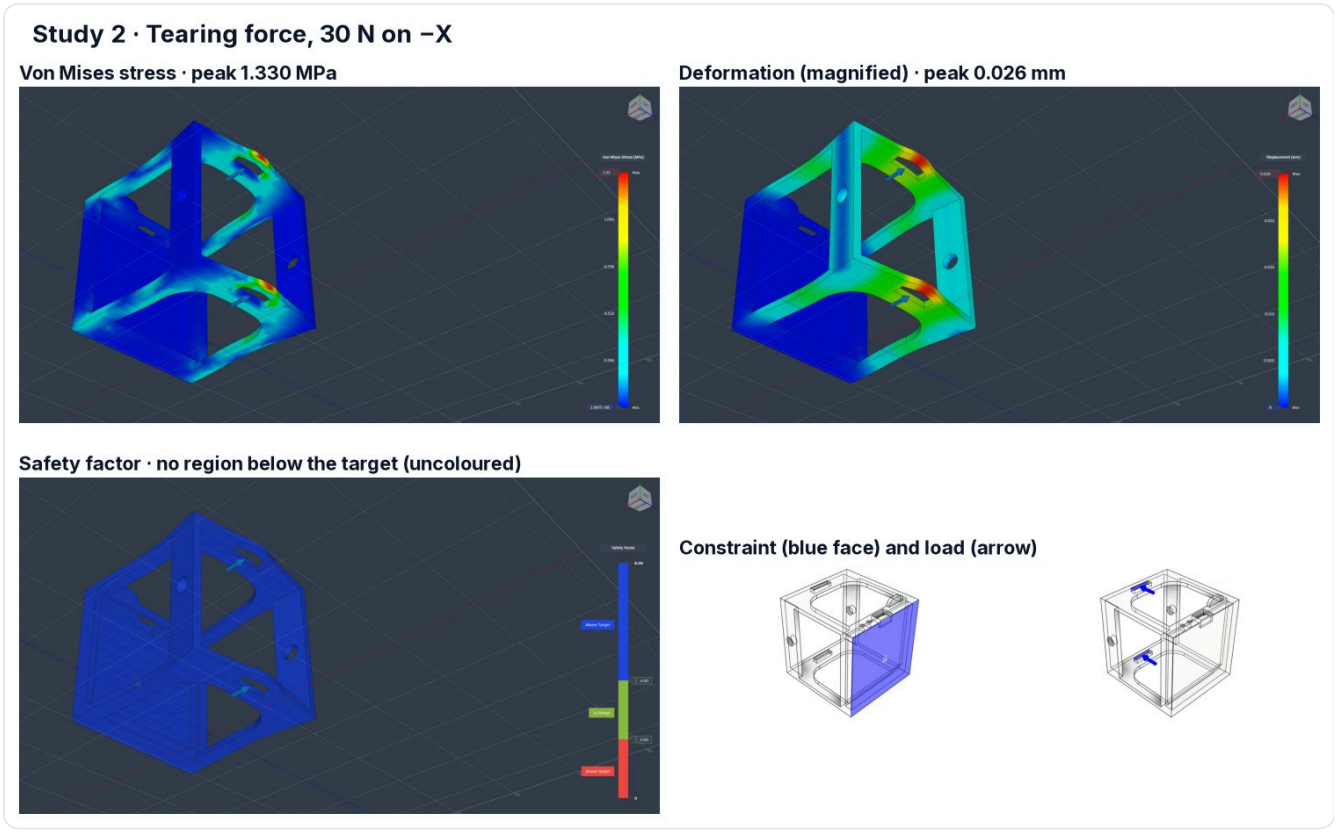


Figure 19 · **Study 2 — tearing force, 30 N on −X.** The lowest stresses of the three studies (peak 1.330 MPa) and a peak displacement of only 0.026 mm: the frame is very stiff against a pull on the harness points.

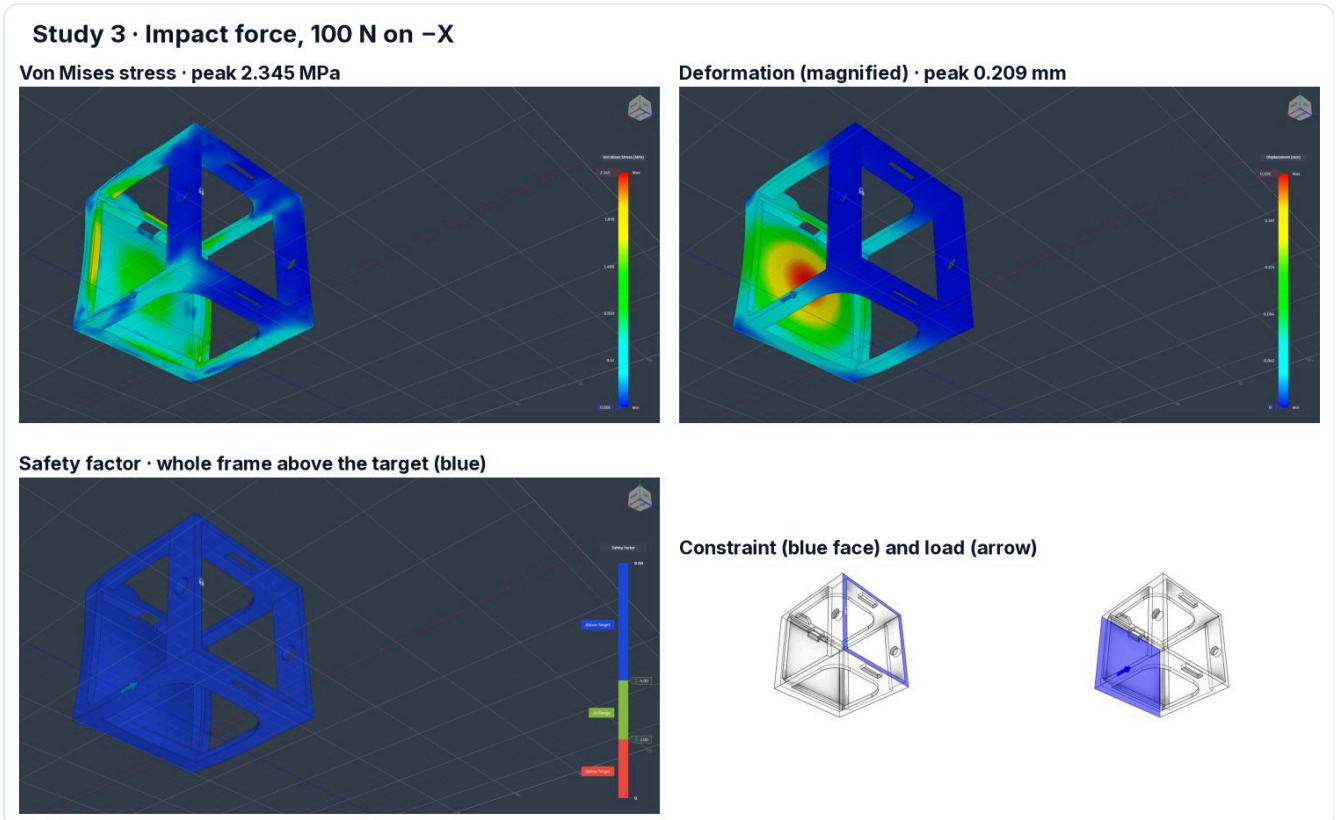


Figure 20 · **Study 3 — impact force, 100 N on −X.** The loaded panel bows (drawn magnified; the true peak displacement is 0.209 mm) and the stress spreads into the base and the neighbouring rails, peaking at 2.345 MPa, 4 % of the 54.4 MPa yield strength used in the study. The safety-factor plot is blue — above the study target — over the whole frame.

How to read them. In the stress plots the colour scale runs from blue (zero) to red (the peak) and the peak is a few megapascals against a yield strength of tens — so even the "red" regions are at a few per cent of yield. In the safety-factor plots Fusion colours only what falls in or below the target band the study was set up with (2 to 4): in studies 1 and 2 nothing does, so the frame is left uncoloured, and in study 3 the whole frame is shown blue, *above target*. In none of the three load cases does any part of the frame enter the band. The deformation plots show where the structure moves — the loaded panel and the free rails — and the numbers are fractions of a millimetre.

9

CHAPTER 9

Descent system and simulation

How the canopy was sized with a closed-form model, why it was sized for the heaviest, hottest case — and how the flights compared with the model.

9.1 What the simulation computes

9.2 Two answers from one model: the minimum canopy, and the canopy flown

9.3 The 6 ft canopy across the mass band

9.4 Three constraints on the canopy that are not the diameter

9.5 The flights against the model

9.1 What the simulation computes

`simulations/descent.py` sizes the canopy and models the fall. It runs in the host test suite, so a change that breaks the physics fails the same build as a change that breaks the firmware, and it is pinned to closed-form limits that can be checked by hand.

- **Canopy area and flat diameter** for a target descent rate: $\mathbf{S = 2\,m\,g\,/\,(\rho \cdot Cd \cdot v^2)}$.
- **The descent rate a canopy you already have will produce**: $v = \sqrt{(2\,m\,g\,/\,(\rho\,Cd\,S))}$.
- **Descent time** from the closed-form solution of $\mathbf{m \cdot dv/dt = m\,g - \tfrac{1}{2}\,\rho\,Cd\,S\,v^2}$, rather than height ÷ rate: the vehicle starts at rest and accelerates into terminal velocity, and over a 30 m drop that transient matters ($v(t) = v_t \cdot \tanh(g\,t\,/\,v_t)$).
- **Air density from the gas law, using the pressure and temperature the vehicle itself measures.** A 35 °C launch day is ~6.5 % thinner than ISA, which is ~7 % more canopy for the same descent rate.
- **The number of telemetry packets** the descent yields at the vehicle's rate.

Deployment delay is modelled as free fall, which is pessimistic on altitude and therefore safe. Canopy opening dynamics, oscillation and wind drift are outside the model; Chapter 14 measures them.

9.2 Two answers from one model: the minimum canopy, and the canopy flown

The model answers two different questions, and both matter.

What is the smallest canopy that is guaranteed to meet the 5 m/s limit? Size it for the worst case — the heaviest vehicle on the hottest day, with the lowest plausible drag coefficient:

Table 18 · **Minimum canopy sizing cases**

Case	Flat diameter	Rate	Time from 30.5 m
500 g, ISA 15 °C, vented flat circular	73.7 cm	5.00 m/s	6.45 s
550 g, 35 °C — the sizing case	**80.0 cm**	**5.00 m/s**	**6.45 s**

Canopy area is linear in mass, so ±10 % of mass is ±10 % of area but only ~5 % of diameter: a canopy sized at 500 g and flown at 550 g **breaks the 5 m/s cap**; one sized at 550 g is compliant across the whole band and costs 6 cm of cloth. The hot day is the sizing day because thinner air is the failure direction. **80 cm is therefore the floor: below it the cap is not guaranteed.**

What canopy was flown? A **6 ft (1.83 m) canopy** — 2.3 times the floor's diameter and 5.2 times its area.

WHY A 6 FT CANOPY, WELL ABOVE THE 80 CM FLOOR

- **The floor is a guarantee at the edge of the model; the mission wants margin inside it.** The model's drag coefficient is the dominant uncertainty — canopy types span Cd 0.75 to 1.40 — and the vehicle's mass is only known to lie in a ±10 % band. A canopy several times the floor turns a prediction that is *just* compliant in its worst case into one that is comfortably compliant in every case.
- **A hand-thrown launch gives the canopy a harder start than a drone release.** The vehicle leaves the hand moving at about 5 m/s with no carrier to clear it, so a large canopy that fills fast and has reserve drag is the safer answer to the question "will it open in time from 30 m?".
- **A slow descent is a benign descent.** At about 2 m/s the vehicle arrives with about 1 J of kinetic energy — the energy of a step of 18–26 cm — which protects the structure, the board and the egg chamber, and gives a longer, steadier flight with more telemetry per metre of altitude (about 45 packets at the max rate).
- **The price is drift and cloth.** A larger canopy drifts further with the wind (measured: ≈ 1.9 m/s of ground speed in Flight 1) and adds mass; at 30 m and a ~15 s descent the drift is a few tens of metres, which the recovery plan absorbs.

9.3 The 6 ft canopy across the mass band

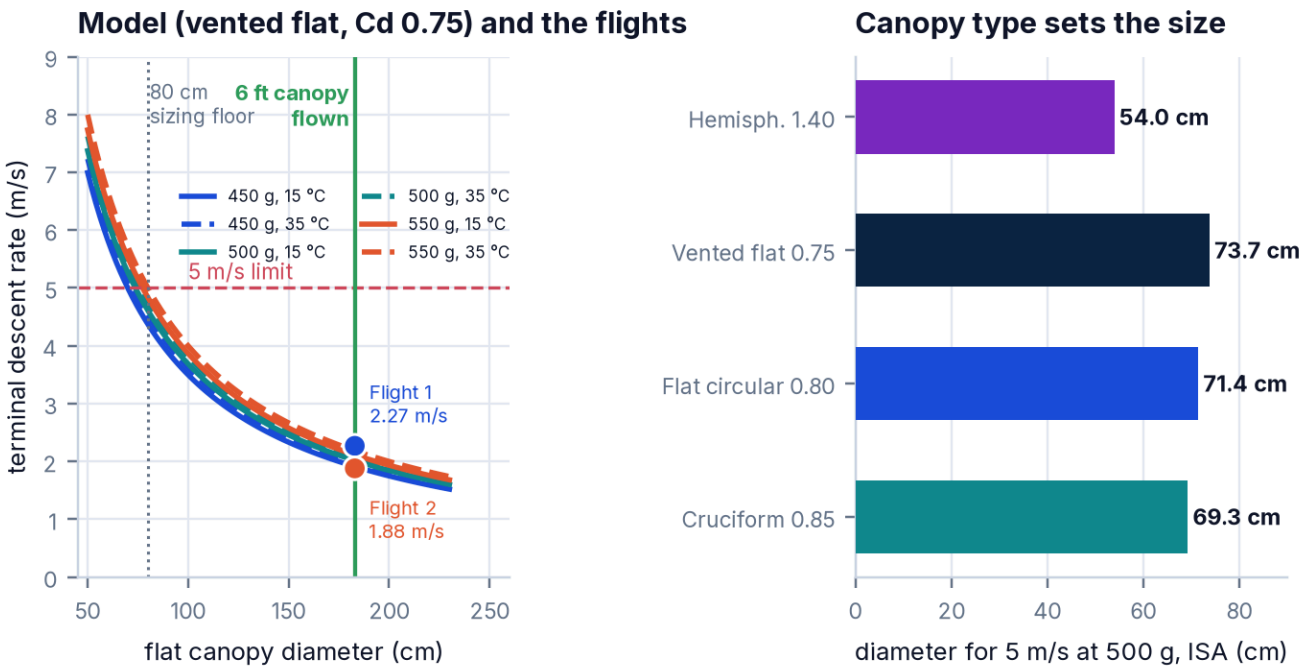


Figure 21 · Left: terminal descent rate against flat canopy diameter for the three masses of the band, on a standard day (solid) and a 35 °C day (dashed). The 80 cm sizing floor and the 6 ft canopy flown are marked, with the two measured flight rates. Right: the diameter needed for 5 m/s at 500 g for each canopy type.

Table 19 · **Model for the 6 ft vented-flat canopy (Cd 0.75) across the band, 29.5 m launch height**

Flight mass	Air	Rate	Descent time	Packets at 0.7 s	Packets at 3.09 Hz
450 g	ISA 15 °C	1.91 m/s	15.6 s	22	~48
450 g	31.4 °C (flight day)	1.97 m/s	15.1 s	21	~47
500 g	ISA 15 °C	2.02 m/s	14.8 s	21	~46
500 g	31.4 °C (flight day)	**2.07 m/s**	**14.4 s**	20	~44
550 g	31.4 °C (flight day)	2.17 m/s	13.7 s	19	~42

9.4 Three constraints on the canopy that are not the diameter

- **It must not be tightly packed** — external or semi-exposed, so it deploys immediately on launch (a rulebook requirement). A canopy stuffed inside the body risks the deployment itself.
- **The descent must be stable**, without tumbling or spinning. An unvented flat circular canopy oscillates; a central vent of roughly 10 % of the diameter costs a little drag and buys a great deal of stability.
- **The structure must be intact after landing and telemetry must continue** — so the antenna and the battery connection have to survive the arrival.

Canopy type is the largest uncertainty in the model

Table 20 · **Drag coefficient and the diameter it implies for 5 m/s at 500 g**

Canopy	Cd	Diameter at 500 g
Cruciform	0.85	69.3 cm
Flat circular	0.80	71.4 cm
Vented flat	0.75	73.7 cm
Hemispherical	1.40	54.0 cm

The sizing uses the *lowest* plausible Cd for a stable canopy (vented flat, 0.75) — the choice that makes the canopy largest and the descent slowest.

9.5 The flights against the model

The model was written before the flights; the 6 ft canopy flew twice. The comparison is a direct test of it:

Table 21 · **Model and measurement**

	Model, 6 ft canopy, Cd 0.75	Flight 1	Flight 2
Steady descent rate, 450 – 550 g	**1.97 – 2.18 m/s** (2.07 at 500 g)	**2.27 ± 0.05 m/s**	**1.88 ± 0.02 m/s**
Deviation from the 500 g prediction	—	+9 %	−9 %
Time to fall 29.5 m	13.7 – 15.1 s (31 °C air)	— (record ends 1.3 m above the ground)	**15.4 s** for 29.6 m
Cd implied for the 6 ft canopy, 450 – 550 g	0.75 assumed	**0.57 – 0.69**	**0.82 – 1.00**

The model predicted both flights to within 10 %. The two flights bracket the 500 g prediction on either side, and the Cd they imply (0.57–1.00) brackets the 0.75 the model assumed — the spread being the same ±10 % uncertainty in the flown mass that the sizing allowed for. A closed-form model with one tabulated coefficient, written before anything had flown, predicted the descent rate of a vehicle thrown from a terrace to within a tenth.

> **HOW THE DRAG COEFFICIENT IS INFERRED**
> At steady descent weight equals drag, so $C_d \cdot S = 2\,m\,g / (\rho\,v^2)$. ρ is computed from the pressure and temperature the vehicle measured in the descent (1.156 kg/m³ and 1.157 kg/m³ for the two flights), v is the least-squares slope of the temperature-corrected height, S is the flat area of a 6 ft disc (2.63 m²) and m is taken across the 450–550 g band. Chapter 14 tabulates the values for each mass.

10

CHAPTER 10

Flight software

The single non-blocking loop, how it reads its sensors, how it decides what state it is in — drawn as flowcharts and quoted from the source.

10.1 The flight loop

10.2 Sensor acquisition

10.3 The mission state machine

10.4 Start-up calibration

10.5 Telemetry generation and the radio

10.6 The fault model

10.7 Code originality and independence

10.8 Design rules, restated as behaviour

10.1 The flight loop

The vehicle runs a **single non-blocking loop on a 2 ms tick.** Nothing waits on hardware, nothing allocates, and every recovery path is bounded.

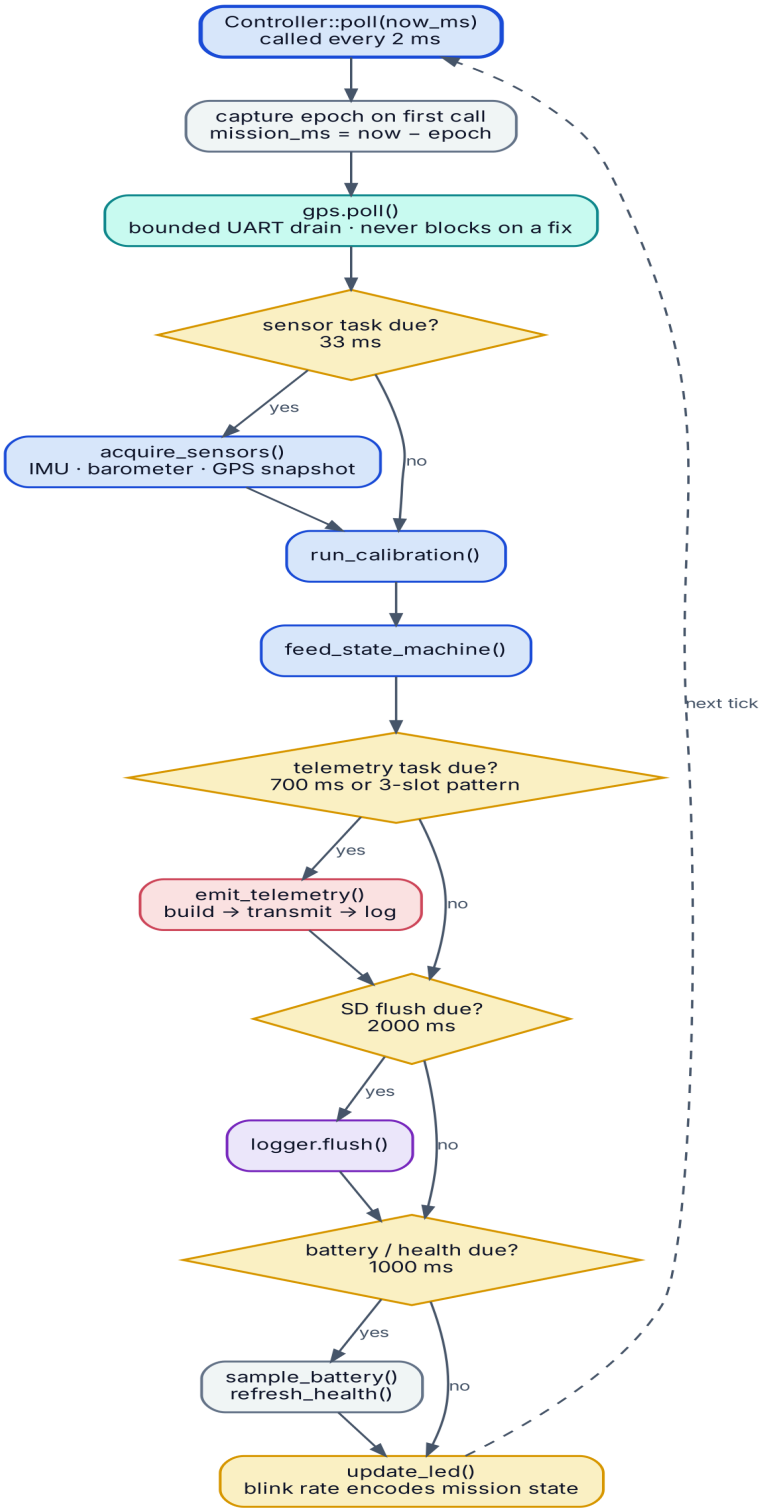


Figure 22 · `Controller::poll()` — the whole of the vehicle's behaviour is composed here.

The ordering is deliberate. GPS is drained first so a full UART buffer can never back up — at 9600 baud the RP2040's 32-byte FIFO fills in 33 ms, and `validate_config()` enforces that the tick is short enough. Calibration and the state machine run *before* telemetry, so every packet carries the state that matches the samples inside it.

CODE · THE SCHEDULER (SRC/SCHEDULER.CPP, COMPLETE)

```
bool PeriodicTask::due(std::uint64_t now_ms) {
    if (period_ms_ == 0) return false;
    if (now_ms < next_ms_) return false;
    next_ms_ += period_ms_;
    if (next_ms_ ⩽ now_ms) {          // fell behind by more than one period:
        next_ms_ = now_ms + period_ms_; // re-anchor instead of firing a catch-up burst
    }
    return true;
}
```

`PeriodicTask` is the only timer in the flight core. Its stall rule matters: after a long SD write or a re-initialisation, the loop **does not** transmit a burst of late packets — it re-anchors to `now + period`, keeping the cadence the ground station sees regular.

10.2 Sensor acquisition

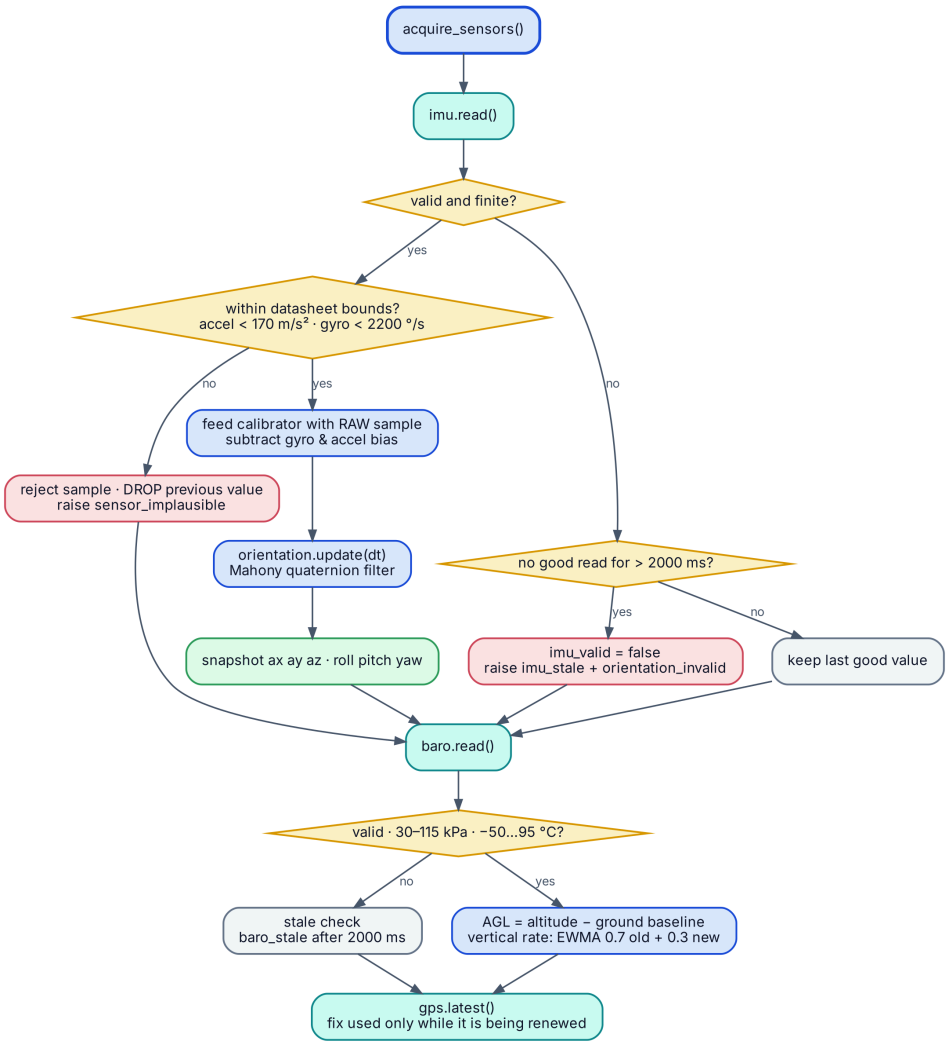


Figure 23 · Sensor acquisition every 33 ms: plausibility gating, staleness and the vertical-rate estimate.

Two rules matter here.

1. **An implausible reading is worse than no reading.** A value outside datasheet-derived bounds (acceleration over 170 m/s², rate over 2200 °/s, pressure outside 30–115 kPa, temperature outside −50…95 °C) is not merely skipped — the previously held value is dropped too, so the vehicle never coasts on data from a sensor that is actively wrong.
2. **Staleness is time-based, not attempt-based.** One failed read changes nothing; two seconds without a good read raises the fault.

Altitude and vertical rate

The barometric altitude is the standard-atmosphere relation against a ground reference captured at calibration:

CODE · SENSOR_MATH.CPP

```cpp
double pressure_altitude_m(double pressure_pa, double reference_pressure_pa) {
    if (!(pressure_pa > 0.0) || !(reference_pressure_pa > 0.0)) return 0.0;
    return 44330.0 * (1.0 - std::pow(pressure_pa / reference_pressure_pa, 1.0 / 5.255));
}
```

The vertical rate is an exponentially weighted average of successive altitude differences (0.7 old + 0.3 new), updated **only when the barometer has produced a new conversion** — re-reading a sensor faster than it converts would otherwise feed repeated values into the derivative and push the estimate toward zero in the middle of a descent. That hold is bounded, because an estimate frozen at a descent rate would stop the landing from ever being detected.

10.3 The mission state machine

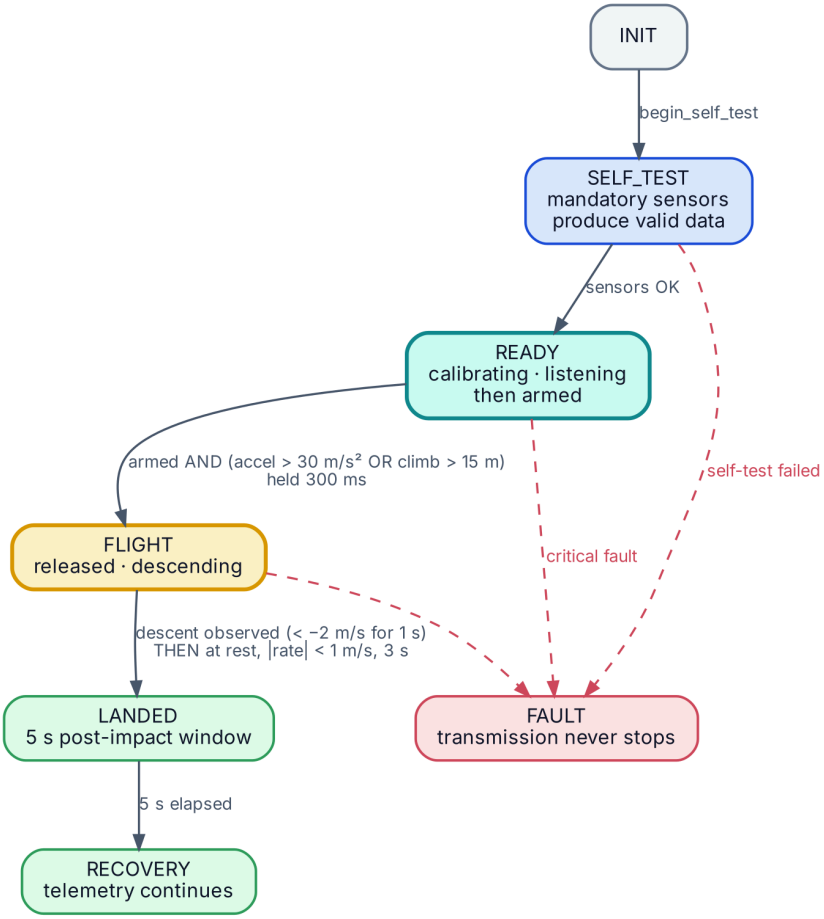


Figure 24 · Every transition and the guard on it. `FAULT` stops state progression, never transmission.

The detection logic itself is short enough to quote. This is the launch test and the descent gate of `state_machine.cpp` (comments abridged):

CODE · STATE_MACHINE.CPP (ABRIDGED)

```
case MissionState::ready: {
    if (!in.armed) { launch_condition_since_ms_ = 0; break; }   // ignore motion until armed
    const bool boost = in.accel_magnitude_mps2 > config_.launch_accel_mps2;
    const bool climb = in.altitude_agl_m      > config_.launch_altitude_gain_m;
    if (boost || climb) {
        if (launch_condition_since_ms_ == 0) launch_condition_since_ms_ = now_ms;
        if (now_ms - launch_condition_since_ms_ ⩾ config_.launch_confirm_ms)
            enter(MissionState::flight, now_ms);
    } else launch_condition_since_ms_ = 0;
    break;
}
case MissionState::flight: {
    // The descent gate: a vehicle cannot land without descending first.
    if (in.altitude_rate_mps < -config_.landing_descent_rate_mps) {
        if (descent_since_ms_ == 0) descent_since_ms_ = now_ms;
        if (now_ms - descent_since_ms_ ⩾ config_.landing_descent_confirm_ms)
            descent_observed_ = true;
    } else descent_since_ms_ = 0;

    if (now_ms - entered_ms_ < config_.min_flight_ms) break;
    const double accel_error = std::fabs(in.accel_magnitude_mps2 - sensors::kStandardGravity);
    const bool at_rest = accel_error < config_.landing_accel_epsilon_mps2 &&
                         std::fabs(in.altitude_rate_mps) < config_.landing_altitude_rate_max_mps;
    if (at_rest && descent_observed_) {
        if (rest_condition_since_ms_ == 0) rest_condition_since_ms_ = now_ms;
        if (now_ms - rest_condition_since_ms_ ⩾ config_.landing_confirm_ms)
            enter(MissionState::landed, now_ms);
    } else rest_condition_since_ms_ = 0;
    break;
}
```

WHY EACH THRESHOLD HAS THE VALUE IT HAS

Constant	Value	Reason
`launch_altitude_gain_m`	15 m	Half the 30 m launch height: far above barometric noise (0.1 m) and anything a hand carries the vehicle through, yet reached with a full 15 m of the climb still ahead — whether the climb is a drone lift or a staircase
`launch_accel_mps2`	30 m/s²	3 g — above any handling, below any real throw or launch
`launch_confirm_ms`	300 ms	Longer than a knock, shorter than any launch
`arming_delay_ms`	3000 ms	Start-up transients (supply settling, filter convergence) are over
`landing_descent_rate_mps`	−2 m/s	Well below the 5 m/s cap, well above hover and noise
`landing_accel_epsilon_mps2`	2.5 m/s²	Within a quarter of 1 g — what “at rest” reads
`landing_altitude_rate_max_mps`	1 m/s	Swing under the canopy sits well below this on the ground
`landing_confirm_ms`	3000 ms	Longer than any bounce, short enough that the 5 s post-impact window starts promptly
`post_impact_transmission_ms`	5000 ms	The rulebook's 5 s; a smaller value will not build

10.4 Start-up calibration

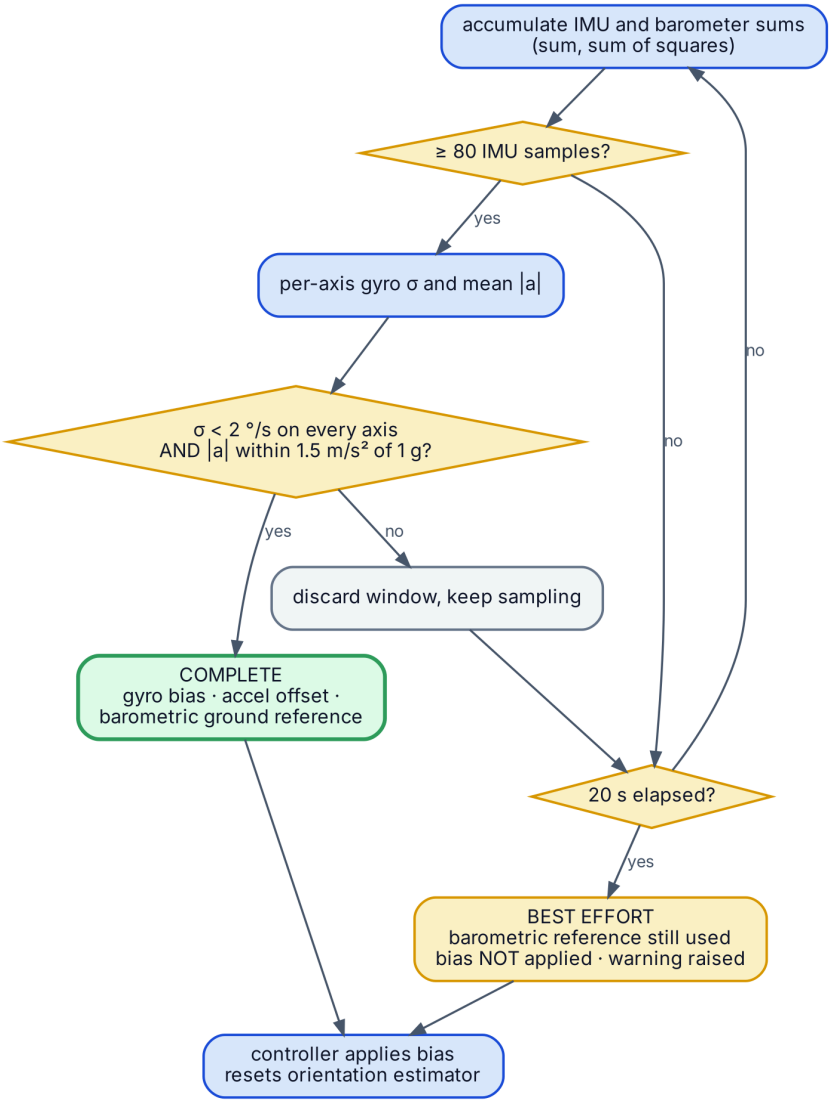


Figure 25 · Pad calibration. Best-effort still produces a usable barometric reference, because averaging pressure does not require stillness.

The asymmetry is intentional. A bias measured while the vehicle was moving would be worse than no correction, so it is discarded — but the **barometric ground reference stays usable** even then. Calibration never blocks the mission. The acceptance test is also two-sided: low variance means "not shaking" but not "not turning" (a constant rotation is perfectly steady), so the mean gyro bias is bounded against the datasheet's zero-rate offset as well.

The post-landing session of Flight 1 shows the calibrator on real data: the transmitted altitude stands at 17.0 m for 5.5 s (the power-on baseline, standard sea level) and then drops to 0.0 ± 0.1 m the instant calibration completes; the yaw channel, which had drifted +0.63 °/s before calibration, is frozen to ±0.1° afterwards.

10.5 Telemetry generation and the radio

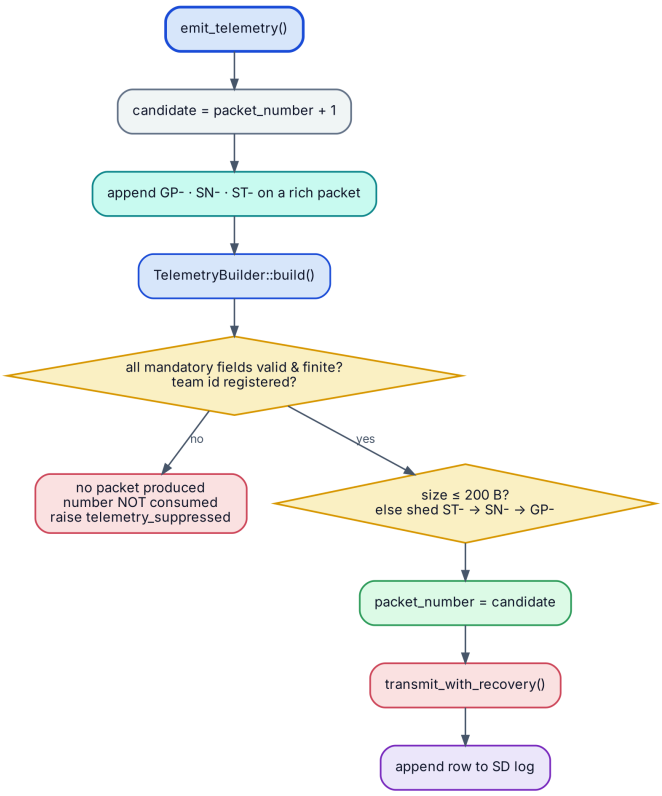


Figure 26 · Building a packet: validity, size shedding and the packet number.

CODE · CONTROLLER.CPP, EMIT_TELEMETRY (ABRIDGED)

```
const std::uint32_t candidate = packet_number_ + 1;
const bool rich = !max_rate_ || max_rate_slot_ == 0;
...
auto built = builder_.build(candidate, mission_ms, snapshot_, extra, rich);
// shed optional content in order of value: status → sound → GPS
if (too_long(built)) { extra.clear(); built = builder_.build(candidate, ...); }
if (too_long(built)) { built = builder_.build(candidate, ..., /*air_sound=*/false); }
if (!built) {                          // mandatory data invalid:
    faults_.report(FaultCode::telemetry_suppressed, ...);
    return;                            // no packet, and the number is NOT consumed
}
packet_number_ = candidate;
service_ground_commands(mission_ms);  // BEFORE the transmit: the receive window is the gap
start_with_recovery(built→packet, mission_ms);
if (max_rate_) telemetry_task_.reschedule(mission_ms, rich ? kMaxRateRichSlotMs : kMaxRateLeanSlotMs);
logger_.append(builder_.sd_line(*built, state, faults_.total_occurrences()));
```

Two details in it carry design decisions. **The command service runs before the transmit** because the receive window is the gap between two packets, and transmitting clears the flags of anything that arrived in it — polling afterwards would read a window that had just been wiped. And **the SD row is written while the packet is still on the air**: the radio sends the packet on its own, so the card write costs no additional time in the cycle.

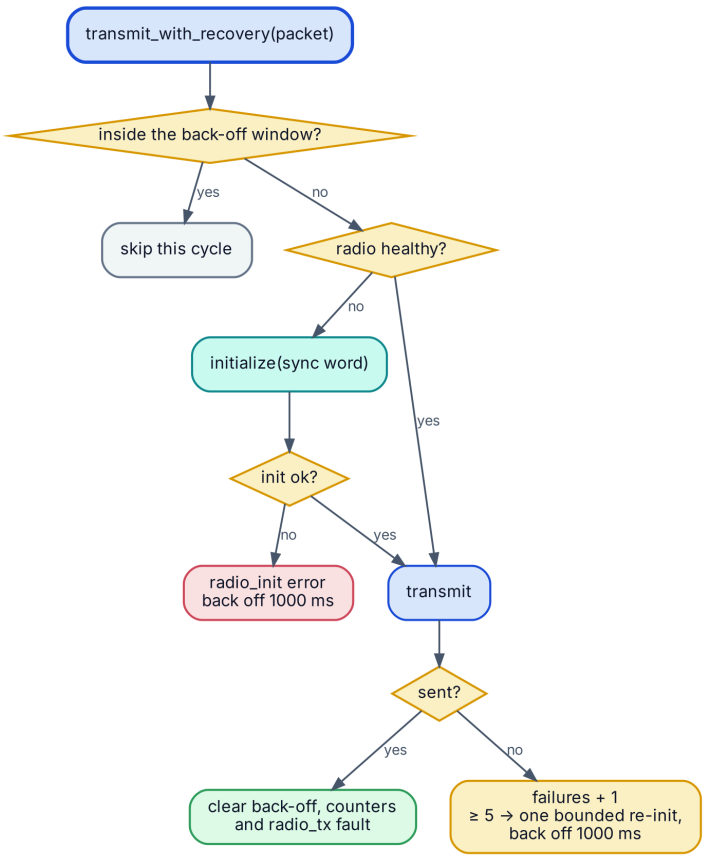


Figure 27 · Transmit with bounded recovery. A dead radio costs one initialisation per back-off window, never a blocking retry loop.

10.6 The fault model

Faults are **classified, not fatal.** Twenty enumerated codes in a fixed-size array — no allocation, constant cost regardless of mission length. A fault carries a severity, an occurrence count and first/last timestamps; severity is monotonic while a fault is active, so a later routine report cannot silently demote one that still applies.

Table 22 · **Fault classes and their effect**

Class	Effect	Examples
Warning	Counted and reported; nothing else changes	Calibration best-effort, low battery, GPS stale, watchdog restart recorded
Degraded	A field or a subsystem is dropped; telemetry continues	SD logging disabled after 10 consecutive write failures, radio re-initialising, sound stale
Critical	`→ FAULT` ; state progression stops — **transmission does not**	Invalid configuration, failed self-test, both IMU *and* barometer stale

The status field makes the fault count visible on the air: `ST-F111` is *flight, armed, calibrated, one fault active*; the watchdog restart after Flight 1's landing is recorded by the same mechanism (`ST-R004` counted four active faults while the vehicle was calibrating, three once it had armed).

10.7 Code originality and independence

Every driver in the flight path is **written for this project against its datasheet and register map**: IMU, BMP280, NEO-6M, SX1278 and the SD protocol. There are no third-party libraries anywhere in the flight path.

- The **BMP280 compensation** reproduces the datasheet's own reference vector.
- The **SD driver** implements CMD0/CMD8/ACMD41 initialisation and SDHC versus SDSC block addressing; a **FAT32 reader** walks an MBR, a BPB and a cluster chain to find the log file.
- The **LoRa driver** is register-level and is driven through a callback struct, so the *same* code runs on the vehicle, on the bridge, and against a fake register bank in tests.
- The **GPS parser** is a streaming NMEA-0183 parser (GGA and RMC) with checksum validation and the satellite/HDOP quality gate.
- The **attitude filter** is a Mahony complementary filter on a unit quaternion — chosen over Euler integration because a tumbling CanSat passes through the ±90° pitch singularity that breaks the Euler form. A test drives it through a 20 s tumble at 100 °/s about all three axes.

Table 23 · **The source tree (lines include tests)**

Area	Files	Lines	Role
`firmware/flight-computer`	52	15,180	Flight core, Pico hardware layer and their tests
`firmware/common`	9	2,244	Telemetry format and SX1278 driver shared by vehicle and bridge
`firmware/ground-station`	6	782	Bridge firmware and CRC framing
`ground-station/software`	17	3,828	Python receive pipeline and tests
`ground-station/web`	3	4,996	Zero-dependency browser console and tests
`simulations`	2	763	Descent model and tests
`tools`	18	5,538	Link-budget calculator, generators, documentation checker, log utilities
`analysis`	8	≈ 3,300	Post-flight analysis, launch-day analysis and figures

18,206 lines of C++ firmware and ≈ 12,800 lines of Python — all written for this project.

10.8 Design rules, restated as behaviour

- **It calibrates itself on the pad — and never lets that block the mission.**
- **A start-up glitch cannot trigger a false launch.** Launch is refused until the vehicle is armed.
- **Wrong data is never transmitted, and suppression never fakes packet loss.**
- **Every peripheral can fail without stopping the mission.** GPS missing → its fields are omitted. SD failing → logging disables itself after ten consecutive failures. Radio down → bounded re-initialisation with a 1 s back-off. Only a total loss of mandatory sensing is critical — and even then telemetry keeps running.
- **It survives its own restarts.** A 2 s hardware watchdog restarts a hung loop; telemetry resumes automatically at the max rate and the restart is reported as a fault so the ground station can see it. Flight 1's post-landing session (Chapter 14) is that path in the log.

11

CHAPTER 11

Telemetry protocol and the radio link

What goes over the air, how big it is allowed to be, why the modem is configured the way it is — and how the link budget compared with the flights.

11.1 The packet

11.2 Two packet shapes and a 200-byte ceiling

11.3 The radio configuration

11.4 Airtime decides the rate

11.5 The link budget, against the flights

11.1 The packet

Every packet is the rulebook string — text, not a binary struct — so any receiver, and any human with a serial terminal, can read it. This is a real packet from Flight 1 (P-1585, received at 18:25:47 IST), split by what each part carries:

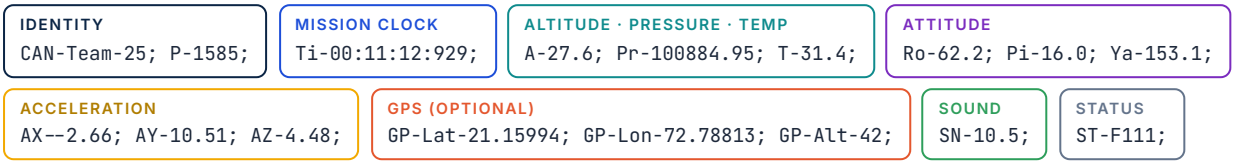


Table 24 · Packet fields

Field	Meaning	Format
`CAN-Team-XX`	Team identifier	Correct identifier in every packet
`P-XXX`	Packet number	Starts at `P-001`, increments sequentially
`Ti-HH:MM:SS:MS`	Mission clock since power-on	Hours wrap at 100 so the field never widens
`A-XXX.X`	Altitude	Metres, 1 decimal, relative to the pad baseline
`Pr-XXXX.XX`	Pressure	Pa, 2 decimals
`T-XX.X`	Temperature	°C, 1 decimal
`Ro-` / `Pi-` / `Ya-`	Roll / pitch / yaw	Degrees, 1 decimal
`AX-` / `AY-` / `AZ-`	Acceleration	m/s², 2 decimals
`GP-Lat` / `GP-Lon` / `GP-Alt`	GPS position, only when a valid fix exists	5 decimals / whole metres
`SN-`	Acoustic level	mV peak-to-peak
`ST-`	Status: state letter, armed, calibrated, active faults — `ST-F110` is FLIGHT, armed, calibrated, none	Dropped from any packet it would push past 200 bytes

Precision is enforced exactly, in the formatter and in all three parsers. The C++, Python and JavaScript implementations are held to **one shared fixture file** (`test-data/protocol-fixtures.tsv`), so they cannot drift apart. A negative value keeps its own minus sign after the separator (`AX--2.66`), which is why a parser must split on the last dash that is not itself a sign; a dedicated fixture holds every parser to that.

11.2 Two packet shapes and a 200-byte ceiling

Table 25 · The two packet shapes

Shape	Carries	Worst case
Rich	The twelve mandatory fields, then `GP-Lat`, `GP-Lon`, `GP-Alt`, then `SN-`, then `ST-`	209 B, held to **200 B** on the air
Lean	The twelve mandatory fields only	147 B

Measured in the flights: **lean packets 118–130 bytes, rich packets 136–188 bytes**, none above 188.

WHY 200 BYTES

The 200-byte ceiling is not the team's. The organizers' receiver — an ESP32 running the arduino-LoRa library — reads into a 201-byte buffer and discards anything longer. Their station is the one that scores, so 200 is the ceiling whatever the team's own bridge can hear. At 6 decimals of latitude and 1 of altitude the GPS block was 55 bytes and mandatory + GPS was 202 — two bytes over. Printed to what the NEO-6M actually resolves (**5 decimals = 1.1 m against its ~2.5 m error, and whole metres**) the block is 51 bytes and the total is 198. The SD log keeps the full precision.

The byte figures are defended **by construction, not by arithmetic**: a test builds the widest packet each shape can produce — packet number 4,294,967,295, a 99:59:59:999 clock, every axis at full scale — and checks it against its constant. That test corrected the design twice, once by two bytes on the lean packet, which crossed a LoRa symbol boundary and cost 5 ms of airtime on every packet.

When a packet would be too long the vehicle sheds optional content in the rulebook's priority order — status field, then sound, then GPS — and never lets the radio truncate silently; if the mandatory block alone overflowed, it would suppress the packet rather than send one that reads as corruption.

11.3 The radio configuration

Table 26 · **Modem parameters**

Parameter	Value	Fixed by
Frequency	433 MHz	Rulebook band
Spreading factor	SF7	Engineering choice — airtime
Bandwidth	125 kHz	Engineering choice
Coding rate	4/5	Engineering choice — lowest overhead
Preamble / CRC	8 symbols / on	SX127x default; corruption must be detectable
TX power	17 dBm (PA_BOOST)	RA-02 maximum without PA_DAC
Sync word	`0xA5`	**Rulebook — and the organizers' station listens on nothing else**

WHY BOTH PICOS FLY THE LAUNCH SYNC WORD, ALWAYS

The rulebook gives `0xF3` for testing and `0xA5` for the launch. On the 10 September range test the vehicle was on the test word and the organizers' station heard only the few packets an SX127x sync filter lets through, while the team's own station heard every one — a silent, near-total loss of telemetry that looks exactly like a dead radio. **Both images now fly `0xA5` for testing as well as for launch**, so what the team's bridge hears is what the organizers' station hears. The 30 September log is the confirmation: the organizers' station recorded both flights.

One definition of the modem parameters is shared by both ends — `link_profile.hpp` — because before it existed the two carried separate copies and agreed only by coincidence.

11.4 Airtime decides the rate

LoRa trades data rate for sensitivity: at a high spreading factor a single packet can occupy the channel for longer than the interval it must fit in. From the Semtech datasheet formula (verified against two published reference vectors, in two languages, and against a radio):

```
Tsym      = 2^SF / BW
Tpreamble = (n_preamble + 4.25) * Tsym
n_payload = 8 + max(ceil((8*PL - 4*SF + 28 + 16*CRC - 20*IH) / (4*(SF - 2*DE))) * (CR + 4), 0)
ToA       = Tpreamble + n_payload * Tsym
```

Table 27 · Airtime of a 255-byte packet at coding rate 4/5

Config	Airtime	Max rate	Rate at 50 % duty	Meets 1 Hz?
SF7 / 125 kHz	**399.6 ms**	2.50 Hz	**1.25 Hz**	**yes**
SF8 / 125 kHz	707.1 ms	1.41 Hz	0.71 Hz	over duty
SF9 / 125 kHz	1250.3 ms	0.80 Hz	0.40 Hz	no
SF7 / 250 kHz	199.8 ms	5.00 Hz	2.50 Hz	yes
SF12 / 125 kHz	9019.4 ms	0.11 Hz	0.06 Hz	no

THE FINDING THAT CHANGED THE DESIGN

The provisional default was SF9 / 125 kHz. At that setting a full packet takes **1,250 ms** of airtime — longer than the interval it was supposed to fit inside. The scheduler had been configured for a 500 ms period, a rate the radio could never have delivered. Re-deriving the rate from airtime moved the modem to **SF7**, which is the only spreading factor that meets 1 Hz with duty margin on a full packet. SF7 costs about 6 dB of sensitivity against SF9 and buys a threefold airtime reduction — a favourable trade for a 30 m descent.

Measured on the delivered module: a 206-byte packet took 333.7 ms against 327.9 predicted, and a 255-byte one 406.9 ms against 399.6 — both about 1.8 % over the model, reproduced to 0.1 ms across four sessions. The slots are therefore sized from **measured** airtime plus a 50 ms guard.

The rate: 1.43 Hz in the window, 3.1 Hz in flight

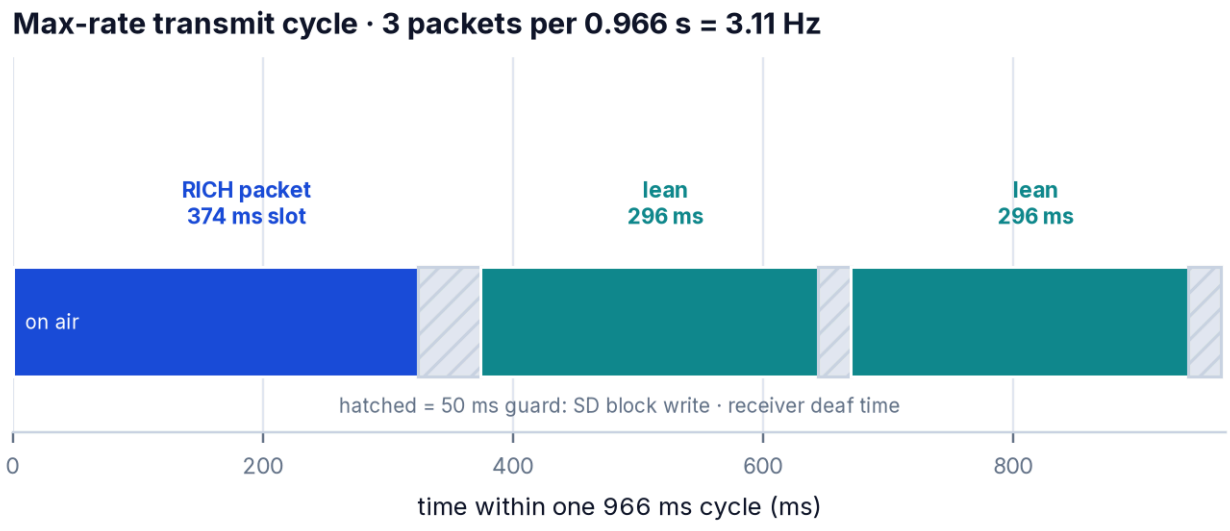


Figure 28 · The max-rate transmit cycle: one rich slot (374 ms) and two lean slots (296 ms each) per 966 ms.

Table 28 · Telemetry rates

	Command window	Flight, after the window closes
Period / cycle	700 ms	966 ms (374 + 296 + 296)
Packet rate	**1.43 Hz**	**3.11 Hz designed · 3.09 Hz measured in Flight 1**

	Command window	Flight, after the window closes
Sensors on the air	1.43 Hz	1.04 Hz (one rich packet per cycle)
Channel duty	46 % measured	84 %

The measured cadence in the flight log — gaps of 0.374 s, 0.296 s and 0.297 s repeating exactly — is the designed pattern to the millisecond (Chapter 14, Figure "Max-rate cadence"). **The 50 ms guard does two jobs in the same gap:** it covers the vehicle's own worst-case SD block write (30 ms), and the organizers' receiver, which is deaf for about 35 ms after every packet while it prints what it just heard at 115200 baud.

The rulebook's 1 Hz is a floor this vehicle cannot be configured onto. Three independent mechanisms enforce it: a `static_assert` refuses to compile a profile above 950 ms, `validate_config()` refuses to run one, and the documentation checker refuses to pass a repository whose text disagrees. A flight build silently running at 1 Hz is exactly the failure the design guards against.

11.5 The link budget, against the flights

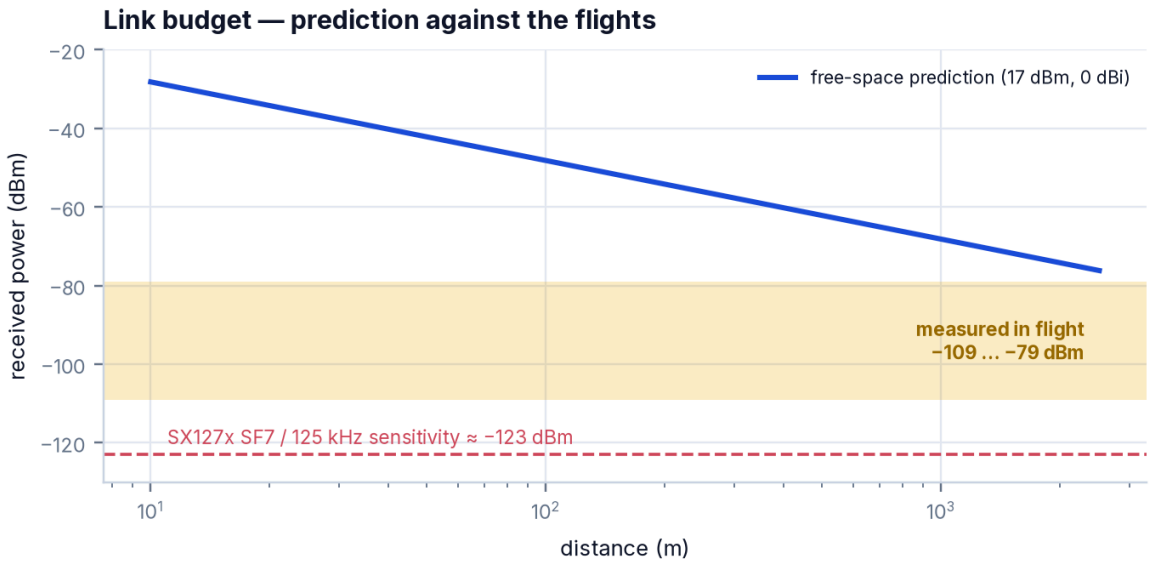


Figure 29 · Free-space prediction (17 dBm, 0 dBi antennas) with the receiver floor, and the band of RSSI the organizers' station actually recorded in flight.

Free-space path loss at 433 MHz is FSPL = 32.45 + 20 $\log_{10}$(433) + 20 $\log_{10}$(d_{km}) dB: 79 dB at 500 m, 85 dB at 1 km, 91 dB at 2 km. Typical SX127x sensitivity at 125 kHz is about **−123 dBm at SF7**. The calculation said SF7 is *not range-limited* for a 30 m descent. The flights say so with measured data:

- **RSSI in flight: −109 to −79 dBm** (mean −92.1 dBm over 102 packets).
- **Link margin over the receiver floor: 14 dB on the weakest packet, 31 dB on average, 44 dB at best.**
- **SNR: −6.75 to +10.75 dB**, and it saturates near +10 dB for any RSSI better than about −90 dBm — the receiver's own noise floor, not the vehicle's.

Chapter 14 analyses the link packet by packet.

12

CHAPTER 12

Onboard logging and the ground station

The card that keeps what the radio can lose, and the three-language pipeline that turns packets into data.

12.1 Onboard logging

12.2 Ground station

12.3 The web console at the competition

12.1 Onboard logging

The onboard log writes into **raw 512-byte blocks with no filesystem**, rewriting its header after every record. A brownout or an impact reset therefore resumes at the correct block instead of overwriting flight data.

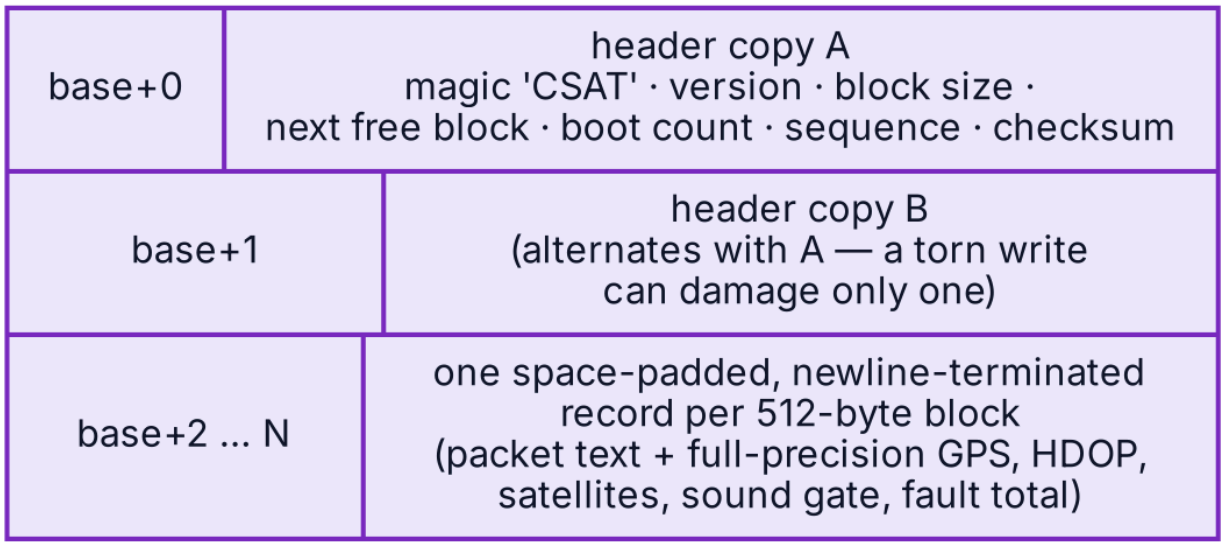


Figure 30 · The raw-block log layout. Two header copies alternate, each with a sequence number and checksum.

The log writes one row per telemetry packet, appended in `Controller::emit_telemetry()` as the packet goes on the air, so it runs at the radio's cadence — 3.11 Hz in flight — not at the 30 Hz sensor rate. Each row carries the packet text plus the satellite count, HDOP and full-precision position that never go on the air, the sound gate percentage, and the fault total.

WHY RAW BLOCKS, AND WHY TWO HEADERS

- **No filesystem to corrupt.** A FAT volume updates a directory entry and an allocation table on every extension; an impact or brownout between those writes can leave a card that mounts as empty. A raw block log has nothing to be inconsistent with.
- **The header is rewritten after *every* record**, so power can fail during a header write. With a single header there would then be no valid resume point and the next boot would restart at the first record block and **overwrite the entire flight it had just recorded.** With two alternating copies, each carrying a sequence number and a checksum, only the copy being written can be damaged; the other still holds the previous complete resume point. A test destroys each copy in turn and checks that the log resumes with its records intact.
- **A full region stops writing rather than wrapping**, so older data is never silently overwritten.
- **A file in a few pieces is not a fault.** The firmware maps log block numbers through an extent list of up to 16 runs; `tools/prepare_sd_card.py` creates and sizes the file and `tools/inspect_sd_log.py` reports its layout.

The SD log is the complete record: it holds every packet the radio sent, and every column the packet leaves out. The post-flight analysis reads it with `tools/read_flight_log.py`, which splits it by power cycle and stops exactly where the data stops.

12.2 Ground station

Three interfaces over one pipeline:

Table 29 · **Ground-station interfaces**

Interface	Use
Web console — `ground-station/web/index.html`	Single file, no build, no dependencies. Demo replay, file replay, or live Web Serial straight from the bridge
Tk dashboard — `python src/main.py live --port COM5 --framed`	Live numeric view, with plots when matplotlib is installed
CLI replay — `python src/main.py replay … --export out.csv`	Offline parse, validate, log and export

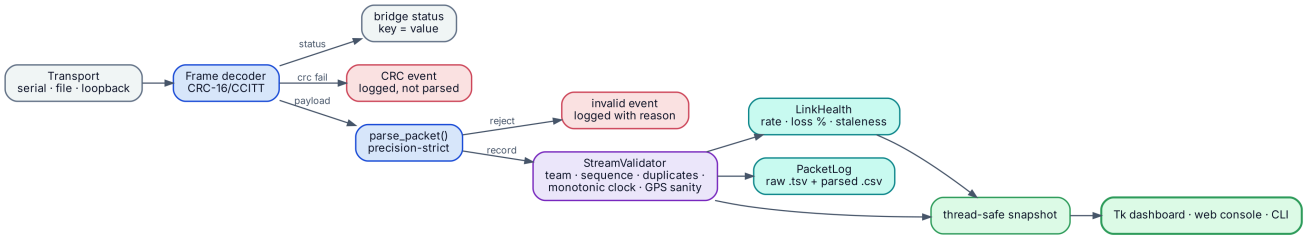


Figure 31 · The receive pipeline. Each stage counts its rejections separately, so a transport fault is never mistaken for a sensor fault.

The bridge firmware is a pure bridge: the SX1278 sits in continuous receive and every received payload is framed onto USB serial as `$<len>,<crc16>,<payload>\n`, with a CRC-16/CCITT-FALSE (polynomial `0x1021`, initial value `0xFFFF`) over the raw payload. It emits a status line at 1 Hz with its own frame and drop counters, re-initialises the radio after 20 consecutive unhealthy polls and runs a 3 s watchdog so a hung bridge reboots and the PC reconnects by itself.

DESIGN DECISIONS IN THE GROUND STATION

- **A CRC on the USB link** separates *transport* corruption from a *malformed packet* — the two need different reactions and look identical without it. The known-answer vector `crc16_ccitt("123456789") == 0x29B1` is asserted in the C++ and Python suites.
- **Three implementations, one fixture file.** The framing and the packet parser exist in C++, Python and JavaScript. They are held to a single shared set of test vectors so they cannot drift, which would otherwise fail silently and totally.
- **Nothing received is ever discarded.** Malformed packets, CRC failures and rejected packets all reach the raw log with their receipt time and the reason.
- **The pipeline runs on a background thread** and hands the interface a snapshot through a bounded queue; when the queue fills, the oldest event is dropped rather than blocking reception, so a slow display can never stall logging.
- **The bridge agreed with the packet stream exactly** on the first closed link: the frame counter incremented once per packet with zero drops — which is how it is known that the radio delivered everything it heard and USB carried everything the radio delivered.

The validator checks team identity, sequence continuity, duplicates, timestamp monotonicity and GPS plausibility; link health is a 5 s sliding window of rate, loss percentage and staleness, and it reports whether the rate that *arrived* cleared 1 Hz — a different question from what the vehicle transmitted.

The vehicle-restart case is handled by design: when the packet counter restarts at `P-001` the ground station recognises a vehicle restart and keeps counting cleanly — the case that appears in Flight 1's logs.

12.3 The web console at the competition

The single-file web console ran live at the competition launch. Its panels — flight profile, attitude, GPS, sound and a raw packet monitor — are driven by the same parser and validator as the Python pipeline.

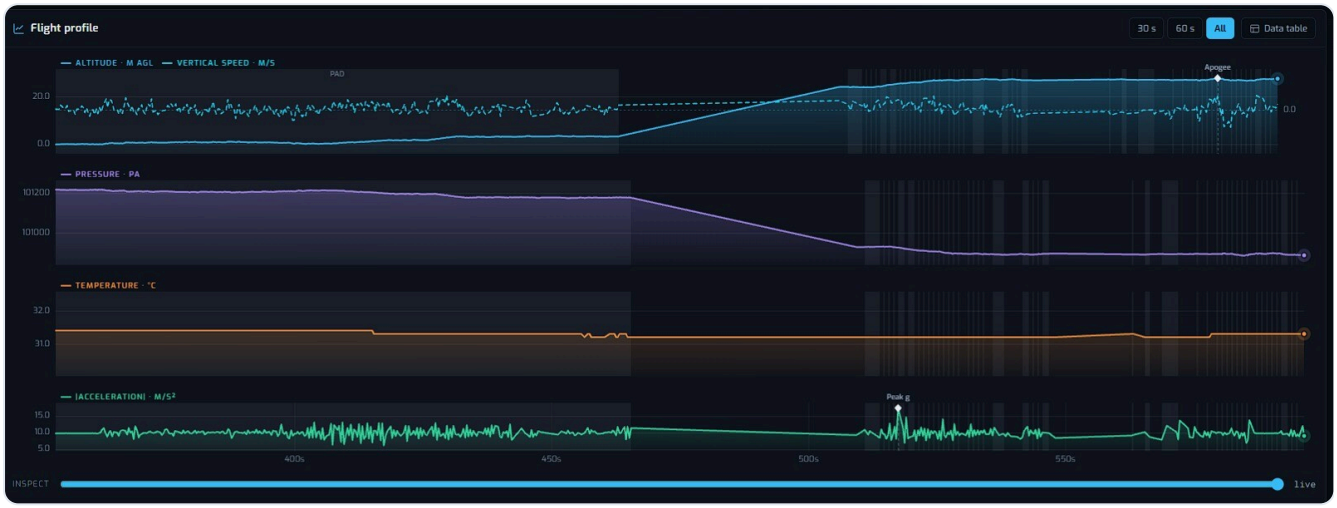


Figure 32 · **Flight profile panel, live.** Altitude above ground (and vertical speed), pressure, temperature and specific force against time, with the pad period shaded, the apogee and peak-g markers, a 30 s / 60 s / all window, a data-table view and an inspect slider that scrubs back through the record. Here the vehicle sits on the pad at 0 m, is carried up the building to about 27 m, and is held at height before the throw.



Figure 33 · **Attitude and GPS panels.** An artificial horizon and compass show roll 131.7°, pitch −15.0° and relative yaw 156.7° (the vehicle held on its side at the terrace edge), with the spin rate; the GPS panel shows the fix, the position relative to the pad (38.2 m at 154°), latitude 21.15995°, longitude 72.78817°, GPS altitude 39 m and the microphone level of 14.5 mV p-p.

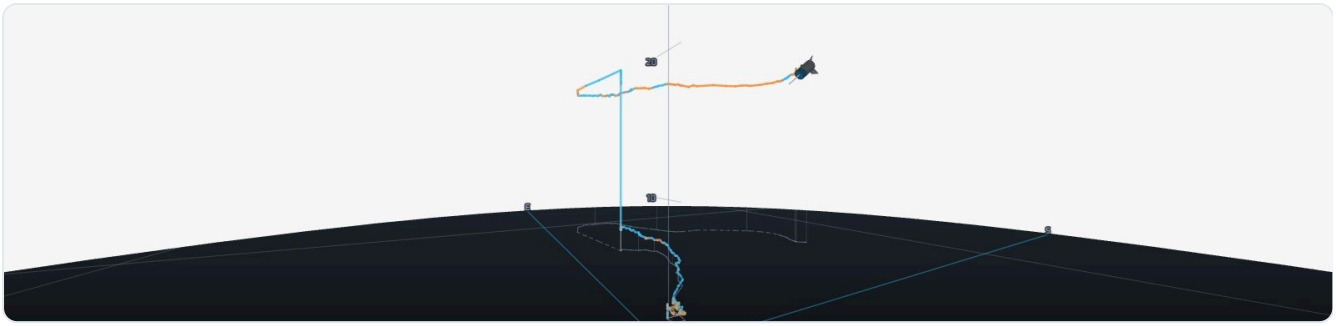


Figure 34 · **Three-dimensional track.** The climb up the building (vertical line), the walk along the terrace (orange: faster segments, blue: slower) and the vehicle's current position, built from altitude, GPS and attitude.

The console shows exactly what the flight analysis of Chapter 14 later confirmed: the carry up the building, the hold at the terrace edge with the vehicle on its side, and a live link throughout.

13

CHAPTER 13

Verification and bring-up

6,132 automated checks on the host, 94 recorded bench measurements — and the prediction written down before each one.

13.1 The host test suite

13.2 Bring-up measurements

13.3 The measurement that characterises the vehicle

13.4 The documentation checker

13.1 The host test suite

```
bash tools/build_host.sh
```

compiles and runs every C++ suite, the Python suites and the Node suite, then checks the documentation against the source. **All of it runs without hardware.**

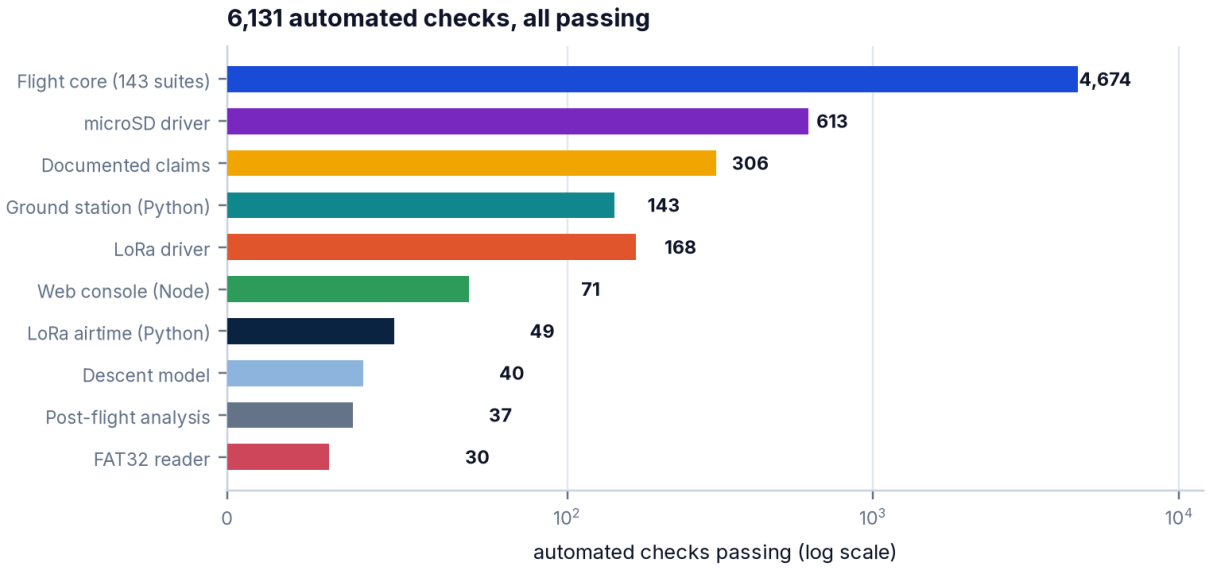


Figure 35 · Automated checks by suite (log scale).

Table 30 · Test suites and what each covers

Suite	Coverage	Result
`flight_tests`	143 suites: packet format, parser, shared fixtures, state machine, orientation and angle wrapping, GPS validation, sensor math, timing, calibration, faults, scheduler, block log and torn-header recovery, controller behaviour, link profile, LoRa airtime	**4674 / 4674**
`sd_card_tests`	microSD init, SDHC vs SDSC addressing, block round trip, bus release, timeouts, write-error paths	**613 / 613**
`sx1278_tests`	LoRa register sequence, TX timeout, RX and CRC handling, RSSI conversion	**168 / 168**
`fat_volume_tests`	FAT32 log lookup: MBR and superfloppy, contiguity, missing file, a card that stops answering	**30 / 30**
`flight_smoke_test`	Boot, first three packets, GPS parse	Passed
`ground_station_tests`	Framing, CRC detection, resync, known-answer vector	Passed
Python — ground station	Parser, validator, transport, health, logging robustness, bridge status, vehicle-restart recovery, cross-language end-to-end trace of real vehicle output	**143 / 143**
Python — tooling	LoRa airtime model pinned to published SX127x reference vectors	**49 / 49**
Python — simulations	Descent model against closed-form limits, ISA density, mass-tolerance argument	**40 / 40**
Python — post-flight analysis	Recovers a synthetic flight's known descent rate, drag coefficient, spin, drift and lost packet; the notebook executed cell by cell	**37 / 37**
Node — web console	Framing, parser, validator, link health, bridge status	**71 / 71**

Suite	Coverage	Result
Documented claims	Numbers in the documentation checked against the source that defines them	**307 / 307**
Pico syntax	11 translation units against SDK stubs	All OK

Total: 6,132 automated checks.

What the tests prove

- The emitted packet matches the rulebook format **byte for byte**.
- The BMP280 compensation reproduces the **datasheet's own reference vector**.
- A boost before arming **cannot** trigger a launch.
- A hovering drone **cannot** be read as a landing.
- Invalid mandatory data suppresses a packet **without consuming its number**.
- `crc16_ccitt("123456789") == 0x29B1` — the standard known-answer vector.
- The vehicle and the bridge are proven to configure **the same radio modem**.
- Three independent parsers in three languages agree on **one fixture file**.

The same tests found **nine real defects in code that already built and passed** during the second review pass — a telemetry rate the radio could not have delivered, three parsers that disagreed, sensors that could not feed their own loop, an attitude filter wrong at the ±180° seam, a packet budget below the real packet, and two SPI drivers that had never executed.

13.2 Bring-up measurements

Every prediction was written down *before* the measurement was taken, and both are recorded side by side in `documentation/testing/bring-up-record.md` — 94 rows, of which 34 were taken at bring-up.

Table 31 · Predicted against measured

Quantity	Predicted	Measured
Radio version register	`0x12`	`0x12`
Airtime, 206-byte packet, SF7/125 kHz	328 ms	**333.7 ms** — +1.8 %, reproduced to 0.1 ms across four sessions
Gyro bias, per axis	within ±25 °/s	**X −3.30, Y +0.86, Z −0.06 °/s**
Barometer output rate	83 Hz	**83.0 Hz**
GPS NMEA output	Clean sentences	**All six, 162 B/s, 0 checksum errors**
SD single block write	—	**2.678 ms mean, 4.9 ms typical worst, 29.8 ms occasional**
SD sustained rate	—	**~300 writes/s**, 3,672 in ten seconds
Rail under 45 transmits	Holds	**3.28 – 3.29 V**
Rail at 100 % write duty	Holds	**3.28 – 3.30 V**
Sensor read cost	Under 33 ms	**0.833 ms worst**
Altitude at rest after calibration	≈ 0.0 m	**−0.5 m to +0.1 m** across 62 packets
First link	—	**66 packets received,** `P-001` **–** `P-066` **, −44 dBm**
`MAX_RATE` at the station	~3.13 Hz	**3.11 Hz** over 544 packets received, RSSI −21 dBm, SNR 10.0 dB

The airtime model survived contact with hardware to within 1.8 %. Chapter 14 adds a second column — what the flights measured — for the quantities that only flight can test.

13.3 The measurement that characterises the vehicle

A 36.8-minute stationary log of 2,209 records was taken as a characterisation of the attitude estimator. The first fifteen minutes were nearly perfect: **±0.8° total in yaw at 0.002 °/s.** The 36.8-minute log gives the vehicle's relative-yaw drift budget, and the pad calibration re-zeroes it at every power-up and again when the command window closes — so the yaw reported in flight is always fresh relative to the pad. Flight 1's post-landing session confirms the effect: yaw frozen to ±0.1° over 7.4 s after the calibration completed.

13.4 The documentation checker

`tools/check_doc_claims.py` has caught real errors repeatedly, including two during the writing of this report. A documentation set that cannot be checked drifts from the code within days; this one fails the build instead. It checks 307 claims, including the numbers in this report's source documents.

14

CHAPTER 14

Flight results and data analysis

Two descents, 102 distinct packets, analysed channel by channel — altitude, temperature, pressure, acceleration, orientation, descent rate, radio link, and how each relates to the others.

14.1 What the ground station recorded

14.2 The telemetry stream itself

14.3 Flight 1

14.4 Flight 2

14.5 The two flights compared

14.6 Attitude and stability

14.7 The pressure–altitude law, and why the transmitted altitude is slightly small

14.8 Descent physics: drag, energy and what the vehicle felt on arrival

14.9 Correlations between the channels

14.10 The radio link in flight

14.11 The sound sensor in flight

14.12 GPS in flight

14.13 The post-landing session: a free calibration experiment

14.14 Predictions against flights

14.15 Summary of the flight analysis

14.1 What the ground station recorded

At the competition launch on 30 September 2026 the organizers' ground station recorded the vehicle's packets and exported them as `Team-25.xlsx` : **180 rows from 10 exported files.** Before any number is quoted, the log has to be made into what it should have been.

Table 32 · From the export to the analysis dataset

Step	Result
Rows in the workbook	**180** — one per received LoRa packet, with host timestamp, packet text, RSSI and SNR
Parsed by the ground station's own `parse_packet()`	**180 / 180** — every row passes the strict parser; `CAN-Team-25` leads every packet
Repeated exports removed (key: packet number + mission clock)	78 duplicate rows removed → **102 distinct packets**
Split at every restart of the mission clock	**4 power sessions** (the rows are not in time order; two exports had been concatenated)
Clock used for dynamics	The **vehicle's mission clock** (`Ti-`), never the host timestamp — the host log buffers: packets P-1622…P-1625 arrived 30 ms apart although they are 0.97 s apart on the vehicle's clock

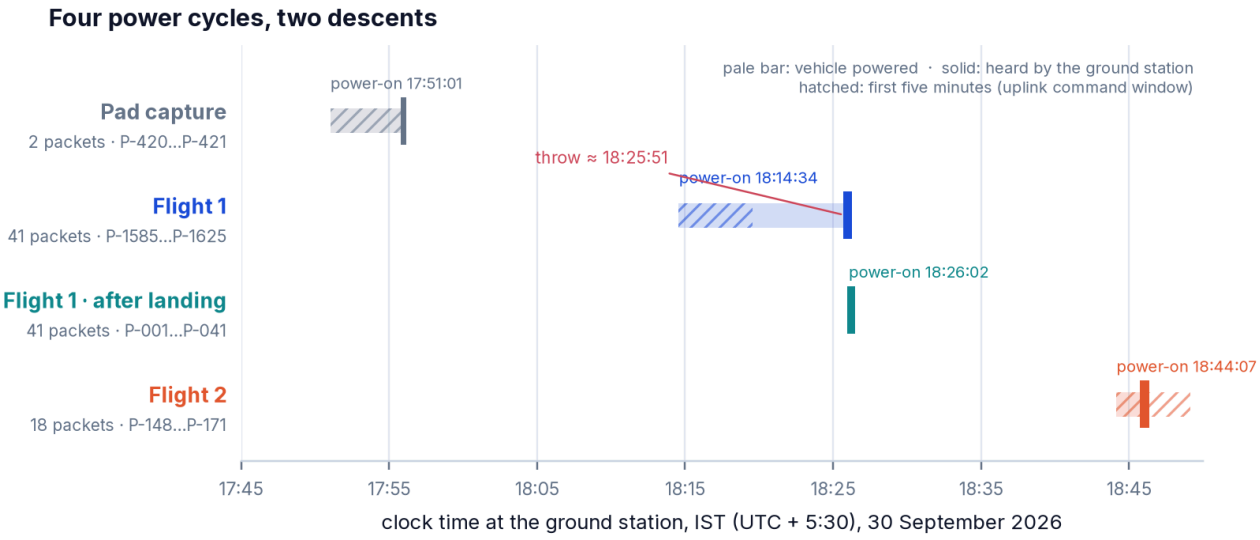


Figure 36 · The four power sessions in the log. Power-on times are reconstructed from the mission clock; the hatched part is the first five minutes (the command window).

Table 33 · The four sessions

Session	Power-on (IST)	Mission clock	Packets	Content
S0 · pad capture	17:51:01	293.3 – 294.0 s	2 (P-420 … P-421)	Standing on the ground, 0 m
F1 · Flight 1	18:14:34	672.9 – 685.9 s	41 (P-1585 … P-1625)	Held at the terrace edge, thrown, descent
G1 · after landing	18:26:02	0 – 12.95 s	41 (P-001 … P-041)	On the ground after Flight 1
F2 · Flight 2	18:44:07	102.9 – 119.0 s	18 (P-148 … P-171)	A 29.6 m descent after a throw from the terrace

Two important conventions follow from the session structure. First, **each power cycle sets its own altitude zero**: the vehicle calibrates a ground baseline at power-up and again when the command window closes, so "0 m" means the pad *of that power cycle*. Second, packet numbers restart at `P-001` at every power cycle — as the rulebook requires — and the ground station's vehicle-restart logic handles it.

14.2 The telemetry stream itself

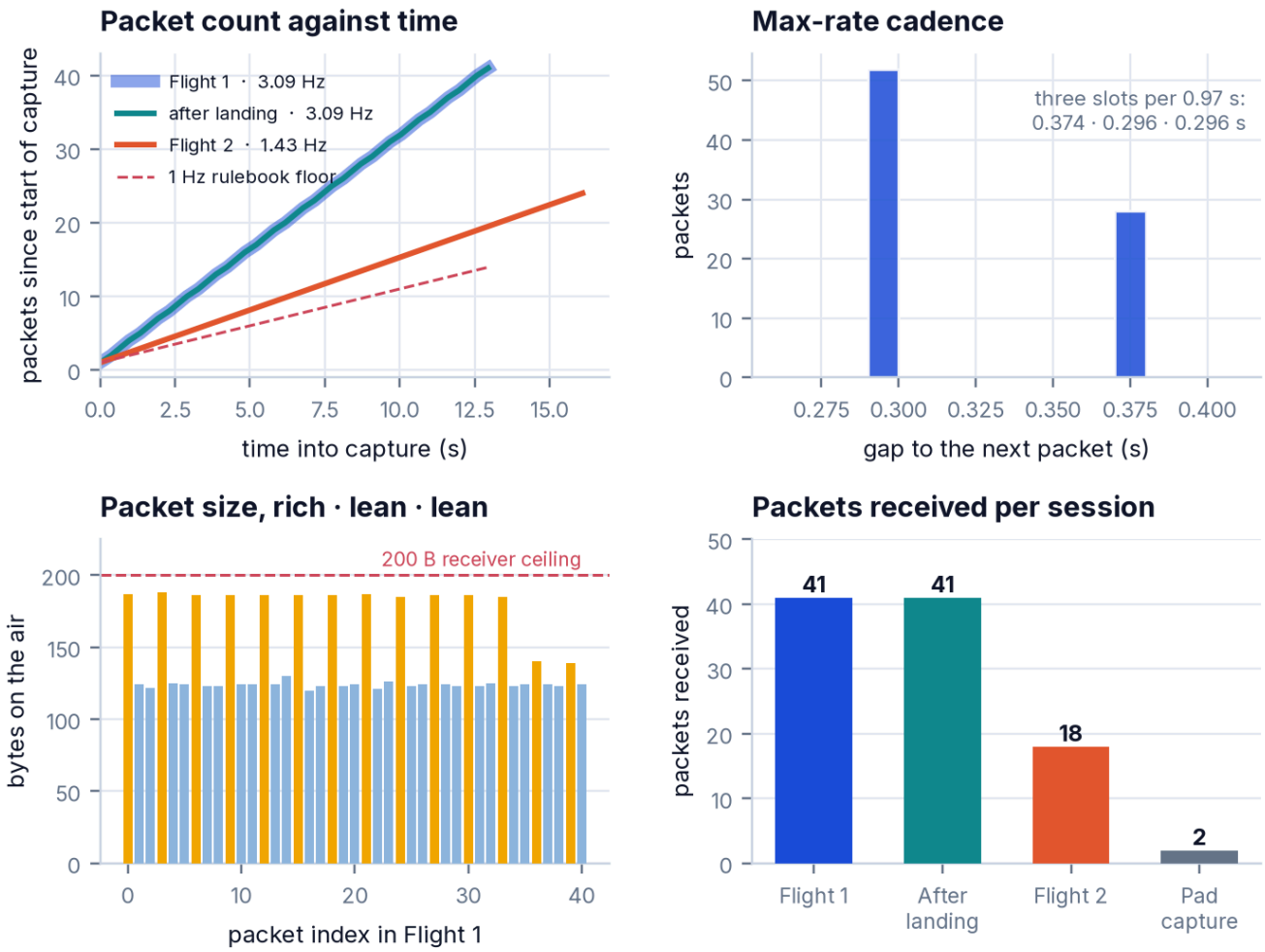


Figure 37 · Top left: packet count against time (the 1 Hz rulebook floor dashed). Top right: gaps between packets in the max-rate pattern. Bottom left: packet size by shape. Bottom right: packets received in each session.

Table 34 · Packet stream statistics

Session	Packets received	Rate on the vehicle's clock	Median gap	Mean packet size
Flight 1	**41**	**3.09 Hz**	0.297 s	143 B
After landing	**41**	**3.09 Hz**	0.297 s	134 B
Flight 2	**18**	**1.43 Hz** transmit cadence	0.700 s	—
Pad capture	2	1.43 Hz	0.701 s	142 B

- **The designed cadence is visible to the millisecond.** In both max-rate sessions the gap to the next packet alternates 0.374 s (after the rich packet), 0.296 s and 0.297 s — the three slots of Section 11.4 — and the pattern `Rll Rll Rll …` (R = rich, l = lean) repeats without a single break across all 41 packets of each session.
- **Packet sizes follow the shapes:** lean packets 118–130 B, rich packets 136–188 B, all comfortably under the organizers' 200-byte ceiling.
- **Flight 1 and the after-landing session each delivered 41 packets**, and the pad capture 2. Flight 2 was captured at the pad-phase cadence of 1.43 Hz (a 0.70 s gap between packets, as in the pad capture) and delivered 18 packets across its 16 s descent.

- **A packet decoded at SNR −6.75 dB** (pad capture, RSSI −105 dBm) — within 1 dB of the SF7 demodulation limit (about −7.5 dB) — shows how much headroom the chosen spreading factor leaves in a weak link.

14.3 Flight 1

Flight 1 was launched from a building terrace — the vehicle was powered on at the ground floor, carried up, held at the terrace edge and **thrown by hand like a projectile**. The vehicle had been powered for 11 min 13 s when the log begins; it had armed after the five-minute command window (`ST-F111` : **F**LIGHT, **1** armed, **1** calibrated), so its state machine recognised the climb while it was carried up and reported `FLIGHT` through the hold, the throw and the descent.

Table 35 · **Flight 1 event log**

Packet	Mission time	Event	Evidence
P-1585 … 1598	672.9 – 677.2 s	**Held at the terrace edge.** Transmitting from 29.4 m (96 ft)	Reported altitude 27.6 – 27.9 m (σ 0.09 m); pressure 1008.8 hPa;
P-1599	677.47 s	**The throw**	Specific force jumps to **50.8 m/s² (5.2 g)** (AX -27.5, AY -13.4, AZ -40.5); RSSI improves from -103 to -88 dBm in one packet
P-1600	677.77 s	**Apex of the throw**	Transmitted altitude 29.1 m (30.7 m corrected = **100.8 ft**);
P-1601 … 1602	678.14 – 678.44 s	**Falling; canopy deploying**	
P-1603	678.74 s	**Canopy opens**	
P-1604 … 1621	679.1 – 684.5 s	**Steady descent**	Linear fall at 2.27 ± 0.05 m/s, R^2 = 0.991
P-1622 … 1625	684.9 – 685.9 s	Arrival	Height 9.7 m above the pad baseline at the last packet; pressure then settles 15 Pa from the ground value measured 2 s later (≈ 1.3 m)

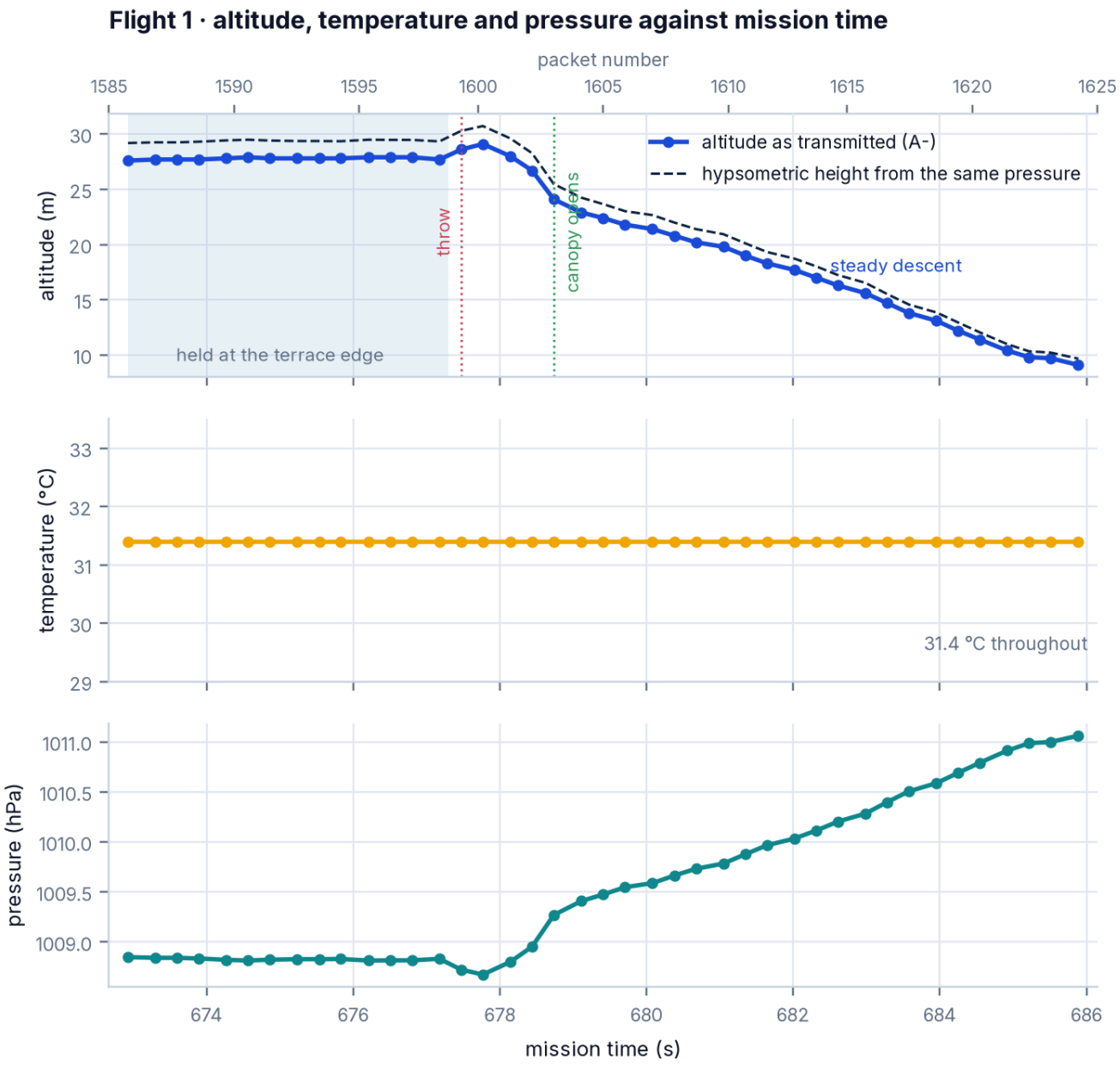


Figure 38 · **Flight 1 — the three mandatory graphs.** Altitude (as transmitted, and re-derived from the same pressure with the hypsometric equation), temperature and pressure against mission time, with packet number along the top.

Reading the three graphs together. Altitude and pressure are one measurement seen twice: the pressure rises by 239 Pa from the hold to the last packet, and the altitude falls by the corresponding 20 m. The temperature channel is a flat 31.4 °C: in a 13-second record the barometer die does not move by a tenth of a degree, which is the right behaviour — a temperature channel that wandered during a 30 m descent (a 0.2 K change in the air) would be reporting electronics, not weather.

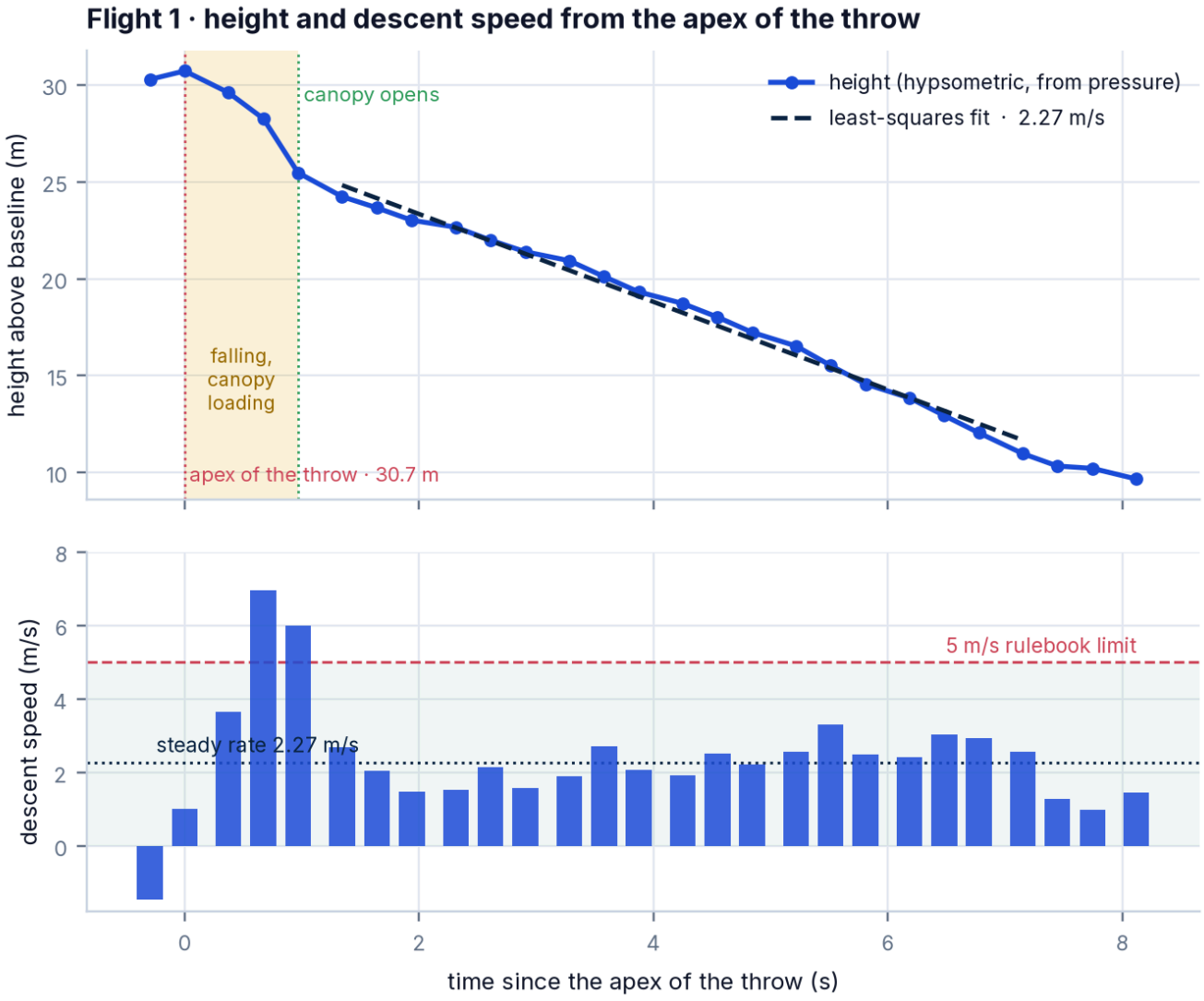


Figure 39 · **Flight 1 — the descent.** Top: height from the apex of the throw, the falling segment shaded, and the least-squares line through the steady descent. Bottom: descent speed from successive packets against the 5 m/s limit.

FLIGHT 1 DESCENT, IN NUMBERS

- **Steady descent rate 2.27 ± 0.05 m/s** (temperature-corrected); 2.16 m/s from the altitude as transmitted — the ISA formula's 5.7 % shortfall at 31 °C (Section 14.7).
- The deployment transient peaks at **7.0 m/s** during the 0.97 s between the apex of the throw and the canopy opening — free fall from rest for one second reaches 9.5 m/s without a canopy; here the canopy was already loading within a second — and the vehicle is on its steady rate **1.3 s after the apex.**
- The vehicle fell 21.1 m in the 8.1 s from the apex to its last packet in the log.

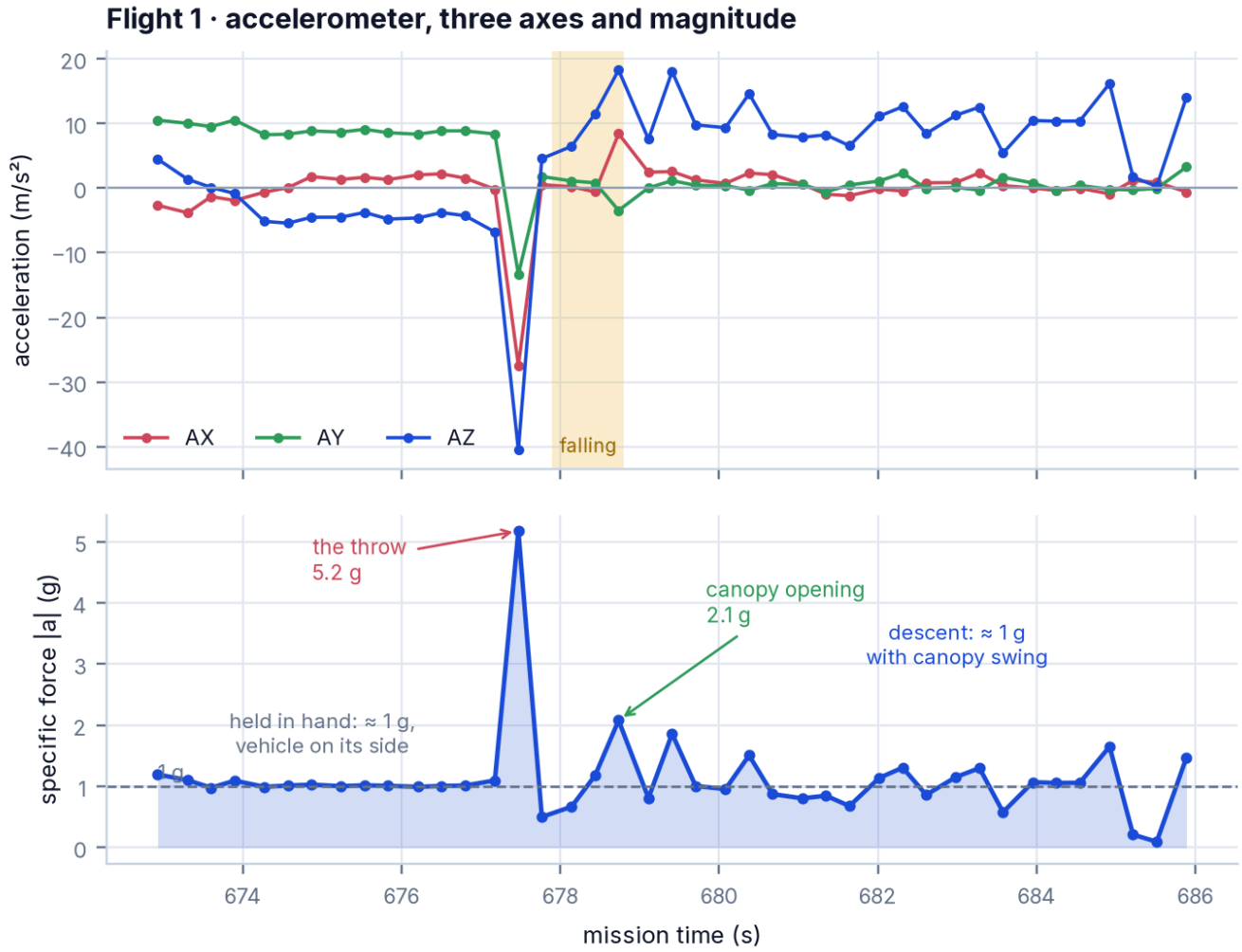


Figure 40 · **Flight 1 — acceleration.** The throw (5.2 g), the dip below 1 g in the air, the canopy-opening load (2.1 g) and the descent at about 1 g with canopy swing.

The accelerometer tells the whole story of the flight. **While held, |a| = 1.0 g with the gravity vector on the vehicle's Y axis — the vehicle is held on its side.** At the throw the three axes swing together to −27, −13 and −41 m/s², a 5.2 g impulse; then |a| collapses to 0.5 g (ballistic flight), recovers through 0.7 and 1.2 g as the canopy fills, and peaks at 2.1 g at the opening. From there the vehicle sits upright — AZ ≈ +9.8 m/s² on average (10.5 ± 3.2) — and swings about it.

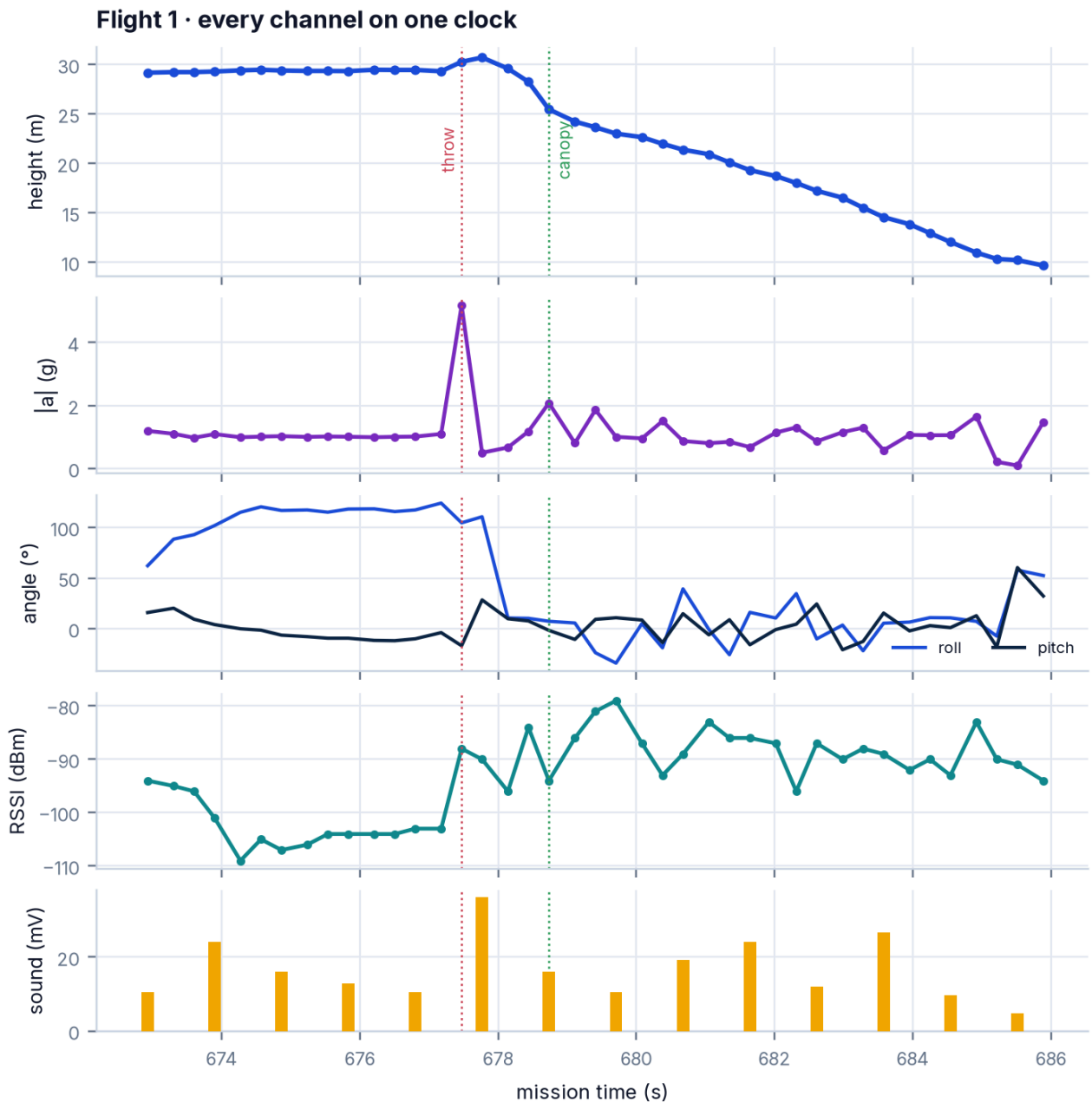


Figure 41 · **Flight 1 on one clock.** Height, specific force, roll and pitch, RSSI and acoustic level — the throw and the canopy opening marked. The same instant shows up in every channel: the acceleration spike, the roll swinging from 110° (on its side) to upright, the radio signal improving by 14 dB, and the loudest microphone packet.

The live view at the competition (Figure Figure 32 in Chapter 12) shows the part of the record that comes *before* the packets in the shared log — the vehicle on the pad, the climb, and the hold: the altitude trace climbs to about 27 m as the vehicle is carried up the building and then holds, pressure falls from about 101,200 Pa to 100,890 Pa, and temperature stays near 31 °C. The attitude panel at the same moment reads roll 131.7°, pitch −15.0° — the vehicle held on its side, as the accelerometer in the shared log shows.

14.4 Flight 2

Flight 2 was launched the same way, but the vehicle was **switched on at the terrace**, so its altitude zero is the terrace: the log begins at the vehicle's first packet (P-148, mission time 102.9 s, transmitted altitude +1.0 m — the top of the throw) and ends after touchdown at −26.9 m. The vehicle was in its 1.43 Hz pad-phase configuration (`ST-R003` throughout), so the log is a clean, uniformly spaced, 0.70 s-cadence record of a complete descent.

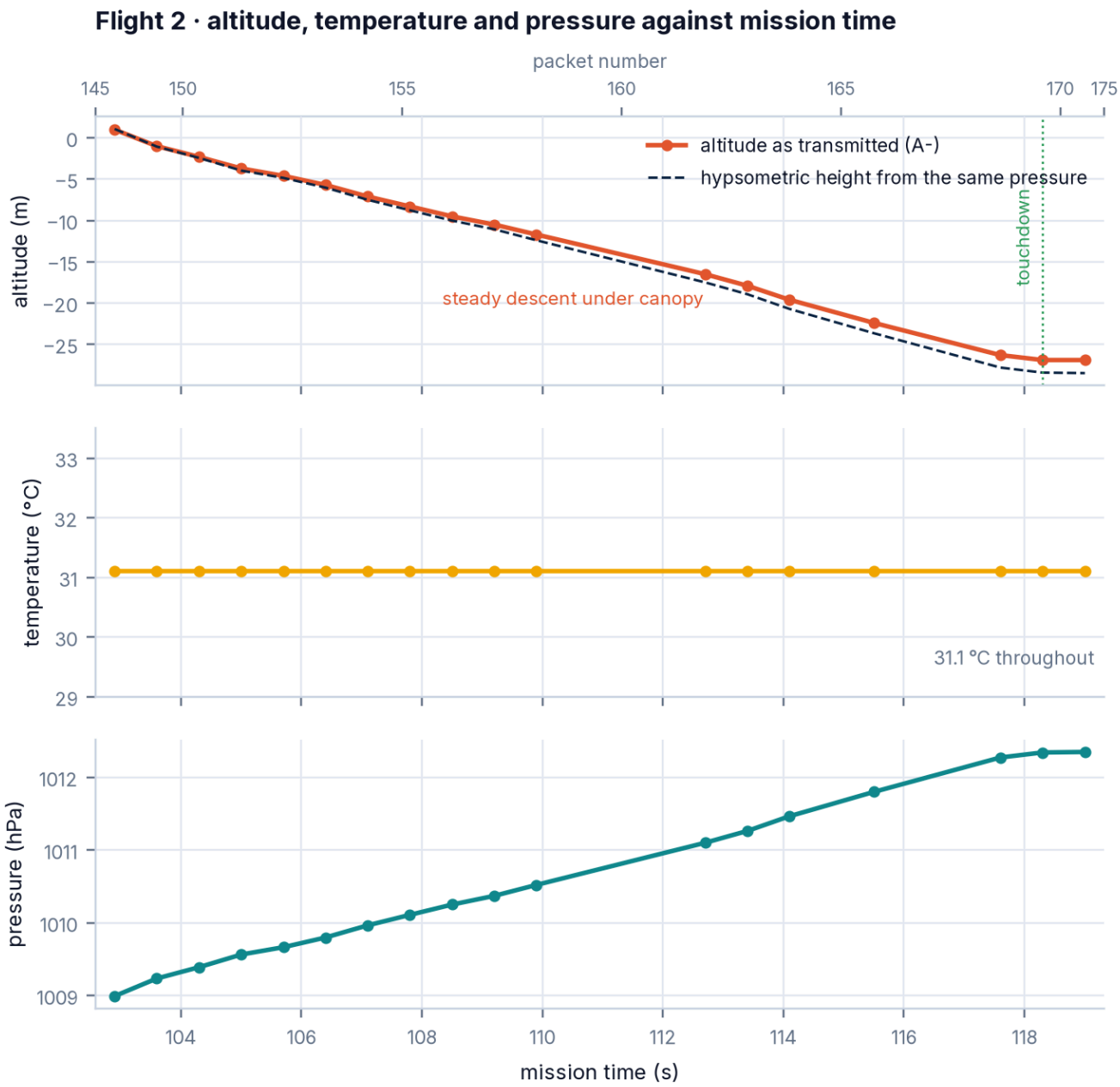


Figure 42 · **Flight 2 — the three mandatory graphs.** The transmitted altitude falls from +1.0 to −26.9 m relative to the vehicle's baseline; the pressure rises 335 Pa; temperature is a constant 31.1 °C.

Table 36 · **Flight 2 in numbers**

Quantity	Value
Drop height, from pressure	**29.6 m = 97.0 ft** (as transmitted: 27.9 m; pressure rise 335 Pa)
Descent time	**15.4 s** from the first packet to touchdown
Steady descent rate	**1.88 ± 0.02 m/s**, R^2 = 0.9988 (1.78 m/s as transmitted)
Mean rate over the whole drop	1.92 m/s; first half 1.79, second half 1.96 m/s — the descent does not accelerate
Peak load during the descent	**1.93 g** at P-150 (mission time 104.3 s)
Touchdown	P-170 at 118.3 s; the next packet reads **1.57 g** and pitch 50° as the vehicle settles
Swing angle (accelerometer vs vertical)	median 13°, 95th percentile 31°

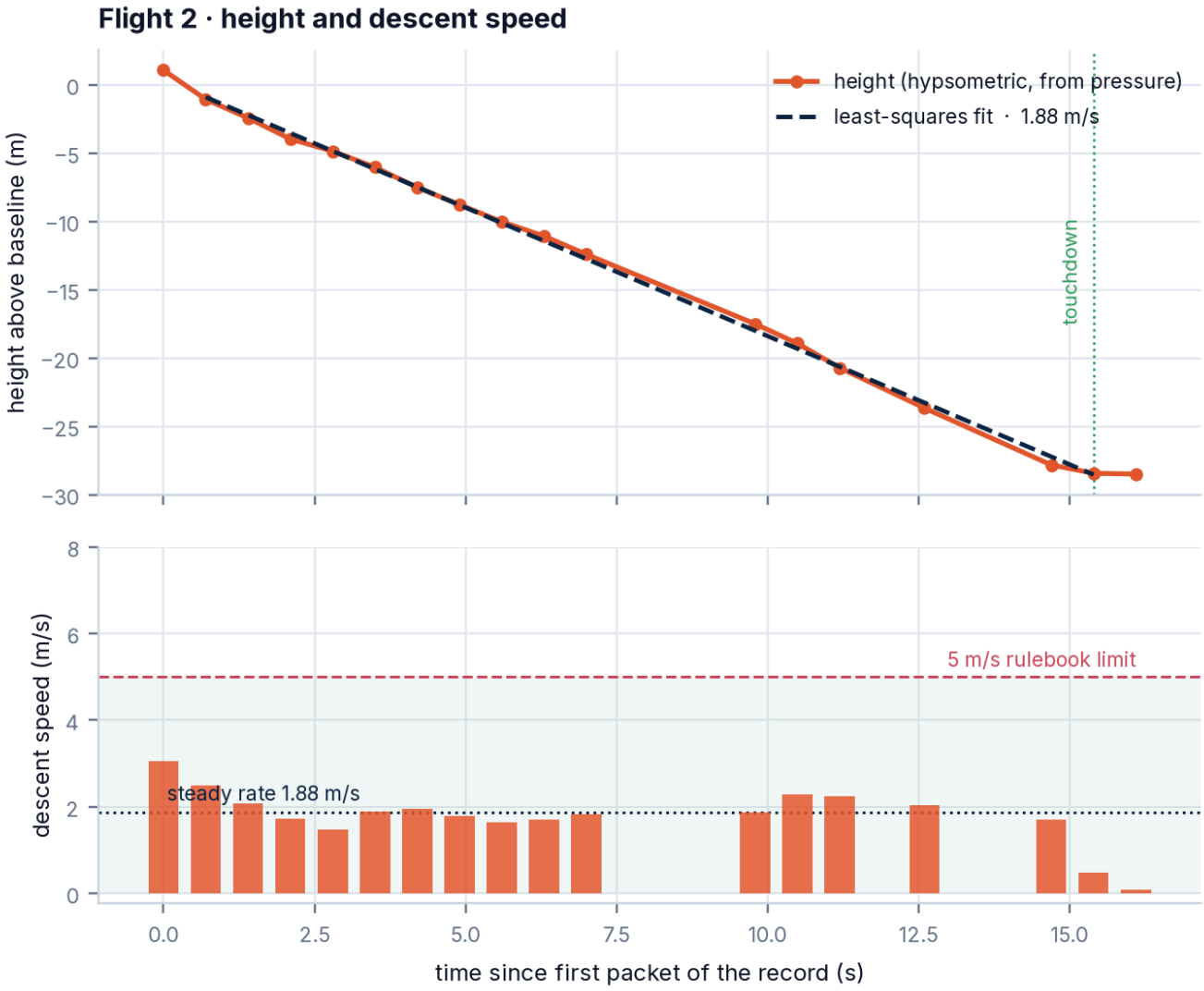


Figure 43 · **Flight 2 — the descent.** The least-squares line is straight to within the noise of the barometer (R^2 = 0.999): the vehicle fell at constant speed from its first sample to touchdown. Bars are speeds between successive received packets.

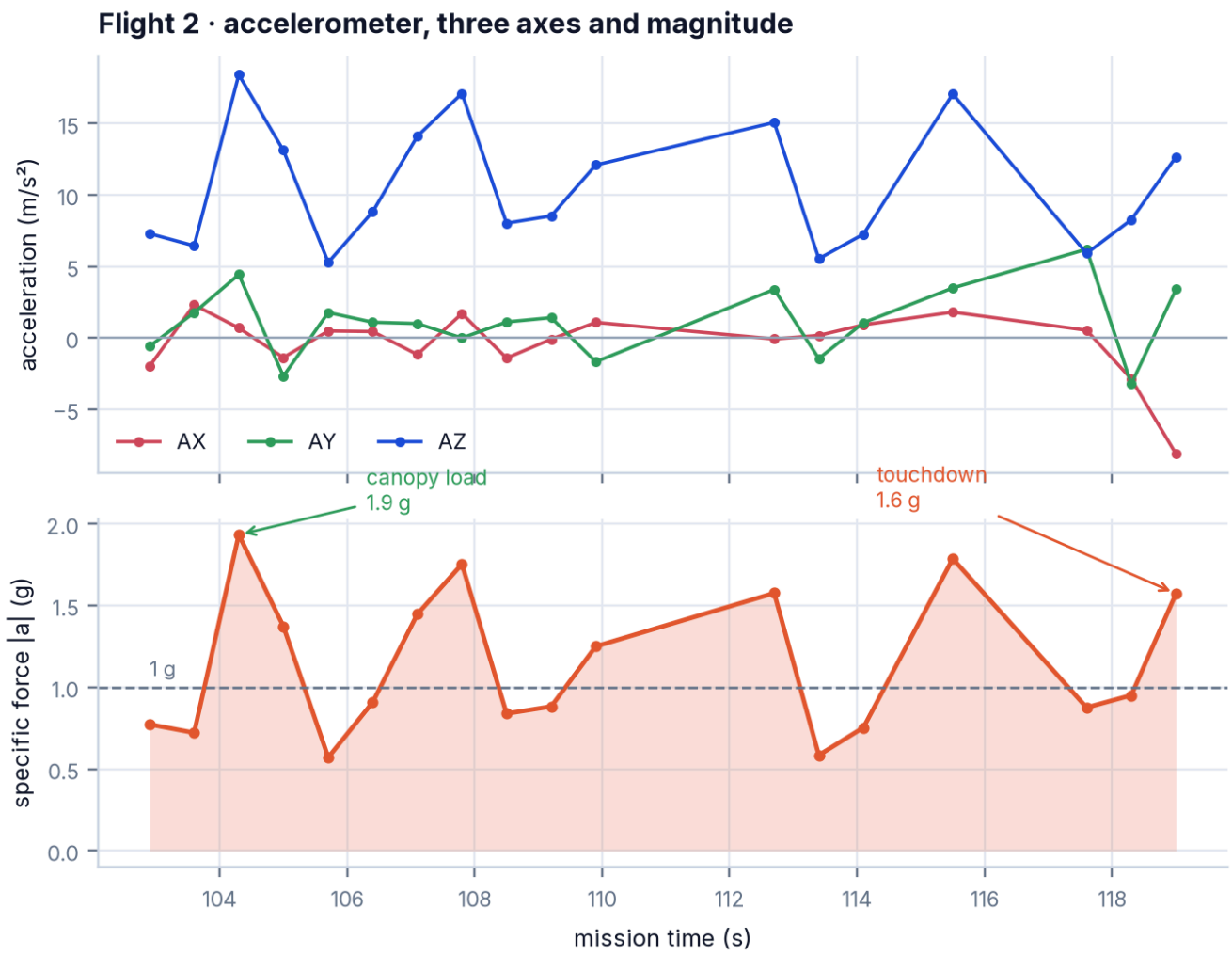


Figure 44 · **Flight 2 — acceleration.** Specific force stays within a band around 1 g, with the canopy load of 1.9 g at P-150 and a touchdown reading of 1.6 g.

14.5 The two flights compared

Table 37 · **Side by side**

	Flight 1	Flight 2
Launch height from pressure	29.4 m held → 30.7 m at the apex (**96 ft → 101 ft**)	**29.6 m (97 ft)**
Steady descent rate	**2.27 ± 0.05 m/s**	**1.88 ± 0.02 m/s**
Margin to the 5 m/s limit	2.2×	2.7×
Peak vertical speed	7.0 m/s (canopy filling)	3.1 m/s
Opening / peak load	2.1 g (the throw 5.2 g)	1.9 g
Swing angle, median / 95th percentile / max	8° / 18° / 18°	13° / 31° / 46°
Roll σ / pitch σ in descent	19° / 12°	26° / 20°
Air density in descent	1.155 kg/m³	1.157 kg/m³
Temperature	31.4 °C	31.1 °C
RSSI mean (min … max)	-93.3 dBm (−109 … −79)	-96.6 dBm (-106 … -87)

	Flight 1	Flight 2
Packets received	**41**	**18**

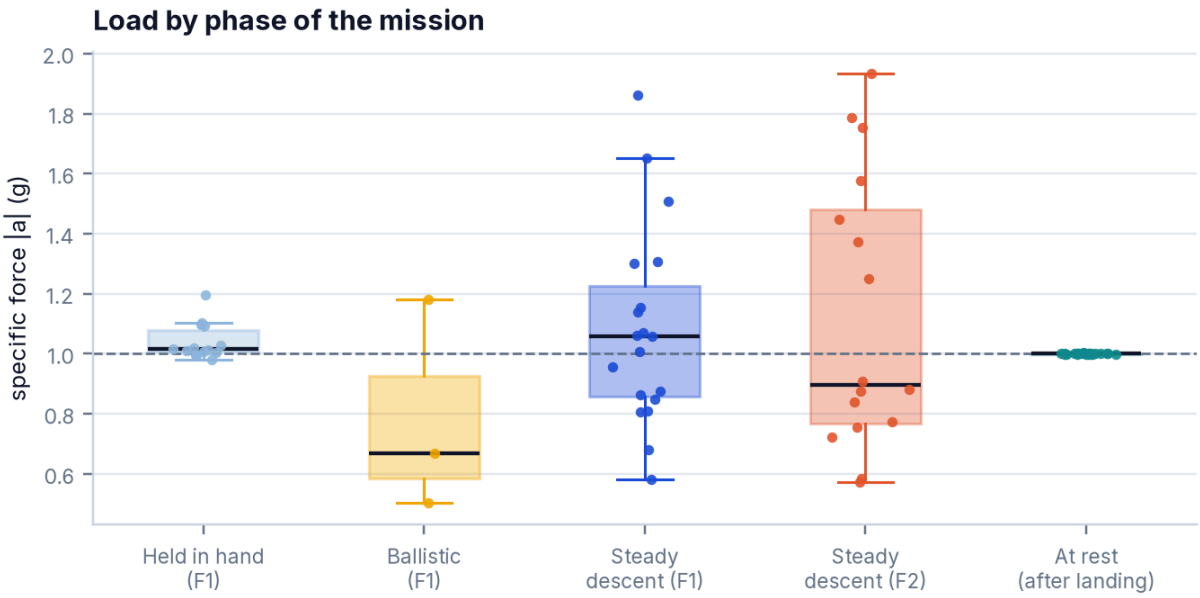


Figure 45 · The load on the vehicle by phase of the mission, in units of g: held in hand, ballistic, steady descent (both flights) and at rest after landing. The resting value (1.000 g, σ 0.0015 g) is the accelerometer's calibration made visible.

Both flights tell the same story from different starting conditions. **They agree on the facts that matter for the rulebook** — a drop of 29–31 m (≈ 97–101 ft), a steady descent at 40–45 % of the permitted rate, a swing that stays inside the 45° cone for 93–100 % of the packets, and an arrival at a speed that corresponds to a fall of 18–26 cm. They differ in the details a descent is expected to differ in: Flight 2 is slower and perfectly straight (1.88 m/s), Flight 1 somewhat faster (2.27 m/s) with a one-second deployment transient at the start of its record. The 0.3 °C difference in air temperature changes the air density by 0.1 % and is irrelevant to the rates.

14.6 Attitude and stability

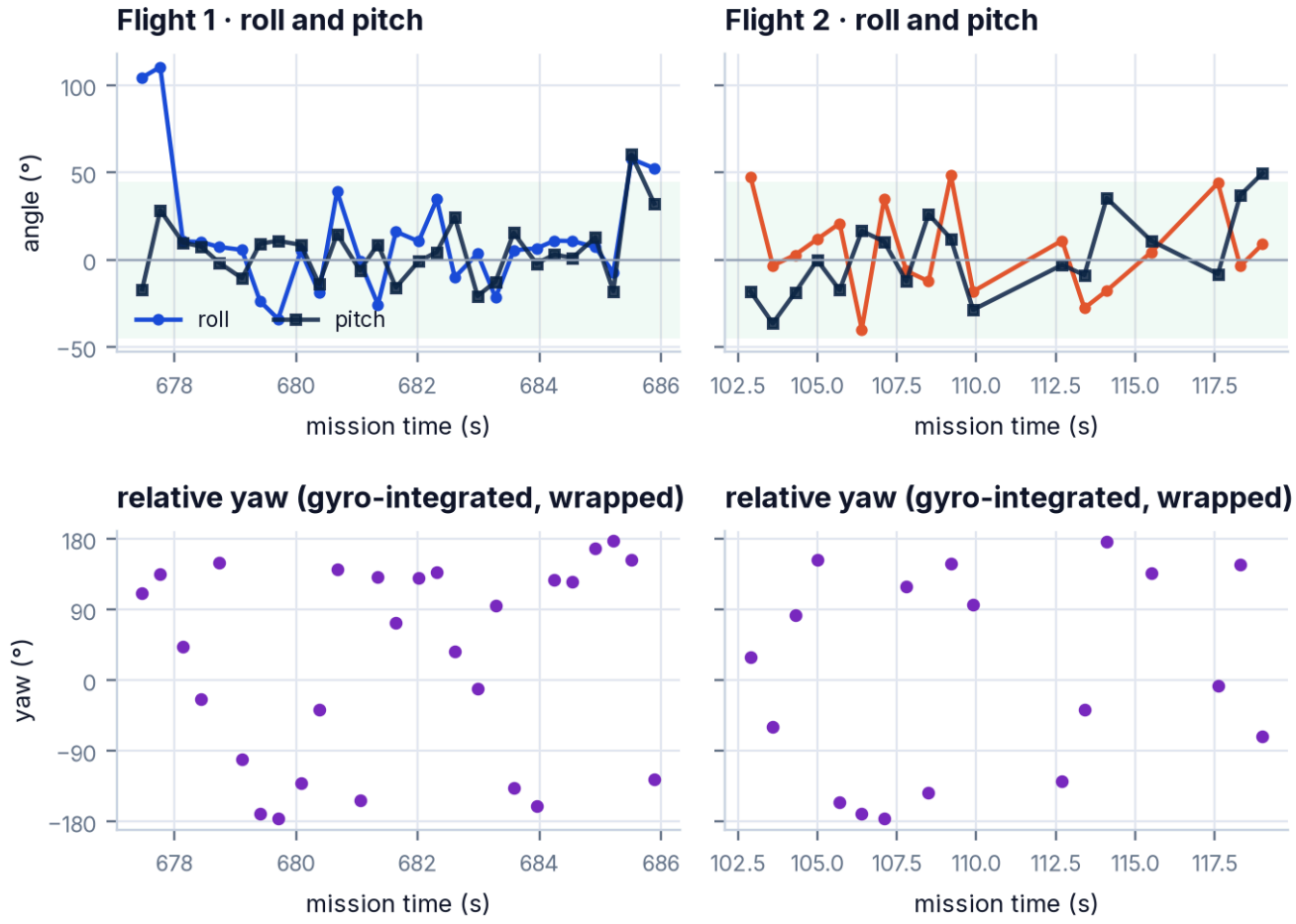


Figure 46 · Roll and pitch (top) and relative yaw (bottom) for both flights. Flight 1 starts at the throw, when the vehicle rotates from the orientation it was held in (roll ≈ 110°) to upright within a packet.

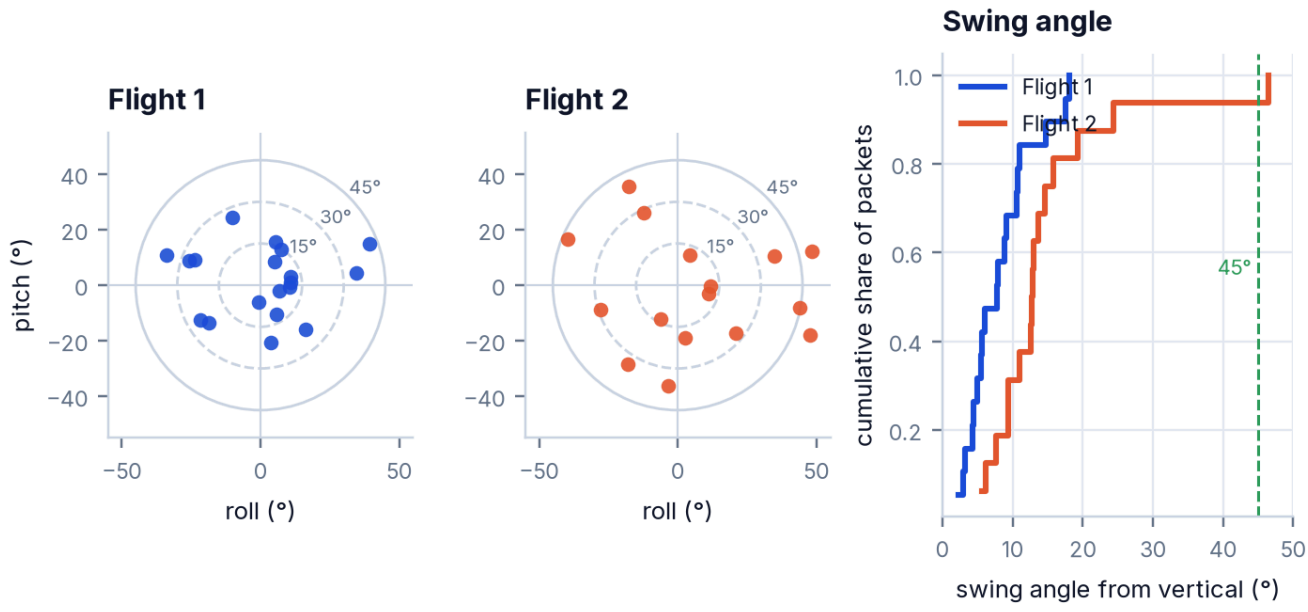
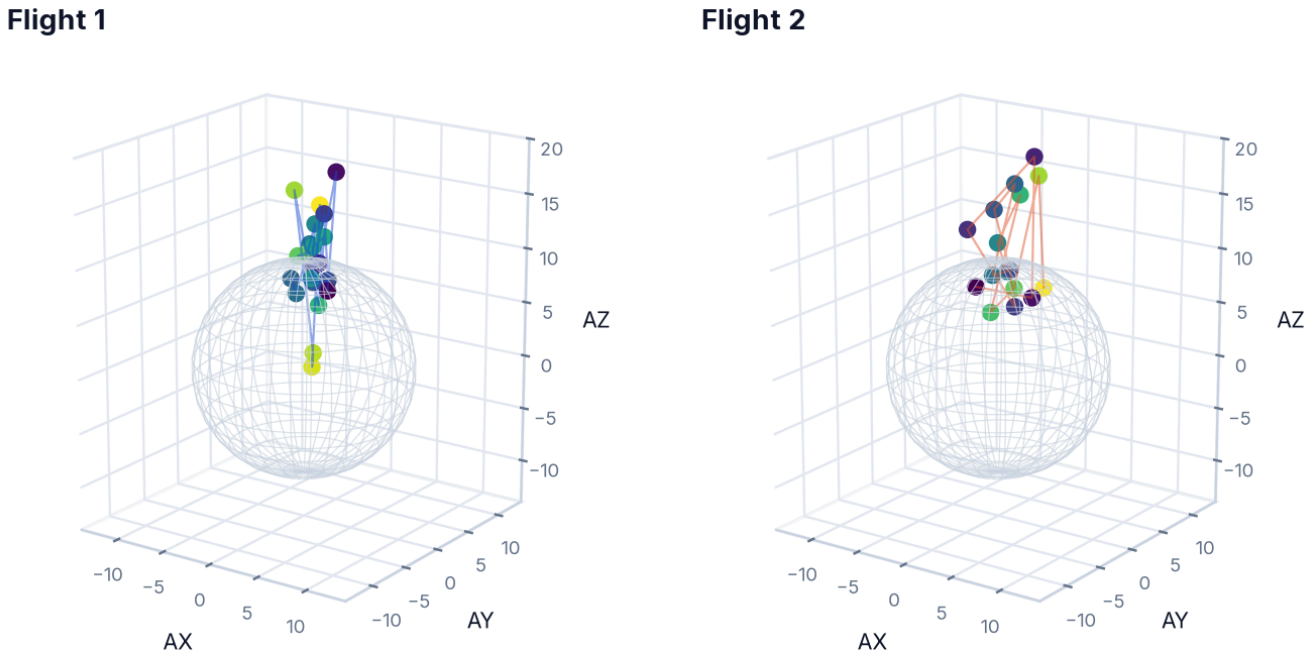


Figure 47 · The swing cone. Each dot is one packet of steady descent: roll against pitch, with 15°, 30° and 45° circles (left, centre); cumulative distribution of the swing angle (right). Flight 1 never leaves the 18° circle; Flight 2 stays inside 45° for 93 % of its packets.

What the attitude data show.

- **Upright and swinging, never tumbling.** After the canopy opens, the vehicle's z axis stays within 18° of vertical in Flight 1 (95th percentile 18°) and 31° (95th percentile) in Flight 2. Roll standard deviation is 19° and 26°, pitch 12° and 20° — a pendulum mode under the canopy, with no sustained drift of the mean attitude (mean roll +1.2° and +3.6°).
- **Roll and pitch swing together in Flight 1** ($r = +0.60$, $p = 0.003$): the canopy swings along one plane that is not aligned with the vehicle's axes — what a single pendulum mode looks like in two body angles.
- **Yaw is a relative angle.** Yaw is integrated from the gyro and is zeroed at calibration; its value is an angle around the vertical from the pad orientation, and its meaning is the *rate*. Because yaw is sampled once per packet (3.1 or 1.4 Hz) the wrapped values jump by more than a half-turn between packets: the transmitted angle is correct, but a spin rate cannot be recovered from samples this sparse.



one dot per packet in steady descent; the grey sphere is 1 g (9.81 m/s²); colour is time

Figure 48 · The acceleration vector of each steady-descent packet in three dimensions. The vectors cluster above the 1 g sphere (grey) along +AZ — the vehicle upright — and spread sideways with the swing.

14.7 The pressure–altitude law, and why the transmitted altitude is slightly small

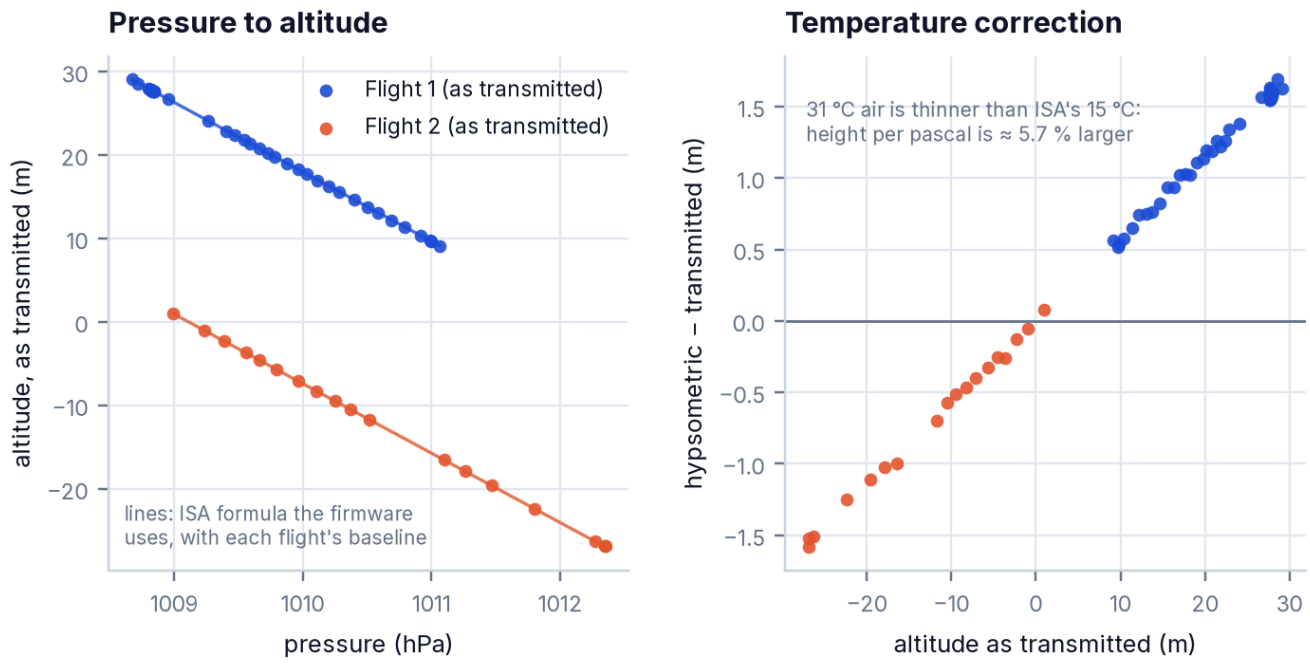


Figure 49 · Left: transmitted altitude against measured pressure for both flights, with the ISA formula (lines) using each flight's recovered baseline. Right: the difference between the temperature-corrected height and the transmitted altitude.

The vehicle converts pressure to altitude with the international standard atmosphere formula, which assumes a 15 °C air column. The analysis recovers the baseline each power cycle used (101,215.8 Pa for Flight 1, switched on at the ground floor; 100,911.5 Pa for Flight 2, switched on at the terrace — the 304 Pa between them is 25.8 m of height) and finds that **the transmitted altitude is reproduced by the ISA formula to ±0.03 m** — the firmware does exactly what it was written to do, to the resolution of the 0.1 m field. Real height per pascal scales with the real air temperature: on a 31 °C day the air is thinner than ISA's, so a given pressure change corresponds to **5.7 % more height** than the ISA formula reports. The right-hand panel shows it: the difference grows linearly with height from the baseline. All descent rates in this chapter use the temperature-corrected height — from the hypsometric equation, $\Delta h = (R\,T / g)\,\ln(p_0 / p)$, with the measured pressure and temperature — and quote the transmitted-altitude rate beside it (2.16 against 2.27 m/s; 1.78 against 1.88 m/s).

Table 38 · **Barometer facts the flights confirm**

Quantity	Value
Baseline pressure, Flight 1 (pad)	101215.8 Pa
Pressure at the terrace edge	100882.9 Pa, i.e. 333 Pa below the pad
Pressure at rest after landing	101121.6 Pa (1.15 Pa σ) — the landing surface is ≈ 8.3 m above the pad baseline
Barometer noise at rest	σ = **1.15 Pa** = 0.10 m
Altitude noise at rest	σ = **0.11 m** (range -0.2 … +0.2 m)
Temperature resolution	0.1 °C; flat to the resolution in both flights

14.8 Descent physics: drag, energy and what the vehicle felt on arrival

At steady descent weight equals drag: **$C_d \cdot S = 2\,m\,g / (\rho\,v^2)$.** The vehicle's mass is only known to lie in the 450–550 g band, so the drag area is given across it:

Table 39 · Drag area, canopy size, energy and loads implied by the measured descent rates

	Mass	$C_d \cdot S$	C_d of the 6 ft canopy (2.63 m²)	Kinetic energy at arrival	Equivalent fall height	Force, 25 ms stop
Flight 1 (2.27 m/s)	450 g	1.49 m²	0.57	1.16 J	26 cm	41 N
	500 g	1.65 m²	0.63	1.28 J	26 cm	45 N
	550 g	1.82 m²	0.69	1.41 J	26 cm	50 N
Flight 2 (1.88 m/s)	450 g	2.16 m²	0.82	0.79 J	18 cm	34 N
	500 g	2.40 m²	0.91	0.88 J	18 cm	38 N
	550 g	2.64 m²	1.00	0.97 J	18 cm	41 N

The vehicle arrives at the speed it would have after stepping off a 26 cm step (Flight 2: 18 cm): about 1 J of kinetic energy for a vehicle that the structural studies loaded with 100 N. The measured Flight 2 touchdown reading of 1.57 g is the accelerometer's own confirmation. Figure Figure 17 in Chapter 8 plots these loads against the studied one.

> **THE MODEL, CHECKED BY THE FLIGHTS**
>
> For the 6 ft canopy (flat area 2.63 m²) the flights imply a drag coefficient of **0.57 – 0.69 (Flight 1)** and **0.82 – 1.00 (Flight 2)** across the 450–550 g band. The descent model of Chapter 9 assumed 0.75 for a vented flat canopy and predicted 1.97 – 2.18 m/s; the flights measured 2.27 and 1.88 m/s — **+9 % and −9 % on the 500 g prediction.** The two flights bracket the model, on either side, by the same ±10 % that the uncertainty in the flown mass allows for. A drag coefficient is the last number a parachute design usually has to guess; here it has been measured, twice, and the guess was in the middle.

14.9 Correlations between the channels

Roughly forty variable pairs were examined for each flight. The heat maps show Pearson r between the channels in the descent proper (Flight 1 from the canopy opening, Flight 2 to touchdown); the pair plot shows the same data as scatter plots with distributions on the diagonal.

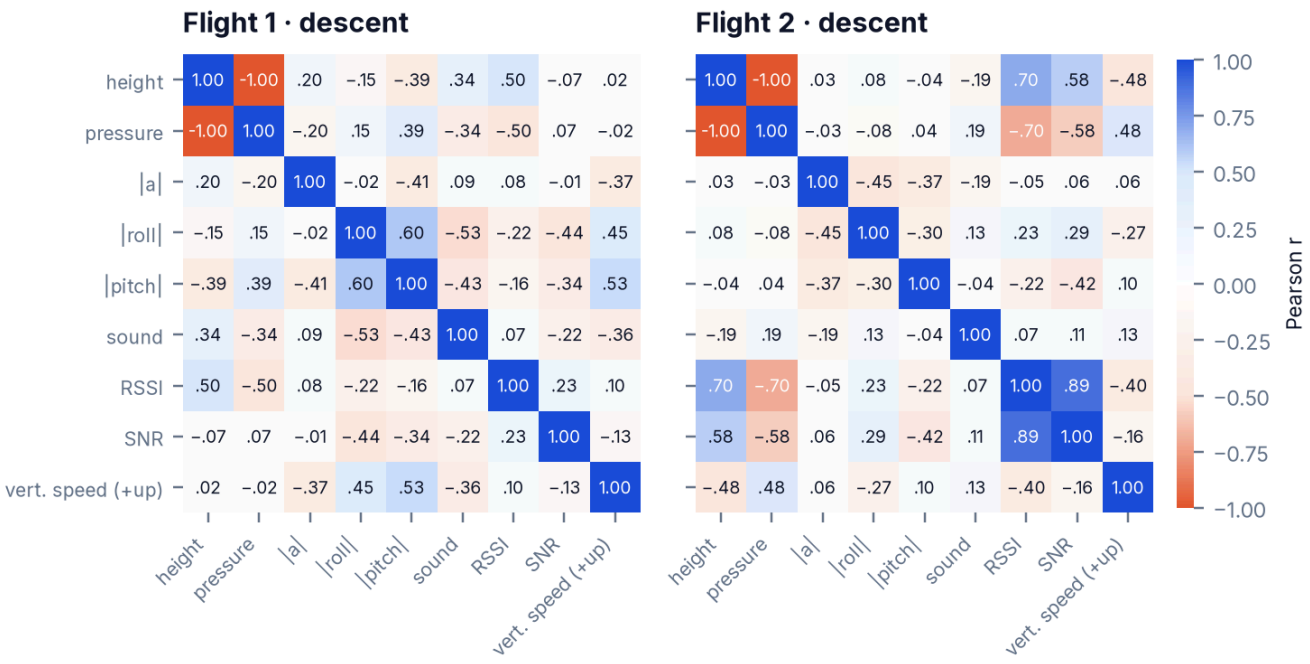


Figure 50 · Pearson correlation matrices for the descent of each flight. Height and pressure are one measurement seen twice (r = −1.00).

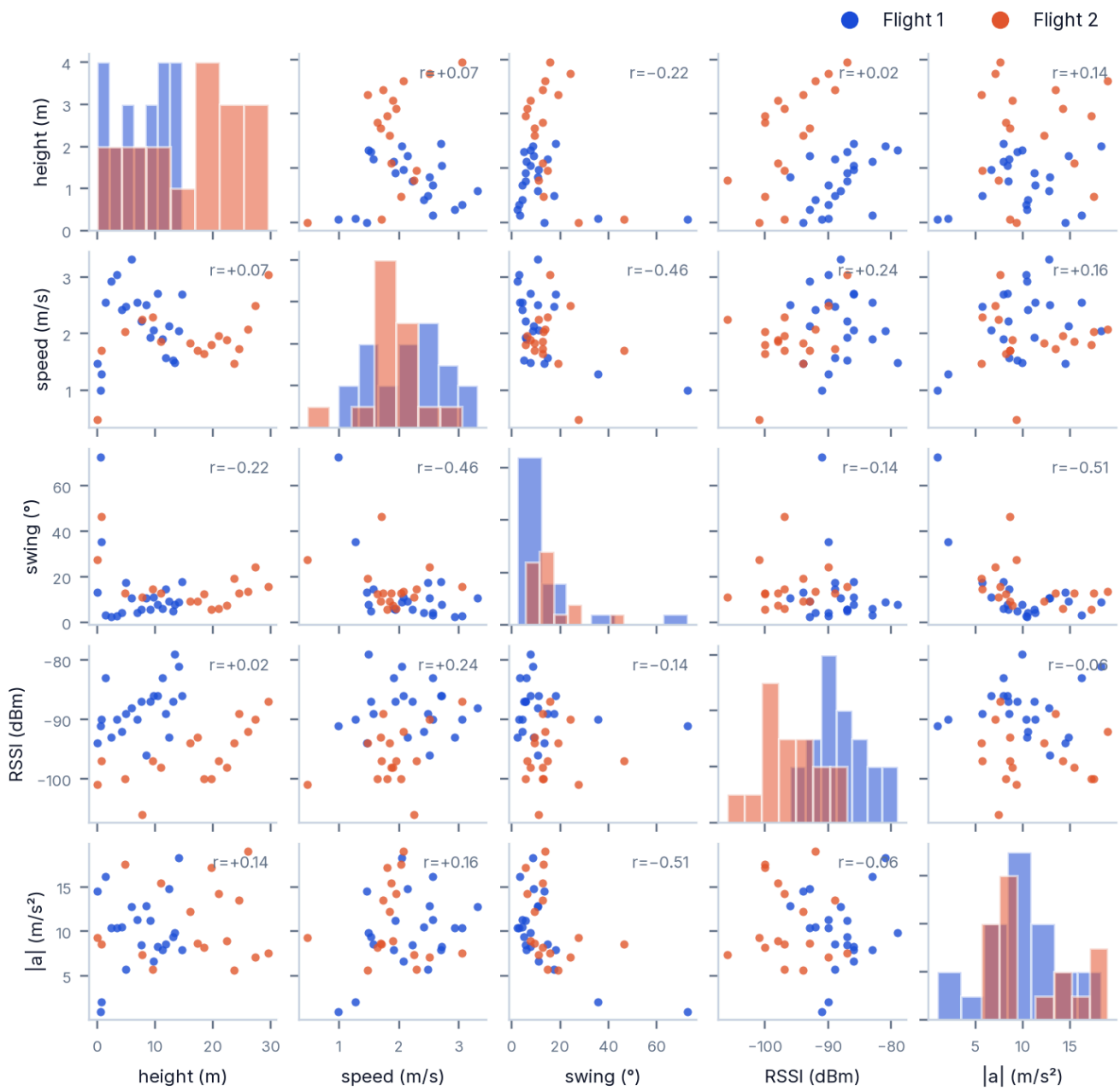


Figure 51 · Pair plot of height above the last sample, descent speed, swing angle, RSSI and specific force, both flights. The diagonal shows each variable's distribution.

Table 40 · The correlations that matter, with significance

Pair	Flight	Pearson r	p	n	Reading
Height ↔ pressure	both	−1.00	< 0.001	22 · 17	The barometer is a perfect altimeter: a pure monotonic law
Height ↔ RSSI	**F2**	**+0.70**	0.002	17	The signal weakens as the vehicle nears the ground: ≈ 3.9 dB per 10 m
Height ↔ RSSI	F1	+0.50	0.019	22	The same trend, shorter range
RSSI ↔ SNR	F2	**+0.89**	< 0.001	17	Signal and signal-to-noise move together while noise is constant
RSSI ↔ SNR	all 102 packets	**+0.77**	< 0.001	102	...until SNR saturates near +10 dB (Figure Figure 52)

Pair	Flight	Pearson r	p	n	Reading
Roll ↔ pitch	F1	+0.60	0.003	22	One swing plane, not two independent oscillations
Height ↔ SNR	F2	+0.58	0.015	17	Follows RSSI
Vertical speed ↔ specific force	both	−0.37 · +0.06	0.090 · 0.820	22 · 17	No relation: speed is steady, so loads come from the swing, not the fall
Vertical speed ↔ sound	both	−0.36 · +0.13	0.432 · 0.624	7 · 18	No relation at constant descent speed (Section 14.12)
Specific force ↔ roll	F2	−0.45	0.067	17	Weak: large roll angles occur at the lighter-loaded points of the swing

Three findings stand out.

1. **Pressure is height.** The −1.00 is not a statistical result but a physical one — and it is why the rulebook's mandatory altitude, pressure and temperature graphs carry the same information, and why the pressure channel is the redundancy for a faulty altitude.
2. **The radio degrades smoothly as the vehicle descends toward the ground, by about 4 dB per 10 m.** Close to the ground the direct path and the ground-reflected path interfere, and the link budget is spent on the ground bounce. The slope is gentle: even the lowest-flying packets keep 17 dB of margin.
3. **Quantities that should not correlate do not.** At steady descent speed there is no relationship between speed and load or speed and sound level: the vehicle is not accelerating, so it does not feel it. A correlation analysis that finds *nothing* where physics predicts nothing is an assurance that the instruments are measuring the vehicle and not each other.

14.10 The radio link in flight

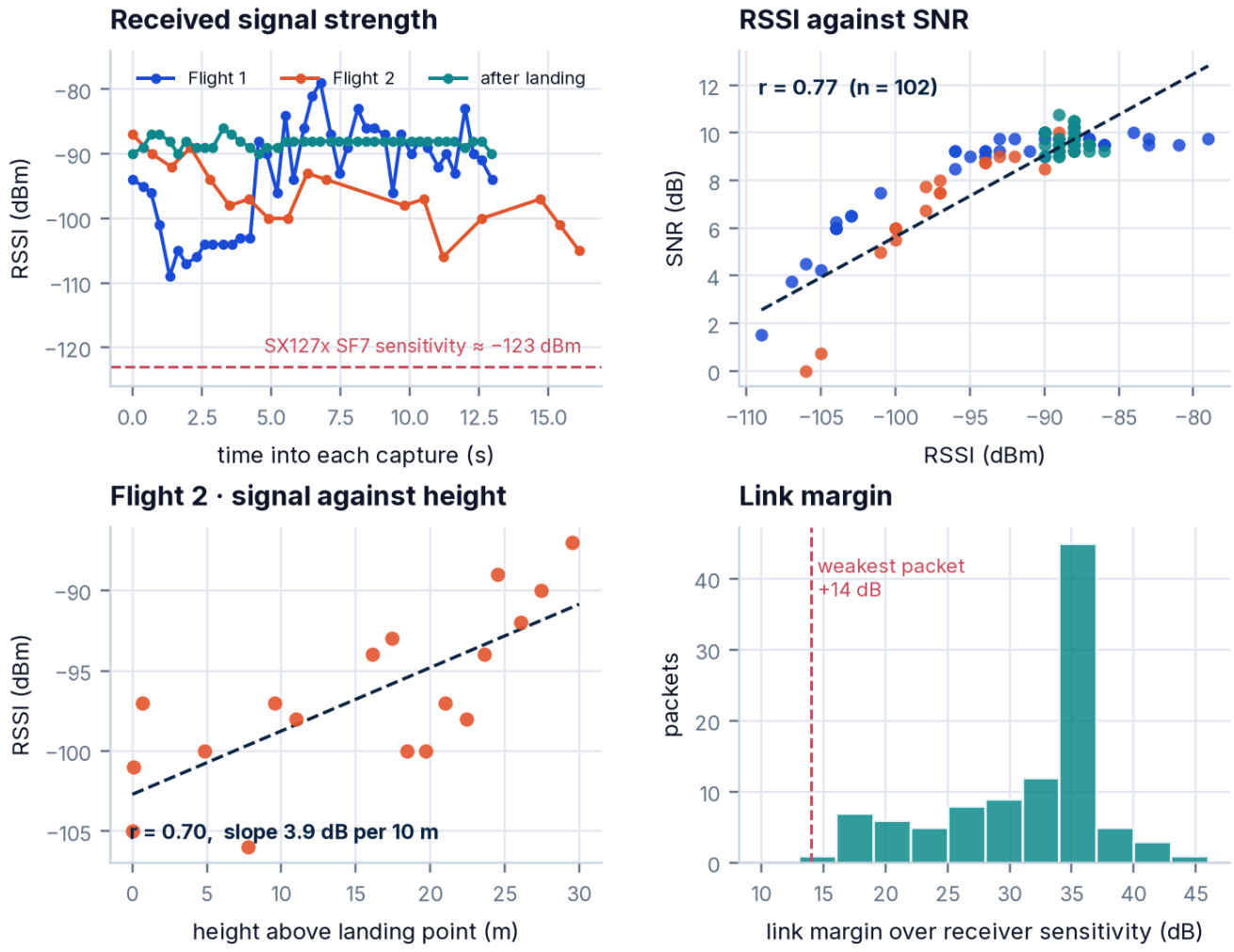


Figure 52 · Top left: RSSI against time into each capture, with the receiver's SF7 sensitivity. Top right: SNR against RSSI. Bottom left: Flight 2 RSSI against height above the landing point. Bottom right: link margin over receiver sensitivity for every packet.

Table 41 · Link statistics, 102 distinct packets

	RSSI (dBm)	SNR (dB)	Margin over −123 dBm
Mean	-92.1	8.3	**31 dB**
Weakest packet	-109	-6.75	**14 dB**
Strongest packet	-79	10.75	44 dB

- **The throw changes the link.** In Flight 1, the mean RSSI while the vehicle was held at the terrace edge was **-102.5 dBm** (SNR 6.1 dB); after the throw and the canopy opening it was **-87.6 dBm** (SNR 9.6 dB) — **14 dB better** — and the step happens in a single packet at P-1599. While it is held in the hand against the thrower's body the antenna is consistent with being shadowed; once thrown it has free air.
- **The weakest packets were still 14 dB above the floor.** Several were decoded while the vehicle was held at the terrace edge (−109 dBm, SNR +1.5 dB). The link has margin for the situation that matters most, the flight itself.
- **After landing the link is steady:** RSSI -88.3 ± 0.8 dBm, SNR 9.7 ± 0.4 dB over 41 packets — the steadiness of a vehicle that is not moving.

14.11 The sound sensor in flight

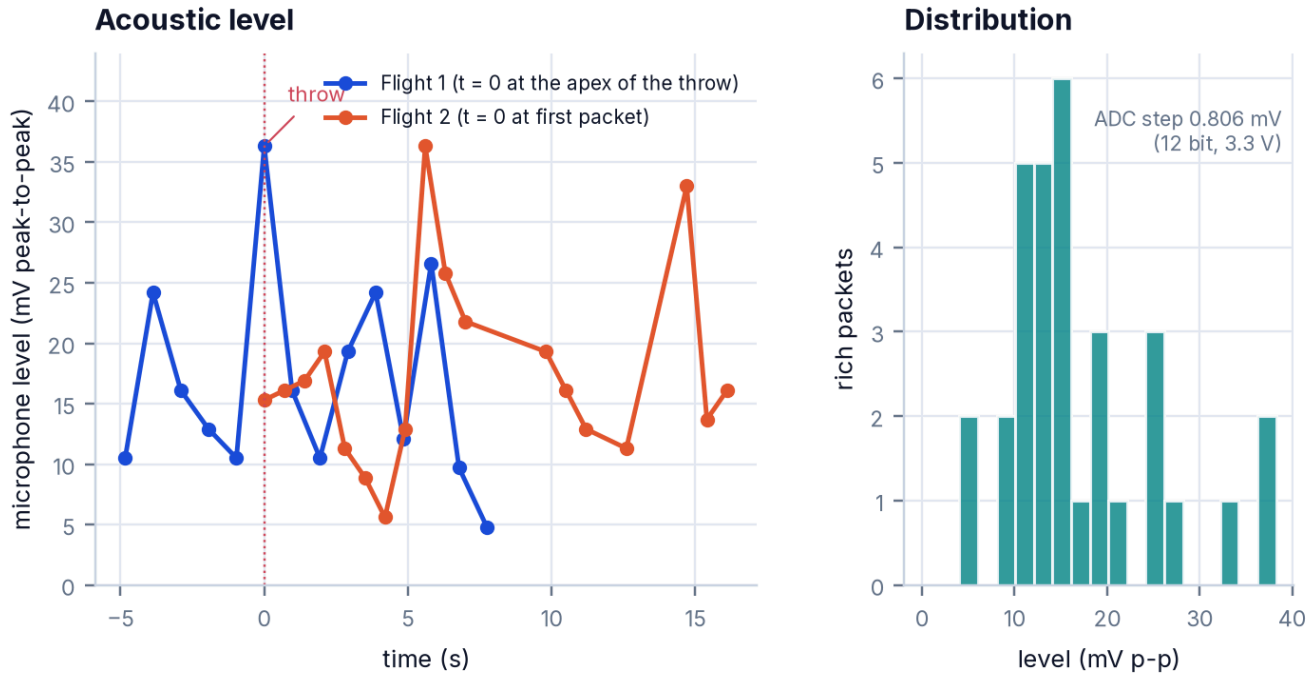


Figure 53 · Acoustic level against time (left) and its distribution (right). The loudest packet of Flight 1 is the throw.

The microphone recorded 14 readings in Flight 1 and 18 in Flight 2 (one per rich packet — GPS, sound and status ride together). Its levels lie between 5 and 36 mV peak-to-peak, quantised in 0.806 mV steps (a 12-bit ADC on 3.3 V). **The loudest reading of Flight 1 (36.3 mV, P-1600) is the packet immediately after the throw** — the impulsive event the microphone was chosen to catch — and the loudest of Flight 2 (36.3 mV, P-156) occurs 5.6 s into the drop. The mean level is the same while held (14.8 mV) and the descent (17.1 mV) of Flight 1, and the level does not correlate with descent speed or with specific force ($|r| < 0.4$, $p > 0.4$): at the steady speed of a canopy descent there is little change in airflow noise to measure.

14.12 GPS in flight

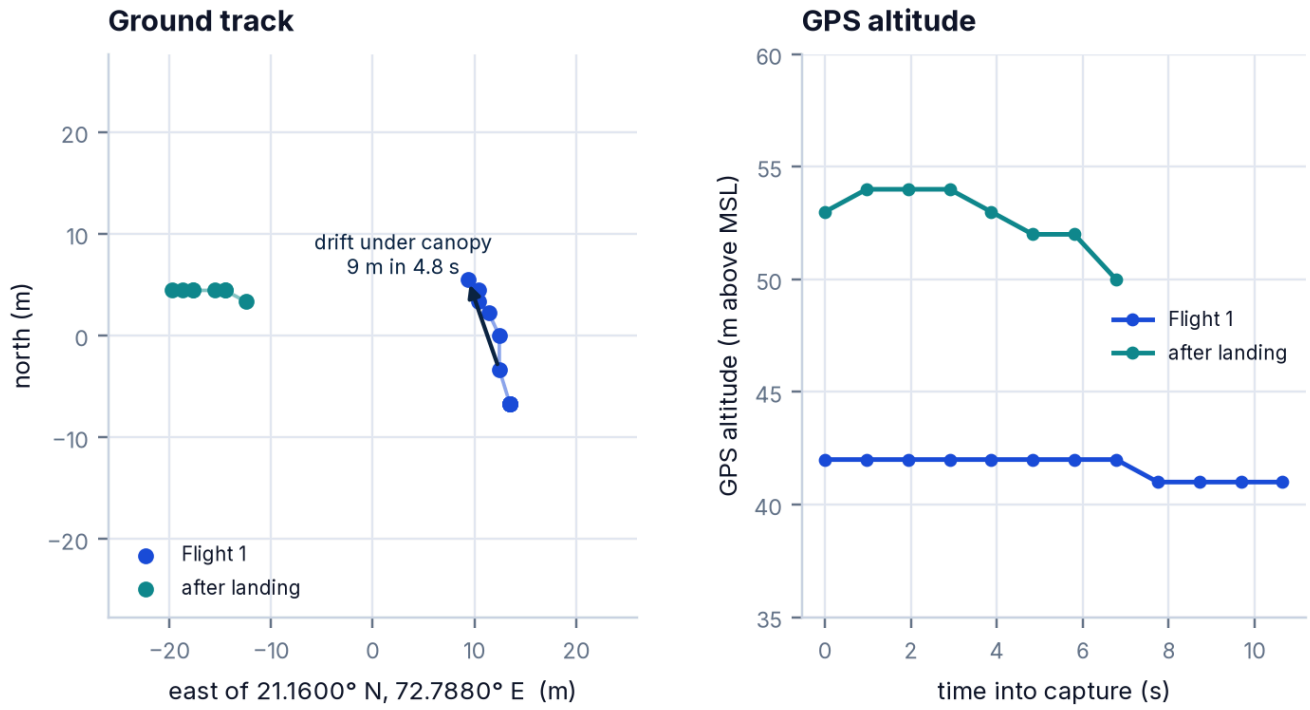


Figure 54 · Left: GPS fixes in the rich packets, in metres from a reference point, for Flight 1 and the after-landing session. Right: GPS altitude.

GPS fixes ride in every rich packet. In Flight 1, 12 fixes cover the hold and the descent: **the position is constant to the fifth decimal while the vehicle is held at the terrace edge, then moves 9.4 m toward 341° (north-north-west) in 4.8 s — a drift of 1.9 m/s** under the canopy — the horizontal speed given by the throw and the wind, carrying the vehicle across the ground as it descends. At rest after landing, 8 fixes scatter with an RMS of **2.3 m**, in line with the receiver's rated ~2.5 m horizontal accuracy. GPS altitude is reported by the receiver as 41–42 m during the hold and descent and 50–54 m at rest after landing: GPS vertical accuracy is several times worse than its horizontal accuracy, which is exactly why altitude in the packet comes from the barometer (resolution 0.1 m) and the GPS altitude is carried as an independent, coarse, optional field.

14.13 The post-landing session: a free calibration experiment

The 13 seconds of Flight 1's aftermath are a clean, stationary record — the vehicle at rest on the landing surface — and a direct view of what the vehicle's power-up sequence does.

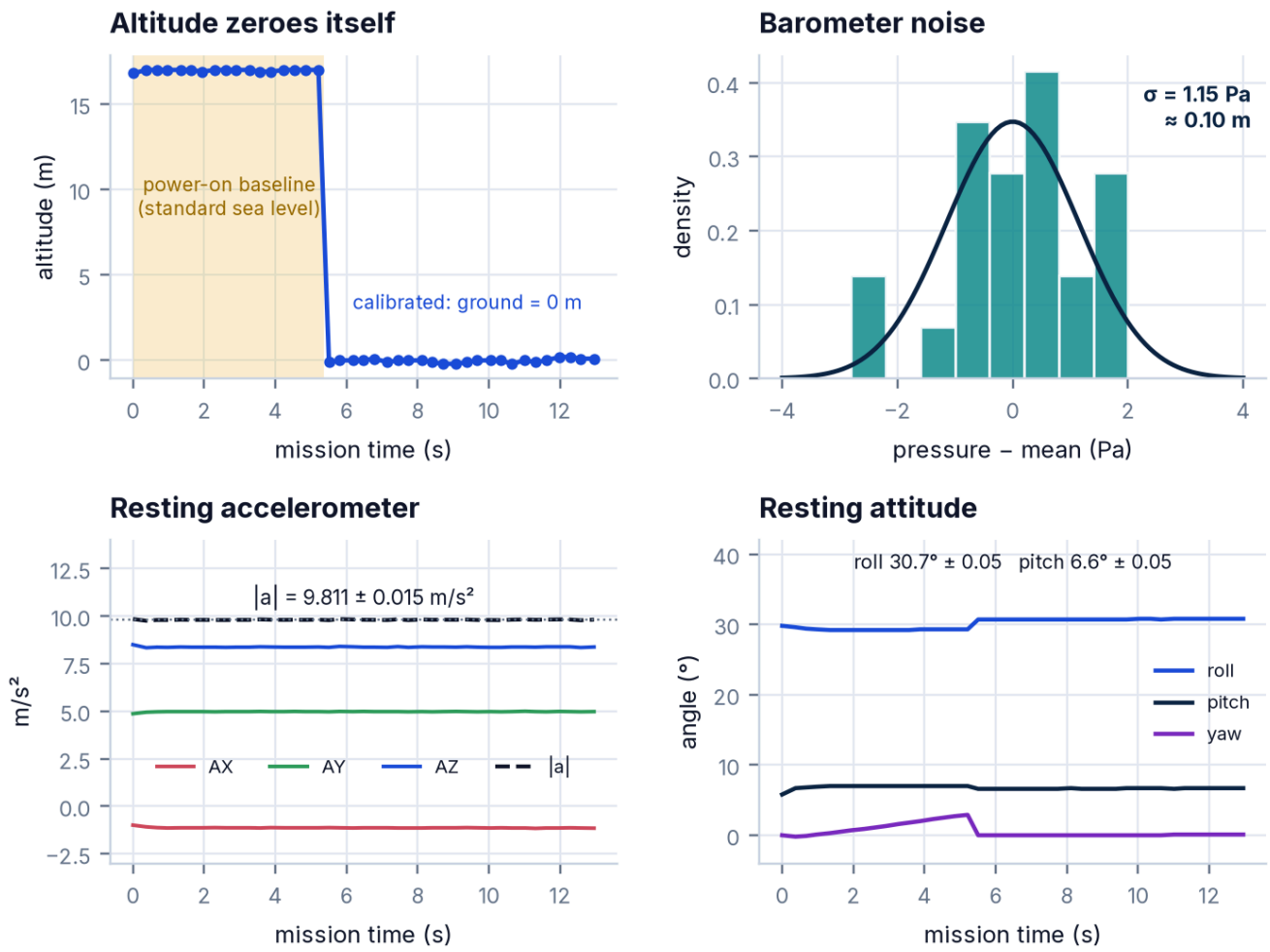


Figure 55 · The after-landing session. Top left: altitude zeroes itself at 5.5 s. Top right: barometer noise at rest. Bottom left: resting accelerometer. Bottom right: resting attitude.

Table 42 · The resting vehicle

Quantity	Value
Telemetry resumed	At mission time 0, **P-001**, already at the max-rate pattern
Calibration completed	**5.5 s** after power-up: status `ST-R004` → `ST-R113`
Altitude before / after calibration	17.0 m (standard-sea-level baseline) → **-0.02 ± 0.11 m**
Specific force at rest	**9.811 ± 0.015 m/s²** against the standard 9.807 — within 0.04 %
Axis noise at rest	AX σ 0.009, AY σ 0.008, AZ σ 0.015 m/s²
Resting attitude	roll 30.7° ± 0.05, pitch 6.6° ± 0.05 — the vehicle lying tilted ≈ 31° from upright
Yaw	drifting at 0.63 °/s before calibration; frozen at 0.0 – 0.1° after
Barometer noise	σ 1.15 Pa (0.10 m)
Telemetry heard after landing	**12.95 s**, 41 packets

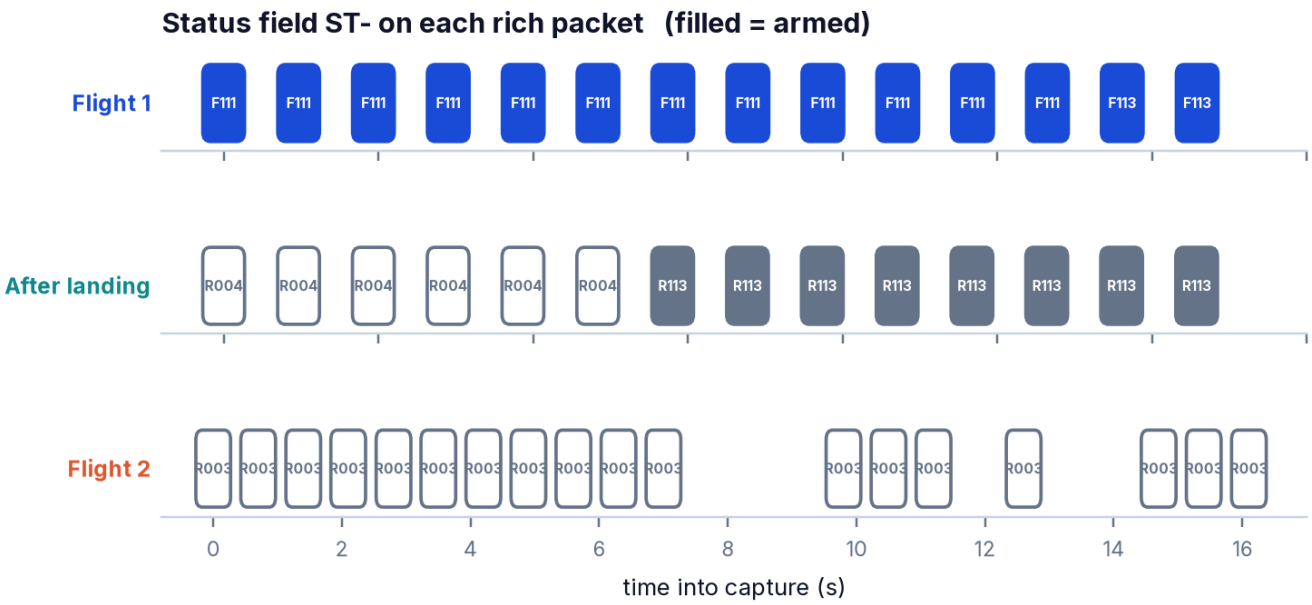


Figure 56 · The status field of every rich packet. Filled = armed. The after-landing session starts unarmed (`R004`) and arms itself at 5.5 s (`R113`).

This session shows five things at once. The vehicle came back on the air by itself within a few seconds of the end of Flight 1's record; it began at `P-001` and at the max-rate pattern — the behaviour designed for a restart in the field; it calibrated on the spot and armed itself in 5.5 s; its barometer and accelerometer resolve at the 0.1 m and 0.015 m/s² level; and its GPS kept a fix at the landing surface. And the telemetry continued for 13 s — 2.6 times the five seconds the rulebook asks for after impact.

14.14 Predictions against flights

Table 43 · **What was predicted, and what the flights measured**

Quantity	Prediction	Flight result
Steady descent rate	≤ 5 m/s; model for the 6 ft canopy: **1.97 – 2.18 m/s** over the mass band	**2.27 and 1.88 m/s** (+9 % / −9 % on the 500 g prediction)
Launch height	100 ft = 30.48 m	**29.4 m held, 30.7 m at the apex of the throw** (Flight 1); **29.6 m** (Flight 2)
Telemetry rate, max rate	3.11 Hz (designed)	**3.09 Hz** measured on the vehicle's clock
Telemetry rate, window	1.43 Hz (700 ms)	**1.43 Hz** (0.700 s gaps in Flight 2 and the pad capture)
Packet size ceiling	≤ 200 B	**≤ 188 B**
Airtime guard	50 ms	cadence reproduced to the millisecond
Link margin	"tens of dB at 30 m" (free-space)	**≥ 14 dB, 31 dB mean**
Altitude at rest after calibration	≈ 0 m	**−0.2 … +0.2 m**
Post-impact telemetry	≥ 5 s	**12.95 s**
Touchdown load	≤ 100 N study	**≈ 34 – 50 N** at a 25 ms stop
Structural margin	SF ≥ 7.6 (derated)	Touchdown loads of 1.6 g

Quantity	Prediction	Flight result
Altitude formula	ISA, ±0.1 m	Reproduced to ±0.03 m; 5.7 % small at 31 °C, as predicted by the hypsometric equation

14.15 Summary of the flight analysis

2.27 m/s
Flight 1 steady descent (R^2 0.991)

1.88 m/s
Flight 2 steady descent (R^2 0.9988)

15.4 s
time of flight, Flight 2 — 29.6 m

5.2 g
the throw, the largest load of either flight

18 °
Flight 1 maximum swing from vertical in descent

14 dB
weakest link margin in 102 packets

41
packets received during Flight 1

13.0 s
telemetry heard after landing

- **The vehicle did what it was designed to do, in the order it was designed to do it**: carried up the building, held at the edge, thrown, opened its canopy within about a second, settled to a steady descent, landed, kept transmitting, restarted itself if needed, calibrated and armed again.
- **The descent was slow, straight and stable** — constant speed to within the noise of a barometer, 40–45 % of the limit, swinging inside a 45° cone.
- **Every channel agrees with every other**: pressure with altitude (r = −1.00), the instant of the throw across acceleration, attitude, radio and sound within one packet, and the instruments' resting values with physics (|a| = 1.000 g to 0.04 %).
- **The telemetry pipeline held**: byte-exact packets, 180 of 180 rows parsed, the designed 0.374/0.296/0.297 s cadence reproduced exactly, 102 distinct packets with no malformed one.

Reproducing every number in this chapter

```
python analysis/flight-2026-09-30/launch_analysis.py Team-25.xlsx --out analysis/flight-2026-09-30
python analysis/flight-2026-09-30/make_chapter.py
```

The first command parses and cleans the log, computes `results.json`, `flight_data_clean.csv` and every figure; the second regenerates this chapter's numbers from `results.json`.

15

CHAPTER 15

Engineering discoveries

The defects and surprises found by running things — each one found, understood and closed — and what each one changed in the design.

15.1 What running the vehicle taught that reading it could not

15.2 The defects a host suite cannot reach

15.3 How each discovery changed the design

15.1 What running the vehicle taught that reading it could not

Every item below came from *running* something rather than from reading it. Each is closed, and most changed a rule that now protects the rest of the design.

Table 44 · Discoveries and their resolution

#	What was expected	What was found	Resolution
D-1	A radio that either works or does not	The same transmit failed 5/5 and then succeeded 45/45 minutes later — same code, same antenna	**Closed** with D-2: a long 3V3 jumper on the RA-02
D-2	A card that reads is a card that writes	With a long 3V3 jumper the card initialised, read its filesystem, then went silent on the first write — five runs running. A short jumper fixed it outright: **100 / 100**	**Closed.** The write-current spike arrives down whatever wire the supply has; reads never draw enough to expose it. Soldered board: short track + 470 µF bulk capacitor
D-3	The card to work under the flight image as under the bring-up image	It failed every time under the flight image while passing twelve times under bring-up	**Root cause found, and it is not the card.** The radio's chip select was configured only by the radio driver, which initialises second, so GP17 floated low (selected) through the whole SD sequence. Fixed by driving both chip selects high with the bus
D-4	Landing detection to need a landing	A landing declared under a **hovering drone, 12 s before release** — the same condition as a vehicle held still at a terrace edge	**Closed by the descent gate** (Section 3.3)
D-5	The vehicle's altitude to be its height above the pad	**5.7 % small on a hot day** — the ISA formula assumes 15 °C	**Corrected in the analysis**; confirmed in flight (Section 14.7)
D-6	A reported GPS fix to be a holdable fix	739 fixes from a stationary receiver: median 8.1 m from the centroid, maximum 50.9 m, largest one-second jump 55.6 m	**Gated.** The parser refuses a fix below 4 satellites or above HDOP 5.0, and parses HDOP, which it previously discarded
D-7	A telemetry rate that the radio can deliver	The provisional SF9 profile needs 1,250 ms of airtime per packet — longer than the 500 ms period it was configured for	**Closed.** The modem moved to SF7; rate re-derived from measured airtime; three layers of guards (Section 11.4)
D-8	The organizers' receiver to hear what the team's bridge hears	It discards anything over 200 bytes, and listens on one sync word only	**Closed.** Packet budget cut to 200 B (GPS printed to its real resolution); both Picos fly `0xA5` always
D-9	Code that builds and passes to be correct	A second review pass found **nine defects** in code that already built and passed	**Closed.** Each got a test
D-10	Mass estimates to be roughly right	Avionics estimate 31 g low; structure estimate 64 g high	**Closed by weighing.** Every mass in the budget is a measurement

15.2 The defects a host suite cannot reach

Five thousand host assertions did not find D-3, D-2 or the supply-sensitivity of the radio: all three needed a powered board. **The tests catch what the logic gets wrong; the bench catches what the wiring gets wrong; the flight catches what the world does.** The three layers are cumulative, and the project's habit was to move a finding

from the outer layer inward — every hardware discovery above became a rule in the code (chip selects together with the bus), a test, and a documented design decision.

15.3 How each discovery changed the design

- **D-3 → rule:** *both chip selects are driven high with the bus.* The order in which drivers start can no longer decide whether the bus is safe.
- **D-4 → rule:** *a vehicle cannot land without descending first.* A physical gate replaced a tuned threshold.
- **D-5 → procedure:** *rates are computed from temperature-corrected height*, and both numbers are always quoted.
- **D-7, D-8 → process:** *airtime is measured, budgets are held to the receiver's real limits, and the same modem definition is shared by both ends.*
- **D-9 → habit:** *a second pass over passing code is part of the work*, not a luxury.

16

CHAPTER 16

Design decision register

Every major decision in one place: what was chosen, what it was chosen over, why, and the evidence.

16.1 How to read the register

16.2 System and software

16.3 Radio and protocol

16.4 Hardware, structure and recovery

16.5 Decisions about how the work was done

16.1 How to read the register

A decision is recorded when it fixes something that would be expensive to change later, and each carries its reason and its evidence. The register is organised by subsystem.

16.2 System and software

Table 45 · **System and software decisions**

#	Decision	Alternatives	Why	Evidence
1	One non-blocking 2 ms loop	RTOS tasks; interrupt-driven	Bounded worst case, trivially analysable, no preemption hazards, and a single scheduler that is fully host-testable	Scheduler tests; no blocking call exists in the loop
2	Flight core with no hardware dependency, six abstract interfaces	Direct SDK calls throughout	Mission logic testable on a laptop against mocks, including failure cases that cannot be produced safely on hardware	5,485 C++ assertions; compile-checked Pico layer
3	Autonomous launch/landing detection; no manual trigger	Arming button; ground command	A human cannot be inside a loop that closes in seconds, whether it is a drone release or a throw; no operator-sized failure mode	Flight 1 reported `FLIGHT` for the carry and the descent
4	Telemetry never stops, in any state	Silence on `FAULT`	Telemetry is what every other judgement depends on	State-machine and fault tests
5	Packet suppressed when a mandatory field is invalid; number not consumed	Send last-good value; send with a flag	Wrong data is worse than no data; sequential numbers make gaps mean radio loss	Controller tests
6	30 Hz acquisition, 3 Hz telemetry	Acquire at the radio rate	The radio is limited by airtime, the sensors are not: decoupling avoids aliasing the vertical-rate estimate	Sensor-rate analysis; barometer 83 Hz measured
7	Vertical rate = EWMA 0.7/0.3, updated only on a fresh conversion	Raw difference; Kalman filter	Smooths 0.1 m barometer noise without lag that would delay landing detection; no state beyond one float	State-machine and sensor-timing tests
8	Launch: armed AND (boost OR 15 m climb), 300 ms	Altitude only; accelerometer only	A drone lift or a carry up a building has no boost; a throw has a boost but no 15 m climb of its own; either must work	Launch-guard tests; Flight 1 (climb met in the carry, 5.2 g at the throw)
9	Landing: descent gate, then rest 3 s	At-rest alone	A hover is at rest too (D-4)	Hover reproduction test; Flight 1
10	5 s post-impact window, enforced at build time	Fixed by convention	A value below the rulebook's 5 s will not compile	`validate_config()`
11	Pad calibration: 80 samples, $\sigma < 2$ °/s, $\lvert a \rvert$ within 1.5 m/s² of 1 g; best-effort after 20 s	Fixed offsets; blocking calibration	Bias is measured where the vehicle sits; a failed window never blocks the mission	Post-landing session: calibrated in 5.5 s
12	Mahony quaternion filter	Euler integration; complementary on angles	A tumbling vehicle passes through the ±90° singularity	20 s tumble test at 100 °/s

#	Decision	Alternatives	Why	Evidence
13	Plausibility gate drops the previous value too	Skip the bad sample only	The vehicle must not coast on data from a sensor that is actively wrong	Gate tests
14	Faults classified (warning / degraded / critical) in a fixed array	Exceptions; dynamic log	No allocation, constant cost, and only total sensing loss is critical	20 fault codes tested
15	2 s hardware watchdog; restart skips the command window	Software watchdog only	A hung loop reboots in hardware, and a mid-flight reset must arm at once	Post-landing restart in the flight log
16	Raw 512-byte block log, alternating headers	FAT file	No filesystem to corrupt; torn header write recoverable	Torn-header test
17	Drivers written in-house from datasheets	Arduino / vendor libraries	Fully understood, testable, no hidden blocking	BMP280 datasheet vector reproduced
18	Five-minute command window, closes by itself	No uplink; permanent uplink	Pad-time control without a flight-time attack surface	Command-window tests

16.3 Radio and protocol

Table 46 · Radio and protocol decisions

#	Decision	Alternatives	Why	Evidence
19	SX1278 LoRa, 433 MHz	nRF24L01 (2.4 GHz)	Same silicon as the organizers' station; better penetration and margin	Organizers' station recorded both flights
20	SF7 / 125 kHz / CR 4/5	SF9 and higher	Only SF7 meets 1 Hz with duty margin (SF9 = 1,250 ms)	Airtime 333.7 ms measured; 1.43 / 3.09 Hz
21	+17 dBm	+20 dBm	Rail stress and current (87 vs 120 mA) for a link that does not need it	Rail 3.28 V under TX; ≥ 14 dB margin
22	Sync word 0xA5 always	0xF3 for testing	One word for test and launch: what the team hears is what the judges hear	10 Sep range test
23	Text packet in rulebook format	Binary	Human-readable; the rulebook defines it	180/180 rows parse
24	Rich + lean packet shapes, 200 B ceiling	Always rich	The organizers' receiver discards > 200 B	Packets ≤ 188 B
25	GPS printed at 5 decimals, whole metres	6 decimals	Matches receiver resolution; saves 4 B	Packet size test
26	700 ms window cadence; 374 + 296 + 296 ms max-rate pattern	1000 ms; uniform 322 ms	Window needs gaps to listen; pattern puts GPS and sound on the air each cycle	Cadence reproduced exactly in flight
27	Slots sized from measured airtime + 50 ms guard	Modelled airtime	The model reads 1.8 % low	Four sessions to 0.1 ms
28	CRC-16 on the USB link	None	Separates transport corruption from malformed packets	Known-answer vector
29	One shared link profile and one shared fixture file	Separate copies	Silent total failure if they disagree	Cross-language tests

16.4 Hardware, structure and recovery

Table 47 · **Hardware, structure and recovery decisions**

#	Decision	Alternatives	Why	Evidence
30	RP2040 Pico (×2)	ESP32, AVR, STM32	RAM, watchdog, ADC, SDK, cost; no stray 2.4 GHz radio	Flight
31	MPU-6500 six-axis IMU	Nine-axis part	Rulebook reference is the six-axis class; roll and pitch are gravity-referenced and absolute	Rest $\|a\|$ = 1.000 g
32	±16 g, ±2000 °/s, 20 Hz filter	±8 g; wider filter	3× headroom over measured peak 5.2 g; no aliasing at 30 Hz	Flight loads
33	BMP280 on I²C	Analogue pressure sensor	The rulebook's named sensor; one chip for P, T, altitude	σ 1.15 Pa = 0.10 m
34	Altitude relative to a ground baseline	Absolute ISA altitude	Rulebook asks 0 m on the pad whatever the weather	Rest ±0.2 m; F1/F2 baselines
35	NEO-6M GPS with quality gate	Raw fixes	A reported fix is not a holdable fix (D-6)	Rest scatter 2.3 m RMS
36	Microphone (LM393 module) as the additional sensor	Humidity, air quality, UV	Flight-proven descent instrument; cheap; impulsive events	The throw is the loudest packet
37	microSD raw log	Flash on the Pico	Removable, large, survives a vehicle fault	100/100 writes; 300/s
38	No external regulator	AMS1117	Dropout at full cell; Pico rail measured 3.28 V	Rail measurements
39	Power LED on the 3.3 V rail	On a GPIO; on the battery	Lights immediately; goes dark if the system is dead	Design review
40	Local capacitors: 470 µF + 100 nF (card), 10 µF + 100 nF (radio)	None; one shared	The card's write spike and the radio's PA key-up each get their own reservoir	Card 100/100 after the fix
41	Universal board, short soldered runs, no sockets	Custom PCB; breadboard	Frozen pin map, short supply paths, same-day changes, no loose sockets	Soldered board passes gates 3–7
42	Sided-box frame 118.5 × 115 × 110 mm	Cylinder	Board fits flat; organizers confirmed a 12 cm sided box	CAD read from STEP
43	Open frame with arched faces	Closed shell	Load path, ambient pressure for the barometer, access	Safety factor ≥ 7.6 derated
44	White PETG, printed on its base	PLA, ABS	Tough, prints without an enclosure, deforms before it breaks; orientation fixed in advance	Structural studies
45	Model minimum 80 cm for 550 g at 35 °C (vented); **6 ft canopy flown**	Fly the minimum	The floor guarantees the cap only at the edge of the model; 5.2× its area adds margin against Cd and mass uncertainty and a harder hand-thrown start, and gives a ~1 J arrival	Model 1.9–2.2 m/s; flights 2.27 and 1.88 m/s
46	Canopy external / semi-exposed	Packed inside	Deploys as soon as the vehicle is launched	Opened about 1 s after the apex of the throw
47	Egg chamber inside the +7 cm allowance	Separate pod	Meets the envelope; part of one printed model	128.7 g including the chamber

16.5 Decisions about how the work was done

#	Decision	Why
48	Every constant names its source	A number with no source cannot be reviewed
49	Predictions written before measurement	Keeps the model honest
50	Documentation fails the build when it disagrees with the code	Stale documentation is believed; a failing build is noticed
51	Photograph and identify every part on arrival; read registers to settle identity	A listing is not a datasheet; a register read is not an opinion
52	Post-flight analysis written and tested against a synthetic flight before the launch	The four-hour analysis window is for interpreting data, not for writing code

17

CHAPTER 17

Project timeline

Twelve days from the first line of code to a built vehicle — and the flights at the end of it.

17.1 The schedule

17.2 The flight-day analysis was ready before the flights

17.1 The schedule

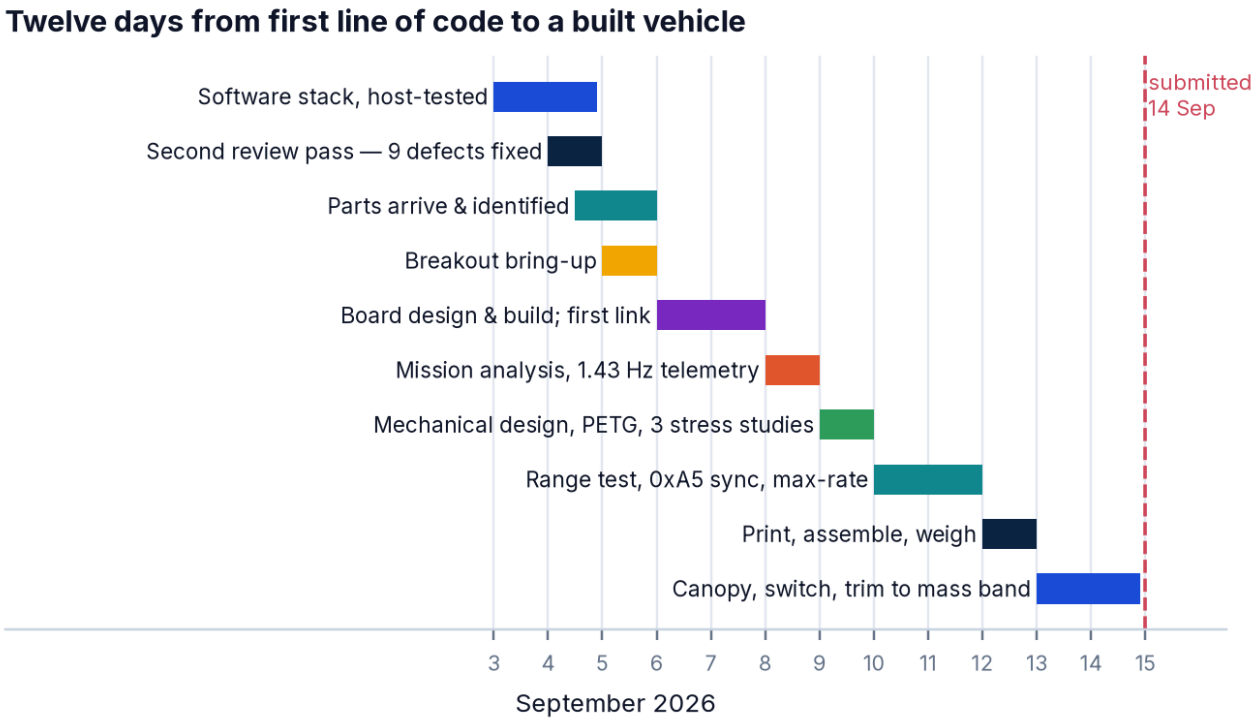


Figure 57 · Development schedule, September 2026.

Table 48 · What happened, day by day

Days	What happened
3 – 4 Sep	The whole software stack, written and tested on the host, then a second pass that found **nine defects in code that already built and passed**
4 – 5 Sep	Parts arrive and are photographed and identified one board at a time; the microSD reader turns out to be a 3.3 V board, closing the power-design question
5 Sep	Breadboard bring-up. The radio answers and transmits, the card initialises and writes, and **two intermittents both turn out to be one long supply jumper**
6 Sep	The board is *designed* rather than assembled: floorplan to scale, coupling analysis, decoupling sized against a failure that actually happened, sixteen gated build steps
7 Sep	The board is built and gates 3–7 pass on it. **The first radio link closes** — 66 packets received. Running the flight image finds the chip-select ordering defect
8 Sep	Telemetry to 1.43 Hz; an authorised erase command; GPS gated on satellite count and HDOP; the first mission analysis, which finds the hover problem
9 Sep	The mechanical design: `Cansat_D1` modelled, 12 cm sided box confirmed, PETG chosen, three stress studies, electronics weighed at 151.299 g
10 – 11 Sep	The first range test; the sync word moves to `0xA5` on both ends; packet budget cut to 200 B; GPS and sound go on the air; `MAX_RATE` measured at 3.11 Hz
12 Sep	The structure is printed; the vehicle becomes a physical object: 280 g on a scale
13 – 14 Sep	Canopy, switch and power LED fitted; mass trimmed into the 450–550 g band; vehicle and report submitted

Days	What happened
30 Sep	**Two flights**, hand-thrown from a building terrace at 29–30 m under the 6 ft canopy. Both descents at 38–45 % of the 5 m/s limit; the analysis in Chapter 14
21 Oct	**First prize** at the SVNIT Physics Club CanSat Competition 2026 (reported by the team). The project is complete and the repository is frozen

17.2 The flight-day analysis was ready before the flights

The rulebook gives four hours after the launch to analyse the data. The analysis pipeline — a one-command tool and a notebook that write the three mandatory graphs, the descent rate by regression, the implied drag coefficient, acceleration, orientation, GPS drift, acoustic level and the comparison of the SD log against the ground log — was written and tested **before there was a flight**, against a synthetic flight whose answers were known: descent rate within 3 %, drag coefficient within 6 %, release time within 0.15 s, and the altitude-formula bias corrected. On the day, the work was to interpret, not to write code; Chapter 14 is that interpretation.

18

CHAPTER 18

Lessons and conclusion

What the project would tell the next team — and what the vehicle achieved.

18.1 Lessons learned

18.2 Conclusion

18.1 Lessons learned

1. Photograph and identify every part on arrival, and trust registers over silkscreen. Chip-ID registers settled the barometer's identity (`0x58` for a BMP280, where a BME280 answers `0x60`) and the IMU's (`0x70`), by the only method that settles them. The inspection of every board paid for itself twice: it closed the power-design question (the SD reader is a 3.3 V board) and it fixed the pin map before the first wire was soldered.

2. The defects that matter most are the ones a test suite cannot reach. Five thousand host assertions did not find the chip-select ordering, the supply jumper, or the modem's real airtime. All needed a powered board. Write the tests — they caught nine real defects in code that already built — but do not mistake them for evidence about hardware.

3. A marginal wire is indistinguishable from a broken component. Two intermittents — a radio that transmitted 5/5 then 0/5, and a card that read perfectly and refused to write — were the same fault on the same kind of wire on two different modules. Shortening the jumper fixed both outright.

4. Estimates are wrong in both directions, and the scale settles it in a minute. The avionics estimate was 31 g low, the structure estimate 64 g high. Both had reasoning behind them and both were labelled as estimates. A weighed number replaced each.

5. Write the operational document early. Writing the concept of operations found the hover problem: a vehicle that declared a landing while hanging under the drone, twelve seconds before release — the same condition as the hold at the terrace edge on 30 September. No test found it because no test described a hover. Describing the mission minute by minute exposed a state the design had never considered — and the fix, the descent gate, is why the state field was correct for every packet of Flight 1.

6. Two systems that must agree should have one definition. The vehicle and the bridge once carried separate copies of the modem parameters; three parsers in three languages disagreed about the packet format. Both were fixed with one shared header and one shared fixture file.

7. Make the documentation fail the build. 307 numbers in the documentation are checked against the source that defines them; the checker caught drift repeatedly, including while this report was written.

8. Write predictions down before the measurement. The bring-up record lists the prediction beside every measurement; the airtime model survived to within 1.8 %, and the descent model's guarantee — that the 5 m/s cap would hold — held with a factor of two to spare.

9. Have the analysis ready before the flight. The four-hour window is short. A tested pipeline turned the organizers' export — duplicated, unordered, four sessions long — into 102 clean packets and a full set of figures without a line of new analysis code.

10. Re-read the code before repeating a number about it. The documentation once said the SD log runs at the 30 Hz sensor rate; the row is appended once per telemetry packet. The 30 Hz figure was true of the acquisition loop and quietly became a claim about the log. A stationary bench log of 2,209 rows in 36.8 minutes had shown the real rate all along, to anyone who divided.

18.2 Conclusion

The CanSat is a complete system: a flight computer that runs one bounded loop and knows nothing of its hardware, a 433 MHz LoRa link held to the organizers' packet format and receiver limits, sensors whose every number says where it came from, a printed structure with measured safety margin, a 6 ft canopy sized from the worst case with margin, and a ground station and an analysis pipeline built alongside it. It was verified in three layers — 6,132 automated checks, 94 bench measurements with predictions recorded in advance, and two flights.

The flights confirmed the design on every quantity they could test. Launched by hand from a terrace like a projectile, both descents were slow (2.27 and 1.88 m/s against a 5 m/s limit, within 10 % of what the model predicted for the 6 ft canopy), straight, and stable; the throw and the canopy opening were visible in every channel at the same instant; Flight 1 delivered 41 packets to the organizers' station at 3.09 Hz; the radio link held a margin of at least 14 dB while held, thrown and descending; the vehicle came back on the air by itself after its landing, calibrated, armed and transmitted for another 13 seconds; and the structure met its arrival at a load below half of what it had been analysed for.

The software was complete before the hardware existed; the hardware was built in twelve days; and the first descents were the first the vehicle ever made. It behaved the way the model said it would. The team won **first prize** at the SVNIT Physics Club CanSat Competition 2026 (reported by the team, 21 October 2026), and the project was closed and its repository frozen that day.

> **THE PROJECT IN ONE LINE**
> A can-sized satellite that senses, decides, transmits and survives — built so that every claim about it can be traced to a measurement.

A

APPENDIX A

Appendices

Flight constants, the repository map, the complete flight dataset, references and a glossary.

A.1 Flight configuration constants

A.2 Repository map

A.3 The complete flight dataset

A.4 References

A.5 Glossary

A.1 Flight configuration constants

From `firmware/flight-computer/src/pico/main.cpp` and `config.hpp`. Every value is validated at start-up by `validate_config()`, which refuses to run a configuration that cannot meet the rulebook.

Table 49 · Flight configuration constants

Constant	Value	Note
`team_id`	`CAN-Team-25`	The formatter refuses the rulebook's `CAN-Team-XX` placeholder
`loop_tick_ms`	2 ms	Bounded twice: scheduling jitter, and draining the GPS UART before its FIFO fills
`sensor_period_ms`	33 ms	30 Hz attitude and altitude-rate update
`telemetry_period_ms`	700 ms	**1.43 Hz.** Hard ceiling 950 ms, enforced three ways
`sd_flush_period_ms`	2000 ms	Appends happen per packet; this is the sync
Watchdog	2000 ms	A hung loop restarts; telemetry resumes automatically
`calib_samples`	80	~2.7 s at 30 Hz
`calib_timeout_ms`	20 000 ms	Then resolve best-effort
`arming_delay_ms`	3000 ms	**and** calibration must have settled
`launch_confirm_ms`	300 ms	Launch condition must hold this long
`launch_altitude_gain_m`	15 m	Or acceleration above 30 m/s²
`min_flight_ms`	3000 ms	Landing detection suppressed until this into flight
`landing_descent_rate_mps`	−2 m/s	The descent gate
`landing_descent_confirm_ms`	1000 ms	Held this long before the gate opens
`landing_confirm_ms`	3000 ms	At-rest condition must hold this long
`post_impact_transmission_ms`	5000 ms	≥ the rulebook's 5 s; a shorter value will not build
`sensor_stale_after_ms`	2000 ms	Time-based, not attempt-based
`sd_max_failures`	10	Consecutive write failures before logging disables itself
`radio_recovery_backoff_ms`	1000 ms	Bounded re-init, never a blocking retry loop
`battery_divider_ratio`	2.0	33 kΩ / 33 kΩ into GP26
`battery_low_voltage`	3.5 V	Raises a warning
`command_window_ms`	300 000 ms	Five minutes from power-on, pre-arm only
`auto_max_rate`	true	Max rate when the window closes, and after a watchdog reset

A.2 Repository map

Repository: github.com/KunjSorathiya/CanSat-Team-Singularity

```
firmware/
  common/              shared telemetry format + SX1278 driver     (cansat::)
  flight-computer/     flight core + Pico HAL + tests              (flight::)
  ground-station/      bridge firmware + USB CRC framing           (ground::)
ground-station/
  software/            Python receive pipeline + tests
  web/                 single-file browser telemetry console
avionics/              per-subsystem summaries against what was measured
electrical/
  schematics/          machine-readable netlist, generated from the firmware
mechanical/            envelope, mass budget, canopy spec, CAD (STEP, F3D), stress studies
simulations/           descent model + tests
analysis/
  flight-2026-09-30/   launch-day analysis: parser, cleaning, statistics, figures, dataset
tools/                 host build, Pico syntax check, LoRa link-budget calculator,
                       netlist and drawing generators, documentation-claim checker,
                       SD-card and flight-log utilities, report figures, SDK stubs
test-data/             shared fixtures the C++, Python and JavaScript parsers all read
documentation/
  requirements/        rulebook, requirement checklist
  mission/             concept of operations
  design/              architecture, protocol, wiring, electrical, link budget
  hardware/            BOM, inspection, assembly, compatibility, GPIO map, datasheets, photos
  project/             timeline, this report and its build (report-2026/)
  testing/             test plan and the bring-up measurement record
  operations/          runbook and launch-day procedure
```

Supporting materials the rulebook asks to be attached or referenced: **schematics and netlist** (`electrical/schematics/`), **wiring diagrams and board layout** (`documentation/hardware/diagrams/`), **CAD** (`mechanical/CAD/Cansat_D1.step` , `.f3d`), **structural studies** (`mechanical/simulation/`), **firmware and ground software** (`firmware/` , `ground-station/`), **the raw flight log and its analysis** (`analysis/flight-2026-09-30/`).

A.3 The complete flight dataset

All 102 distinct packets from the organizers' export, in vehicle order within each session. **S0** pad capture · **F1** Flight 1 · **G1** after landing · **F2** Flight 2. `A` is the altitude as transmitted; `h` is the temperature-corrected height above that power cycle's baseline; `Pr` in Pa; `T` in °C; angles in degrees; acceleration in m/s²; `SN` in mV p-p; `ST` the status field; RSSI in dBm and SNR in dB are the receiving station's measurements. GPS coordinates are in `flight_data_clean.csv` .

Ses	P	Ti (s)	A	h	Pr	T	Ro	Pi	Ya	AX	AY	AZ	SN	ST	RSSI	SNR
S0	420	293.300	-0.7	-0.7	101200.16	31.5	-52.6	-27.1	123.1	4.40	-7.00	5.42	14.5	R003	-105	-6.50
S0	421	294.001	-0.7	-0.7	101200.16	31.5	-51.2	-24.6	121.7	4.21	-6.97	5.59	4.8	R003	-105	-6.75
F1	1585	672.929	27.6	29.2	100884.95	31.4	62.2	16.0	153.1	-2.66	10.51	4.48	10.5	F111	-94	9.00
F1	1586	673.305	27.7	29.2	100884.28	31.4	88.5	20.3	175.4	-3.82	10.03	1.37			-95	9.00
F1	1587	673.601	27.7	29.2	100884.28	31.4	92.9	9.4	166.2	-1.31	9.51	0.11			-96	9.25
F1	1588	673.898	27.7	29.3	100883.60	31.4	101.9	4.1	150.0	-1.91	10.53	-0.84	24.2	F111	-101	7.50
F1	1589	674.272	27.8	29.4	100882.25	31.4	115.0	-0.0	121.8	-0.63	8.26	-5.14			-109	1.50
F1	1590	674.569	27.9	29.5	100881.57	31.4	120.4	-1.4	102.7	0.07	8.34	-5.43			-105	4.25
F1	1591	674.865	27.8	29.4	100882.43	31.4	116.8	-6.3	92.6	1.78	8.85	-4.51	16.1	F111	-107	3.75
F1	1592	675.239	27.8	29.4	100882.92	31.4	117.3	-7.8	89.8	1.34	8.67	-4.50			-106	4.50
F1	1593	675.536	27.8	29.4	100882.92	31.4	115.1	-9.3	87.6	1.68	9.10	-3.76			-104	6.00
F1	1594	675.832	27.8	29.4	100883.11	31.4	118.2	-9.3	85.3	1.32	8.57	-4.79	12.9	F111	-104	6.25
F1	1595	676.208	27.9	29.5	100881.57	31.4	118.4	-11.5	83.9	2.05	8.34	-4.62			-104	6.00
F1	1596	676.505	27.9	29.5	100881.76	31.4	115.7	-11.8	84.2	2.21	8.86	-3.79			-104	6.00
F1	1597	676.801	27.9	29.5	100881.76	31.4	117.3	-9.9	82.4	1.54	8.86	-4.27	10.5	F111	-103	6.50
F1	1598	677.176	27.7	29.3	100883.29	31.4	124.1	-3.8	79.0	-0.18	8.37	-6.78			-103	6.50
F1	1599	677.472	28.6	30.3	100872.48	31.4	104.6	-16.8	111.5	-27.48	-13.36	-40.54			-88	9.50
F1	1600	677.768	29.1	30.7	100867.57	31.4	110.6	28.5	135.3	0.54	1.79	4.56	36.3	F111	-90	9.00
F1	1601															

678.142 | 28.0 | 29.6 | 100880.22 | 31.4 | 10.6 | 9.9 | 42.7 | 0.21 | 1.11 | 6.47 | | | -96 | 8.50 | | F1 | 1602 | 678.439 | 26.7 | 28.3 | 100895.39 | 31.4 | 10.3 | 7.7 | -24.4 | -0.58 | 0.82 | 11.53 | | | -84 | 10.00 | | F1 | 1603 | 678.735 | 24.1 | 25.5 | 100926.96 | 31.4 | 7.4 | -1.6 | 149.6 | 8.41 | -3.45 | 18.27 | 16.1 | F111 | -94 | 9.25 | | F1 | 1604 | 679.111 | 22.9 | 24.2 | 100940.97 | 31.4 | 5.7 | -10.5 | -101.0 | 2.46 | 0.04 | 7.56 | | | -86 | 9.50 | | F1 | 1605 | 679.409 | 22.4 | 23.7 | 100947.54 | 31.4 | -23.7 | 9.3 | -170.0 | 2.56 | 1.16 | 18.05 | | | -81 | 9.50 | | F1 | 1606 | 679.706 | 21.8 | 23.0 | 100954.79 | 31.4 | -33.9 | 10.9 | -175.3 | 1.28 | 0.43 | 9.79 | 10.5 | F111 | -79 | 9.75 | | F1 | 1607 | 680.081 | 21.4 | 22.7 | 100958.84 | 31.4 | 5.1 | 8.6 | -130.7 | 0.72 | 0.40 | 9.35 | | | -87 | 9.50 | | F1 | 1608 | 680.379 | 20.8 | 22.0 | 100966.46 | 31.4 | -18.8 | -13.7 | -37.2 | 2.33 | -0.32 | 14.60 | | | -93 | 9.25 | | F1 | 1609 | 680.675 | 20.2 | 21.4 | 100973.21 | 31.4 | 39.4 | 14.9 | 141.0 | 2.07 | 0.69 | 8.30 | 19.3 | F111 | -89 | 9.75 | | F1 | 1610 | 681.050 | 19.8 | 20.9 | 100978.44 | 31.4 | -0.7 | -6.0 | -152.2 | 0.60 | 0.57 | 7.86 | | | -83 | 9.50 | | F1 | 1611 | 681.347 | 19.0 | 20.1 | 100987.74 | 31.4 | -25.6 | 8.9 | 131.9 | -0.95 | -0.60 | 8.25 | | | -86 | 9.50 | | F1 | 1612 | 681.643 | 18.3 | 19.3 | 100996.68 | 31.4 | 16.2 | -15.8 | 73.8 | -1.17 | 0.49 | 6.55 | 24.2 | F111 | -86 | 9.50 | | F1 | 1613 | 682.017 | 17.7 | 18.7 | 101003.43 | 31.4 | 10.6 | -0.8 | 130.8 | -0.15 | 1.09 | 11.12 | | | -87 | 9.75 | | F1 | 1614 | 682.314 | 17.0 | 18.0 | 101011.39 | 31.4 | 34.7 | 4.5 | 137.8 | -0.45 | 2.32 | 12.60 | | | -96 | 9.25 | | F1 | 1615 | 682.610 | 16.3 | 17.2 | 101020.32 | 31.4 | -10.0 | 24.5 | 36.3 | 0.82 | -0.05 | 8.43 | 12.1 | F111 | -87 | 9.75 | | F1 | 1616 | 682.984 | 15.6 | 16.5 | 101028.28 | 31.4 | 3.6 | -20.7 | -10.2 | 0.86 | 0.13 | 11.28 | | | -90 | 10.00 | | F1 | 1617 | 683.281 | 14.7 | 15.5 | 101039.77 | 31.4 | -21.7 | -12.5 | 95.5 | 2.35 | -0.33 | 12.53 | | | -88 | 9.25 | | F1 | 1618 | 683.578 | 13.8 | 14.6 | 101050.58 | 31.4 | 5.5 | 15.5 | -137.4 | 0.38 | 1.68 | 5.43 | 26.6 | F111 | -89 | 9.25 | | F1 | 1619 | 683.953 | 13.1 | 13.9 | 101058.69 | 31.4 | 6.6 | -2.1 | -160.3 | 0.02 | 0.79 | 10.46 | | | -92 | 9.75 | | F1 | 1620 | 684.249 | 12.2 | 12.9 | 101068.98 | 31.4 | 11.0 | 3.1 | 128.2 | -0.37 | -0.39 | 10.36 | | | -90 | 10.00 | | F1 | 1621 | 684.545 | 11.4 | 12.0 | 101079.11 | 31.4 | 10.8 | 1.1 | 125.6 | -0.03 | 0.42 | 10.39 | 9.7 | F113 | -93 | 9.75 | | F1 | 1622 | 684.920 | 10.4 | 11.0 | 101091.27 | 31.4 | 7.3 | 12.9 | 167.6 | -0.88 | -0.18 | 16.18 | | | -83 | 9.75 | | F1 | 1623 | 685.216 | 9.8 | 10.3 | 101098.55 | 31.4 | -7.3 | -17.9 | 177.6 | 1.18 | -0.29 | 1.70 | | | -90 | 9.75 | | F1 | 1624 | 685.514 | 9.7 | 10.2 | 101099.90 | 31.4 | 57.9 | 60.4 | 153.2 | 0.90 | -0.06 | 0.28 | 4.8 | F113 | -91 | 9.25 | | F1 | 1625 | 685.889 | 9.1 | 9.7 | 101106.14 | 31.4 | 52.5 | 32.3 | -125.7 | -0.60 | 3.27 | 14.08 | | | -94 | 9.25 | | G1 | 1 | 0.000 | 16.8 | -0.1 | 101123.18 | 31.4 | 29.8 | 5.8 | -0.0 | -0.99 | 4.87 | 8.50 | 10.5 | R004 | -90 | 9.00 | | G1 | 2 | 0.375 | 17.0 | 0.0 | 101121.15 | 31.4 | 29.6 | 6.7 | -0.2 | -1.08 | 4.95 | 8.34 | | | -89 | 9.50 | | G1 | 3 | 0.672 | 17.0 | 0.0 | 101121.34 | 31.4 | 29.4 | 6.8 | -0.1 | -1.12 | 4.97 | 8.37 | | | -87 | 9.50 | | G1 | 4 | 0.968 | 17.0 | 0.0 | 101121.52 | 31.4 | 29.3 | 6.9 | 0.1 | -1.14 | 4.98 | 8.36 | 11.3 | R004 | -87 | 9.25 | | G1 | 5 | 1.342 | 17.0 | 0.1 | 101120.85 | 31.4 | 29.2 | 7.0 | 0.3 | -1.13 | 4.98 | 8.38 | | | -88 | 9.75 | | G1 | 6 | 1.638 | 17.0 | 0.0 | 101121.34 | 31.4 | 29.2 | 7.0 | 0.5 | -1.13 | 4.98 | 8.37 | | | -90 | 9.50 | | G1 | 7 | 1.935 | 16.9 | -0.0 | 101122.02 | 31.4 | 29.2 | 7.0 | 0.7 | -1.13 | 4.98 | 8.37 | 6.4 | R004 | -88 | 9.50 | | G1 | 8 | 2.309 | 17.0 | 0.0 | 101121.34 | 31.4 | 29.2 | 7.0 | 0.9 | -1.12 | 4.97 | 8.37 | | | -89 | 9.75 | | G1 | 9 | 2.605 | 17.0 | 0.0 | 101121.34 | 31.4 | 29.2 | 7.0 | 1.1 | -1.13 | 4.98 | 8.36 | | | -89 | 9.50 | | G1 | 10 | 2.902 | 17.0 | 0.0 | 101121.34 | 31.4 | 29.2 | 7.0 | 1.3 | -1.13 | 4.98 | 8.37 | 24.2 | R004 | -89 | 9.00 | | G1 | 11 | 3.276 | 17.0 | 0.0 | 101121.34 | 31.4 | 29.2 | 7.0 | 1.6 | -1.13 | 4.98 | 8.37 | | | -86 | 9.25 | | G1 | 12 | 3.573 | 16.9 | -0.1 | 101122.20 | 31.4 | 29.2 | 7.0 | 1.8 | -1.14 | 4.99 | 8.39 | | | -87 | 9.50 | | G1 | 13 | 3.869 | 16.9 | -0.1 | 101122.20 | 31.4 | 29.3 | 7.0 | 2.0 | -1.12 | 4.98 | 8.38 | 6.4 | R004 | -88 | 9.75 | | G1 | 14 | 4.243 | 17.0 | 0.0 | 101121.34 | 31.4 | 29.3 | 7.0 | 2.3 | -1.13 | 4.98 | 8.37 | | | -89 | 9.00 | | G1 | 15 | 4.539 | 17.0 | 0.1 | 101120.66 | 31.4 | 29.3 | 7.0 | 2.5 | -1.13 | 4.99 | 8.37 | | | -90 | 10.00 | | G1 | 16 | 4.836 | 17.0 | 0.0 | 101121.34 | 31.4 | 29.3 | 7.0 | 2.7 | -1.13 | 4.98 | 8.37 | 5.6 | R004 | -89 | 10.75 | | G1 | 17 | 5.211 | 17.0 | 0.0 | 101121.52 | 31.4 | 29.3 | 7.0 | 2.9 | -1.13 | 4.98 | 8.38 | | | -89 | 9.25 | | G1 | 18 | 5.507 | -0.1 | -0.1 | 101122.20 | 31.4 | 30.7 | 6.6 | -0.0 | -1.12 | 4.97 | 8.36 | | | -88 | 9.75 | | G1 | 19 | 5.804 | 0.0 | 0.1 | 101120.85 | 31.4 | 30.7 | 6.6 | -0.0 | -1.13 | 4.99 | 8.41 | 12.1 | R113 | -88 | 9.50 | | G1 | 20 | 6.178 | -0.0 | 0.0 | 101121.52 | 31.4 | 30.7 | 6.6 | -0.0 | -1.14 | 4.98 | 8.39 | | | -88 | 9.50 | | G1 | 21 | 6.475 | 0.0 | 0.0 | 101121.00 | 31.4 | 30.7 | 6.6 | 0.0 | -1.13 | 4.99 | 8.37 | | | -88 | 10.25 | | G1 | 22 | 6.772 | 0.1 | 0.1 | 101120.32 | 31.4 | 30.7 | 6.6 | 0.0 | -1.13 | 4.98 | 8.37 | 7.3 | R113 | -88 | 10.50 | | G1 | 23 | 7.146 | -0.1 | -0.1 | 101122.35 | 31.4 | 30.7 | 6.6 | 0.0 | -1.13 | 4.98 | 8.36 | | | -88 | 9.75 | | G1 | 24 | 7.442 | 0.0 | 0.0 | 101121.00 | 31.4 | 30.7 | 6.6 | 0.0 | -1.14 | 4.98 | 8.40 | | | -88 | 9.50 | | G1 | 25 | 7.739 | -0.0 | -0.0 | 101121.68 | 31.4 | 30.7 | 6.6 | 0.0 | -1.14 | 4.97 | 8.36 | 13.7 | R113 | -88 | 9.50 | | G1 | 26 | 8.113 | -0.0 | 0.0 | 101121.52 | 31.4 | 30.7 | 6.7 | 0.0 | -1.14 | 4.98 | 8.39 | | | -88 | 10.00 | | G1 | 27 | 8.409 | -0.1 | -0.1 | 101123.03 | 31.4 | 30.7 | 6.6 | 0.0 | -1.13 | 4.97 | 8.38 | | | -88 | 9.75 | | G1 | 28 | 8.706 | -0.2 | -0.1 | 101123.21 | 31.4 | 30.7 | 6.6 | 0.0 | -1.13 | 4.98 | 8.38 | 6.4 | R113 | -88 | 10.00 | | G1 | 29 | 9.080 | -0.2 | -0.1 | 101123.21 | 31.4 | 30.7 | 6.6 | 0.0 | -1.13 | 4.99 | 8.37 | | | -88 | 9.50 | | G1 | 30 | 9.376 | -0.1 | -0.1 | 101122.54 | 31.4 | 30.7 | 6.6 | 0.0 | -1.12 | 4.98 | 8.39 | | | -88 | 9.75 | | G1 | 31 | 9.673 | -0.0 | -0.0 | 101121.86 | 31.4 | 30.7 | 6.7 | 0.0 | -1.13 | 4.97 | 8.37 | 5.6 | R113 | -88 | 10.50 | | G1 | 32 | 10.048 | -0.0 | -0.0 | 101121.86 | 31.4 | 30.8 | 6.7 | 0.0 | -1.14 | 4.98 | 8.36 | | | -88 | 9.25 | | G1 | 33 | 10.344 | -0.0 | -0.0 | 101121.86 | 31.4 |

30.8 | 6.7 | 0.0 | -1.13 | 4.97 | 8.36 | | | -88 | 9.50 | | G1 | 34 | 10.640 | -0.2 | -0.2 | 101123.40 | 31.4 | 30.7 | 6.7 | 0.0 | -1.14 | 4.98 | 8.38 | 5.6 | R113 | -88 | 9.75 | | G1 | 35 | 11.015 | -0.0 | -0.0 | 101121.86 | 31.4 | 30.8 | 6.6 | 0.1 | -1.14 | 5.00 | 8.38 | | | -88 | 9.75 | | G1 | 36 | 11.311 | -0.1 | -0.1 | 101122.72 | 31.4 | 30.8 | 6.7 | 0.1 | -1.16 | 4.98 | 8.37 | | | -88 | 9.25 | | G1 | 37 | 11.608 | 0.0 | 0.0 | 101121.38 | 31.4 | 30.8 | 6.7 | 0.1 | -1.14 | 4.97 | 8.39 | 13.7 | R113 | -88 | 10.25 | | G1 | 38 | 11.984 | 0.2 | 0.2 | 101119.16 | 31.4 | 30.8 | 6.7 | 0.1 | -1.14 | 4.99 | 8.39 | | | -89 | 9.25 | | G1 | 39 | 12.280 | 0.2 | 0.2 | 101119.16 | 31.4 | 30.8 | 6.7 | 0.1 | -1.13 | 4.98 | 8.39 | | | -88 | 9.75 | | G1 | 40 | 12.578 | 0.1 | 0.1 | 101120.70 | 31.4 | 30.8 | 6.7 | 0.1 | -1.14 | 4.97 | 8.35 | 6.4 | R113 | -88 | 10.25 | | G1 | 41 | 12.953 | 0.1 | 0.1 | 101120.70 | 31.4 | 30.8 | 6.7 | 0.1 | -1.15 | 4.98 | 8.38 | | | -90 | 10.00 | | F2 | 148 | 102.901 | 1.0 | 1.1 | 100899.29 | 31.1 | 47.7 | -18.1 | 29.6 | -1.98 | -0.58 | 7.30 | 15.3 | R003 | -87 | 9.50 | | F2 | 149 | 103.600 | -1.0 | -1.1 | 100923.45 | 31.1 | -3.3 | -36.4 | -59.3 | 2.32 | 1.77 | 6.45 | 16.1 | R003 | -90 | 8.50 | | F2 | 150 | 104.301 | -2.3 | -2.4 | 100938.98 | 31.1 | 2.8 | -18.8 | 83.0 | 0.70 | 4.44 | 18.42 | 16.9 | R003 | -92 | 9.00 | | F2 | 151 | 105.000 | -3.7 | -4.0 | 100956.35 | 31.1 | 11.9 | -0.2 | 153.0 | -1.37 | -2.67 | 13.13 | 19.3 | R003 | -89 | 10.00 | | F2 | 152 | 105.704 | -4.6 | -4.8 | 100966.48 | 31.1 | 20.9 | -17.3 | -155.0 | 0.49 | 1.78 | 5.29 | 11.3 | R003 | -94 | 8.75 | | F2 | 153 | 106.401 | -5.7 | -6.0 | 100979.80 | 31.1 | -39.8 | 16.6 | -169.4 | 0.45 | 1.10 | 8.83 | 8.9 | R003 | -98 | 7.75 | | F2 | 154 | 107.101 | -7.1 | -7.5 | 100996.50 | 31.1 | 34.9 | 10.5 | -176.1 | -1.16 | 1.01 | 14.11 | 5.6 | R003 | -97 | 8.00 | | F2 | 155 | 107.801 | -8.3 | -8.8 | 101010.87 | 31.1 | -6.3 | -12.3 | 119.4 | 1.71 | 0.00 | 17.11 | 12.9 | R003 | -100 | 5.50 | | F2 | 156 | 108.501 | -9.5 | -10.0 | 101025.05 | 31.1 | -12.3 | 26.2 | -143.5 | -1.41 | 1.12 | 8.03 | 36.3 | R003 | -100 | 6.00 | | F2 | 157 | 109.202 | -10.5 | -11.1 | 101037.02 | 31.1 | 48.5 | 12.2 | 149.0 | -0.08 | 1.42 | 8.54 | 25.8 | R003 | -93 | 9.00 | | F2 | 158 | 109.901 | -11.7 | -12.4 | 101052.07 | 31.1 | -18.1 | -28.6 | 95.8 | 1.09 | -1.67 | 12.10 | 21.8 | R003 | -94 | 8.75 | | F2 | 162 | 112.700 | -16.5 | -17.5 | 101109.96 | 31.1 | 11.1 | -2.9 | -128.0 | -0.07 | 3.40 | 15.08 | 19.3 | R003 | -98 | 6.75 | | F2 | 163 | 113.400 | -17.9 | -18.9 | 101126.17 | 31.1 | -27.7 | -8.7 | -37.5 | 0.18 | -1.44 | 5.55 | 16.1 | R003 | -97 | 7.50 | | F2 | 164 | 114.100 | -19.6 | -20.7 | 101146.43 | 31.1 | -17.7 | 35.4 | 176.8 | 0.92 | 1.07 | 7.26 | 12.9 | R003 | -106 | 0.00 | | F2 | 166 | 115.501 | -22.4 | -23.7 | 101179.86 | 31.1 | 4.4 | 11.0 | 136.5 | 1.81 | 3.50 | 17.07 | 11.3 | R003 | -100 | 6.00 | | F2 | 169 | 117.600 | -26.3 | -27.8 | 101227.13 | 31.1 | 44.1 | -8.1 | -7.0 | 0.54 | 6.21 | 5.92 | 33.0 | R003 | -97 | 7.50 | | F2 | 170 | 118.301 | -26.9 | -28.4 | 101234.08 | 31.1 | -3.6 | 37.0 | 147.2 | -2.87 | -3.22 | 8.27 | 13.7 | R003 | -101 | 5.00 | | F2 | 171 | 119.003 | -26.9 | -28.5 | 101234.75 | 31.1 | 9.0 | 49.7 | -70.8 | -8.12 | 3.42 | 12.67 | 16.1 | R003 | -105 | 0.75 |

A.4 References

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18. Software used for the analysis and this report: Python, NumPy, SciPy, pandas, Matplotlib, Python-Markdown, Pygments, Graphviz, and Chromium for PDF output. Typefaces: Inter, Space Grotesk and JetBrains Mono (SIL Open Font License).

A.5 Glossary

Term	Meaning
AGL	Altitude above ground level — here, above the pad baseline of the current power cycle
Armed	The vehicle has completed its arming delay and calibration, so launch detection is live (`ST-R11x`)
BMP280	Bosch barometric pressure and temperature sensor
$C_d \cdot S$	Drag coefficient times reference area: the drag area of the descending vehicle
Command window	The first five minutes after power-on, during which the vehicle listens between packets and has not yet armed
CRC	Cyclic redundancy check; here CRC-16/CCITT-FALSE on the USB link and the LoRa payload
Descent gate	The rule that a vehicle cannot be declared landed until it has been observed descending
EWMA	Exponentially weighted moving average
FSPL	Free-space path loss
HDOP	Horizontal dilution of precision, a GPS fix-quality figure
Hypsometric equation	$\Delta h = (R\ T / g)\ \ln(p_o/p)$: height from a pressure ratio at a given air temperature
IMU	Inertial measurement unit: accelerometer and gyroscope
ISA	International Standard Atmosphere (15 °C at sea level)
Lean / rich packet	Mandatory-fields-only packet / packet that also carries GPS, sound and status
LoRa	A chirp-spread-spectrum radio modulation
Max rate	The 3.11 Hz three-slot transmit pattern the vehicle adopts when the command window closes
RSSI / SNR	Received signal strength (dBm) and signal-to-noise ratio (dB) measured by the receiving radio
SF	LoRa spreading factor
Specific force	What an accelerometer measures: 1 g at rest, 0 in free fall
ST-	The status field: state letter, armed, calibrated, active-fault count
Swing angle	The angle between the vehicle's z axis and the measured specific force
Watchdog	A hardware timer that restarts the flight computer if the loop stops feeding it